

Closing the Loop:
Fully Automated Radiotherapy Planning
for Head and Neck Cancer

Ruochen Gao

Colophon

About the cover:

This cover was created with the assistance of ChatGPT 5.5. It was inspired by the aurora that I was fortunate to witness in Abisko, northern Sweden. While the aurora is best known for its spectacular display of light, the interaction between energetic particles and the Earth's atmosphere also produces invisible low-energy X-rays. Unlike these naturally occurring X-rays, those used in radiotherapy are generated and delivered in a controlled manner, allowing them to be precisely shaped to treat cancer while sparing healthy tissue. The luminous VMAT treatment arc is woven into the aurora, creating a visual bridge between nature and medicine. It symbolizes how the same underlying physical principles can, through scientific understanding and engineering, be transformed into technology that improves patient care.

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Closing the Loop: Fully Automated Radiotherapy Planning for Head and Neck Cancer

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Contents

1	Introduction	1
1.1	External beam radiotherapy	1
1.2	Challenges of head and neck radiotherapy	2
1.3	Treatment planning as a multi-criteria optimization problem	3
1.4	Automated radiotherapy planning	5
1.5	Thesis outline	5
2	MCP-MedSAM: A Powerful Lightweight Medical Segment Anything Model Trained with a Single GPU in Just One Day	7
2.1	Introduction	9
2.2	Related work	12
2.2.1	Medical image segmentation	12
2.2.2	SAM	12
2.2.3	Prior knowledge in medical image analysis	13
2.2.4	SAM for medical segmentation	13
2.3	Materials and methods	14
2.3.1	Datasets	14
2.3.2	MCP-MedSAM	15
2.3.3	Training strategy	21
2.4	Experiments and results	22
2.4.1	Experimental setup	22
2.4.2	Evaluation metrics	23
2.4.3	Results	23
2.5	Discussion and conclusion	27
3	On Factors that Influence Deep Learning-Based Dose Prediction of Head and Neck Tumors	33
3.1	Introduction	35
3.2	Materials and methods	37

3.2.1	Patient dataset	37
3.2.2	Data preprocessing and training strategy	38
3.2.3	Evaluation metrics	38
3.2.4	Input and dose grid resolution	40
3.2.5	Input type	40
3.2.6	Loss function	40
3.2.7	Model structure	41
3.2.8	Noise	41
3.3	Results	43
3.3.1	Input and dose grid resolution	43
3.3.2	Input type	43
3.3.3	Loss function	44
3.3.4	Model structure	44
3.4	Discussion	46
3.5	Conclusion	52
4	Clinical DVH Metrics as a Loss Function for 3D Dose Prediction in Head and Neck Radiotherapy	53
4.1	Introduction	55
4.2	Materials	57
4.2.1	Dataset	57
4.2.2	Evaluation metrics	59
4.3	Methods	60
4.3.1	CDM loss formulation	60
4.3.2	Bit-mask encoding	64
4.4	Experiments and results	66
4.4.1	Experiment setup	66
4.4.2	Results	67
4.5	Discussion	69
5	Automated Head and Neck Radiotherapy Planning from Deep Learning-Based Dose Prediction via Dose Mimicking	73
5.1	Introduction	75
5.2	Materials and methods	77
5.2.1	Patient cohort	77
5.2.2	Deep learning dose prediction	77
5.2.3	Dose mimicking and plan generation	78
5.2.4	Evaluation metrics	78
5.2.5	Statistical analysis	81

5.3	Results	82
5.3.1	DVH metrics comparison	82
5.3.2	NTCP comparison	83
5.3.3	Plan complexity comparison	83
5.4	Discussion and conclusion	84
6	Summary and Future Work	89
6.1	Summary	89
6.2	Discussion and future work	90
6.2.1	Toward end-to-end radiotherapy planning	92
6.2.2	Personalized and interactive planning through human preferences and human–AI collaboration	93
6.3	General conclusions	94
7	Nederlandse samenvatting	97
7.1	Samenvatting	97
7.2	Algemene conclusies	98
	References	101
	List of Publications	111
	Portfolio	113
	Acknowledgements	115
	Curriculum Vitae	117

1

Introduction

This thesis focuses on advanced techniques in radiotherapy, with particular emphasis on generalizable medical image segmentation models, deep learning-based dose prediction, and automated treatment plan generation through dose mimicking. This chapter introduces the clinical and methodological background relevant to this thesis, including:

- The clinical principles, workflow, and technological evolution of external beam radiotherapy, with emphasis on photon-based techniques such as VMAT;
- The anatomical and planning challenges associated with head and neck radiotherapy;
- The formulation of radiotherapy treatment planning as a multi-criteria optimization problem;
- Recent advances in automated radiotherapy planning, including deep learning-based dose prediction and the generation of clinically deliverable treatment plans;
- An overview of the thesis structure and the main contributions presented in the subsequent chapters.

1.1 External beam radiotherapy

Cancer treatment typically involves multiple approaches, including surgery, systemic therapy (e.g., chemotherapy and immunotherapy), and radiotherapy (RT). Among these, radiotherapy plays an important role in modern oncology, with approximately half of all cancer patients receiving RT at some point during the course of their disease [1, 2]. Unlike surgery, which removes tumor tissue directly, or systemic therapies that affect the whole body, radiotherapy is a local treatment that delivers ionizing radiation to carefully defined target areas. The goal is to cause irreversible DNA damage in tumor cells while minimizing dose to surrounding healthy tissues [3]. In clinical practice, radiotherapy is mainly delivered in two ways: external beam radiotherapy, where radiation is delivered from outside the patient, and brachytherapy,

where a radiation source is placed inside or close to the tumor. This thesis focuses on external beam radiotherapy.

A typical external beam radiotherapy workflow consists of several sequential steps, including patient intake and consultation, planning CT acquisition, anatomical delineation of target volumes and organs-at-risk (OARs), treatment planning, irradiation, and post-treatment follow-up [4]. These steps are closely interconnected: the planning CT and delineated structures provide the anatomical basis for treatment planning, the resulting plan must be delivered accurately during irradiation, and follow-up is needed to assess tumor response and treatment-related toxicity. This thesis mainly focuses on automating the delineation and treatment planning components within this workflow.

External beam radiotherapy primarily relies on photon or proton beams. Photon beams, generated by medical linear accelerators, deposit dose gradually along their path with both entrance and exit dose. Proton beams, in contrast, exhibit a Bragg peak, allowing dose deposition to be concentrated at a defined depth with minimal exit dose [5]. While proton therapy offers potential dosimetric advantages, photon therapy remains the most widely available and clinically used modality worldwide [6]. This thesis focuses exclusively on photon radiotherapy.

Technological advances have significantly improved photon delivery. Intensity-Modulated Radiation Therapy (IMRT) introduced fluence modulation through multi-leaf collimators (MLCs), enabling highly conformal dose distributions. Volumetric Modulated Arc Therapy (VMAT) further extended this concept by delivering radiation continuously during gantry rotation, dynamically modulating MLC positions, dose rate, and gantry speed, as shown in Figure 1.1. Compared to static-field IMRT, VMAT typically achieves similar or improved plan quality while reducing treatment delivery time [7].

These technological developments have substantially increased plan quality, but they have also increased planning complexity. In this thesis, we mainly focus on VMAT, which represents current clinical practice.

1.2 Challenges of head and neck radiotherapy

Head and neck cancer (HNC) represents one of the most challenging disease sites in radiotherapy. This region contains many OARs densely distributed around tumors that often exhibit complex and irregular geometries [9]. Accurate anatomical delineation is therefore essential for treatment planning, as segmentation of planning target volumes (PTVs) and OARs defines the geometric basis for subsequent dose optimization.

However, segmentation in the head and neck region remains challenging. Many critical structures are small, thin, or poorly visible on CT, the primary imaging modality used in the radiotherapy workflow. Multimodal imaging such as MRI or PET is frequently incorporated to improve visualization, but this also introduces additional

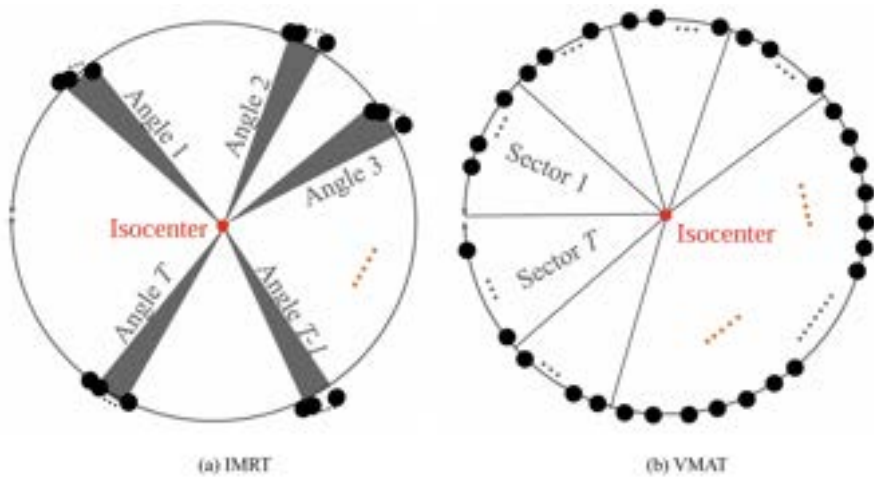


Figure 1.1: Comparison of IMRT (a) and VMAT (b). IMRT plans consist of multiple fixed beams, each containing several control points at identical gantry angles. In contrast, VMAT delivers radiation continuously along a designated arc, where control points are evenly distributed and may be grouped into multiple sectors during planning and optimization. Figure adapted from [8].

complexity in image registration and cross-modality interpretation. Substantial inter-observer variability has therefore been reported in clinical contouring practice [10]. These challenges have motivated the development of robust and generalizable segmentation approaches that can operate reliably across imaging modalities and anatomical variations, providing a consistent foundation for downstream automated workflows. Recent advances in general-purpose segmentation models, such as the Segment Anything Model (SAM) [11], have demonstrated strong generalization across domains, although adaptation to medical imaging remains an active area of research [12, 13, 14, 15, 16].

Even with accurate segmentation, treatment planning in this region remains intrinsically difficult. Effective treatment requires sharp dose falloff between the PTVs and nearby OARs, as even small dose deviations may lead to clinically significant side effects, including xerostomia, dysphagia, or neurological injury. For example, irradiation of the salivary glands may reduce saliva production and cause chronic dry mouth [17]. The presence of multiple competing dose constraints and the need for rapid dose falloff further increase the complexity of treatment planning for HNC.

1.3 Treatment planning as a multi-criteria optimization problem

Radiotherapy treatment planning is formulated as an inverse problem. Given a patient's anatomy and prescribed dose, the planner must determine machine parameters

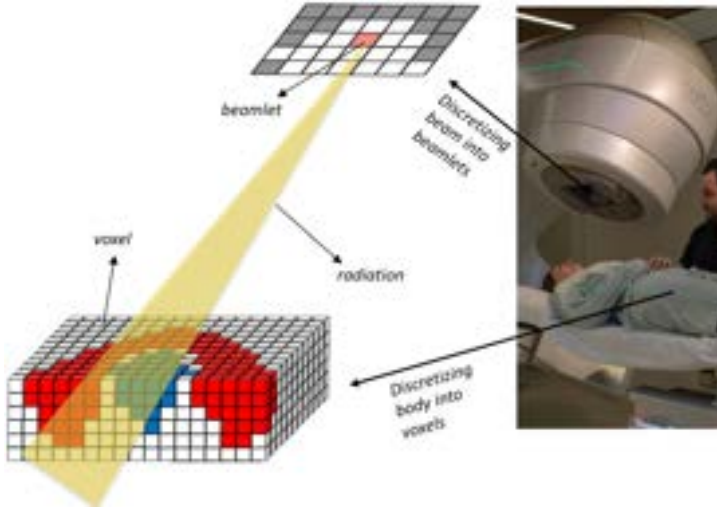


Figure 1.2: Mathematical modeling of radiotherapy dose calculation, with the beam decomposed into beamlets and the patient represented by voxels. Figure reproduced from <https://github.com/PortPy-Project/PortPy>.

that produce an optimal dose distribution. Mathematically, dose calculation can be expressed as $d = Ax$, where d represents voxel dose, x denotes beamlet intensities, and A is the dose influence matrix describing the physical relationship between beamlets and voxel dose deposition [18, 19] (Figure 1.2). This discretized beamlet-voxel model forms the computational foundation of radiotherapy planning algorithms.

Because treatment goals involve competing clinical priorities, radiotherapy planning is inherently a multi-criteria optimization (MCO) problem [18, 19]. Typical objectives include maximizing tumor target coverage while minimizing dose to multiple OARs, which are often conflicting. The set of optimal trade-off solutions is commonly described by the Pareto surface [18], where no objective can be further improved without degrading at least one other objective. Clinical plan evaluation frequently relies on dose-volume histogram (DVH) metrics that reflect these trade-offs between target coverage and organ sparing.

In practice, planners iteratively adjust objective weights and constraints to navigate the Pareto surface of achievable solutions. As delivery techniques such as VMAT introduce a larger number of decision variables and delivery constraints, the search space becomes even more complex.

Furthermore, an optimal dose distribution must ultimately be deliverable, meaning it can be translated into physically feasible MLC leaf trajectories and machine parameters for VMAT delivery. Therefore, automated planning must address not only optimality in dose space, but also feasibility in machine parameter space [8, 19].

1.4 Automated radiotherapy planning

The increasing complexity of modern radiotherapy planning has led to growing interest in automated solutions. Advanced delivery techniques such as VMAT involve high-dimensional parameter spaces, and treatment planning often requires iterative adjustment of optimization objectives to balance target coverage and OAR sparing. This trial-and-error process is time-consuming, cognitively demanding, and dependent on planner experience, which may lead to variability in plan quality [20, 21]. Automated approaches therefore have the potential to improve efficiency, consistency, and accessibility of high-quality radiotherapy planning.

Recent advances in deep learning have enabled three-dimensional dose prediction directly from patient anatomy [22, 23, 24, 25, 26, 27, 28]. Rather than manually navigating the multi-criteria optimization process, neural networks can learn the mapping from anatomical structures to clinically acceptable dose distributions using historical treatment plans. Such models can implicitly capture trade-offs between target coverage and normal tissue sparing, offering a potential pathway toward efficient and consistent automated planning.

However, predicted dose distributions exist only in voxel space and are not directly deliverable. Practical VMAT delivery requires explicit machine parameters, including multi-leaf collimator trajectories, gantry motion, and dose rate modulation, all subject to mechanical and physical constraints. Bridging the gap between predicted dose distributions and physically deliverable treatment plans is therefore essential for clinical implementation.

One practical strategy is dose mimicking [29, 30], in which the predicted dose distribution is incorporated into a treatment planning system as a reference objective to guide inverse planning and generate deliverable machine parameters. By combining dose prediction with dose mimicking, it becomes possible to move from patient anatomy to clinically deliverable VMAT treatment plans within an automated planning framework.

1.5 Thesis outline

This thesis investigates automated radiotherapy treatment planning for HNC, with a primary focus on deep learning-based dose prediction and the generation of clinically deliverable VMAT plans through dose mimicking. In this thesis, automated radiotherapy planning mainly refers to the stages from dose prediction to deliverable plan generation.

In addition to the radiotherapy planning components, this thesis also explores generalizable medical image segmentation methods. Although the segmentation study presented in Chapter 2 is developed for broader medical imaging scenarios, robust and scalable segmentation models are highly relevant to radiotherapy workflows,

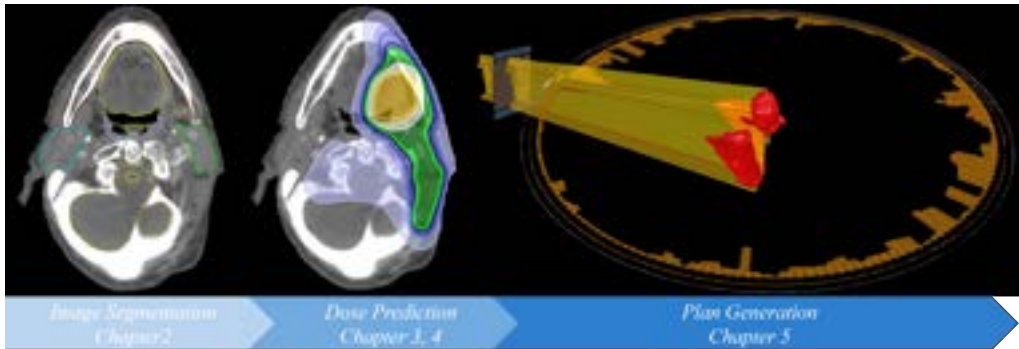


Figure 1.3: Overview of research topics in this thesis.

where accurate anatomical delineation forms the basis for reliable treatment planning and automation. Figure 1.3 provides an overview of the research topics and their relationships.

The individual contributions are structured as follows:

- **Chapter 2** presents MCP-MedSAM, a lightweight modality-agnostic medical image segmentation foundation model. The chapter addresses multi-modality medical segmentation challenges and establishes a scalable delineation framework applicable across anatomical sites and imaging modalities, including but not limited to radiotherapy.
- **Chapter 3** investigates key factors influencing deep learning-based 3D dose prediction for head and neck photon therapy, including input and dose-grid resolution, input representation, loss functions, model architecture, and the impact of noise on model performance.
- **Chapter 4** proposes a clinical DVH metric loss (CDM loss) that jointly incorporates differentiable formulations of common *D-metrics* and differentiable surrogates of *V-metrics*. Additionally, a lossless bit-mask encoding scheme is introduced to efficiently represent a large number of overlapping regions of interest, significantly improving training efficiency.
- **Chapter 5** integrates the trained dose prediction model into a commercial treatment planning system (RayStation) and performs automated dose mimicking to generate clinically deliverable VMAT plans.
- **Chapters 6 and 7** summarize the overall contributions, implications, and future research directions in English and Dutch, respectively.

2

MCP-MedSAM: A Powerful Lightweight Medical Segment Anything Model Trained with a Single GPU in Just One Day

This chapter was adapted from:

Ruochen Gao, Donghang Lyu, and Marius Staring, "Swin-LiteMedSAM: A lightweight box-based segment anything model for large-scale medical image datasets", in Medical Image Segmentation Challenge, pp. 70–82, Cham: Springer Nature Switzerland, 2024.

Donghang Lyu, **Ruochen Gao**[†], and Marius Staring, "MCP-MedSAM: A Powerful Lightweight Medical Segment Anything Model Trained with a Single GPU in Just One Day", Machine Learning for Biomedical Imaging, vol. 3, pp. 135-151, May 2025.

Abstract

Medical image segmentation involves partitioning medical images into meaningful regions, with a focus on identifying anatomical structures and lesions. It has broad applications in healthcare, and deep learning methods have enabled significant advancements in automating this process. Recently, the introduction of the Segment Anything Model (SAM), the first foundation model for segmentation task, has prompted researchers to adapt it for the medical domain to improve performance across various tasks. However, SAM’s large model size and high GPU requirements hinder its scalability and development in the medical domain. To address these challenges, research has increasingly focused on lightweight adaptations of SAM to reduce its parameter count, enabling training with limited GPU resources while maintaining competitive segmentation performance. In this work, we propose MCP-MedSAM, a powerful and lightweight medical SAM model designed to be trainable on a single A100 GPU with 40GB of memory within one day while delivering superior segmentation performance. Recognizing the significant internal differences between modalities and the need for direct segmentation target information within bounding boxes, we introduce two kinds of prompts: the modality prompt and the content prompt. After passing through the prompt encoder, their embedding representations can further improve the segmentation performance by incorporating more relevant information without adding significant training overhead. Additionally, we adopt an effective modality-based data sampling strategy to address data imbalance between modalities, ensuring more balanced performance across all modalities. Our method was trained and evaluated using a large-scale challenge dataset, compared to top-ranking methods on the challenge leaderboard, MCP-MedSAM achieved superior performance while requiring only one day of training on a single GPU.

2.1 Introduction

As a key task in the field of medical image analysis, medical image segmentation serves as the foundation for numerous clinical applications, such as disease diagnosis, treatment planning, and surgical interventions [31]. Accurate medical image segmentation can precisely identify anatomical structures and pathological regions, leading to more informed medical decisions. Over the last decade, deep learning-based segmentation models like nnU-Net [32] have achieved significant success in medical image segmentation. These models are tailored to specific imaging modalities (e.g., MRI, CT, ultrasound, as shown in Figure 2.1) and particular diseases (e.g., prostate tumors, pulmonary nodules). This specialization means that different scenarios require different models. Additionally, their optimization for specific datasets limits their generalization across diverse data distributions [13]. Therefore, the development of a universal model for medical segmentation becomes increasingly promising in this context, offering the potential to unify approaches across various imaging modalities and clinical applications.

The introduction of the Segment Anything Model (SAM) [11], the first foundation model for image segmentation, marked a significant breakthrough by offering a framework capable of generalizing across a wide variety of natural images. SAM's robust generalization and zero-shot capabilities have paved the way for the potential development of general medical segmentation models, enabling adaptation across diverse medical image modalities while achieving strong performance on each. Building on SAM's foundation, MedSAM [13] was developed using a large-scale medical dataset covering 10 different imaging modalities and 30 types of diseases. This model has demonstrated superior overall performance compared to traditional, modality-specific specialist models, making a paradigm shift in medical image segmentation. By reducing the need to develop separate models for each imaging modality, it significantly streamlines the entire segmentation process.

Similar to many foundational models, SAM demands significant computational resources, including large-scale GPU clusters and long training time. These requirements limit the applicability and research on the SAM model, especially for smaller research groups and academic institutions that lack enough computational resources. Many previous studies on applying SAM to medical imaging concentrate on freezing the image encoders weights for direct inference while adding supplementary components for adaptation [33, 34], or introducing trainable adapters into the frozen image encoder [35, 16]. However, the simplicity of these introduced components limits their effectiveness, making it challenging for models to achieve optimal performance, especially when dealing with new imaging modalities or segmentation targets. Re-

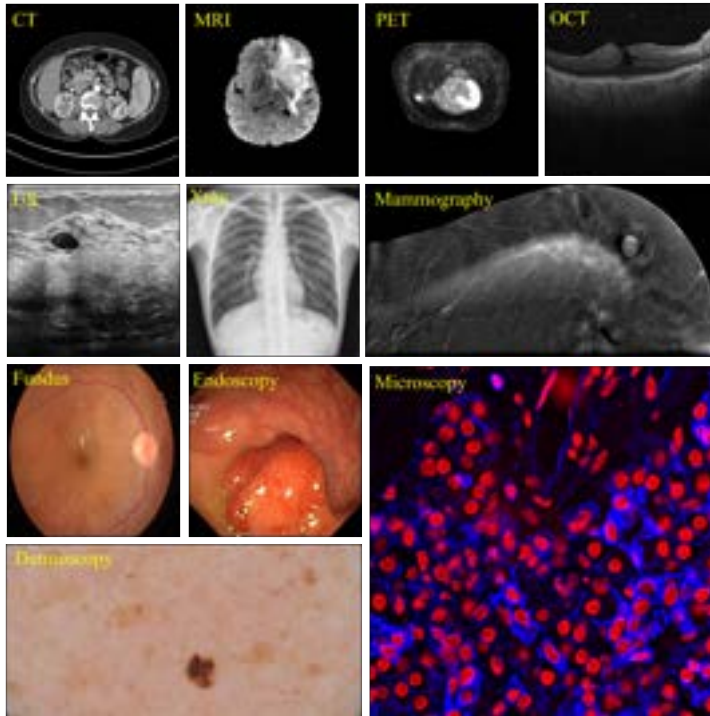


Figure 2.1: Examples of various medical imaging modalities.

cently, some efforts have been made by LiteMedSAM¹, which distills the heavy image encoder into a lightweight version. This significantly reduces model parameters and curtails computational resource consumption, making MedSAM become more accessible for broader research and application. However, this efficiency comes at the expense of segmentation accuracy. Additionally, due to LiteMedSAM’s original training strategies, its performance is susceptible to imbalances in imaging modalities within the training data, further limiting its robustness across diverse medical imaging scenarios. Therefore, maintaining optimal overall performance of the MedSAM with a limited computational resource consumption still remains a critical goal.

MedSAM’s structure consists of three key components: an image encoder, a prompt encoder, and a mask decoder. While most prior research has primarily focused on optimizing the image encoder and mask decoder, the prompt encoder is usually left unchanged, typically relying on widely used prompt types such as boxes, points, and scribbles. Although these prompts offer valuable spatial hints to help the MedSAM segment targets more accurately, they still require manual effort for annotation, adding to the labor costs. Moreover, determining the optimal quantity and placement of points

¹<https://github.com/bowang-lab/MedSAM/tree/LiteMedSAM>

and scribbles within the given box remains challenging, impacting both model training and final segmentation performance [12]. Therefore, optimizing the prompt encoder with some more effective and stable prompts could be beneficial, as it aims to enhance segmentation accuracy consistently without significantly increasing computational resources and labor costs. In this paper, we introduce **MCP-MedSAM**, which builds on the LiteMedSAM architecture to reduce computational resource costs and accelerate the training process. It incorporates two new types of prompts, a modality prompt and a content prompt, into the prompt encoder, generating sparse embedding and dense embedding representations and integrating them within the mask decoder to further improve segmentation accuracy. For the modality prompt, it is a learnable prompt and its embedding representation aims to take into account the differences among various imaging modalities, enriching the modality-specific information and helping the model adapt to diverse input characteristics. Meanwhile, the embedding representation of content prompt has two types: a sparse one and a dense one, both of which aim to leverage the direct information about the target within the given box but from different perspectives. Additionally, considering the strong zero-shot capability and extensive applications of the CLIP [36] model, it is effectively integrated into the processing networks of both prompts to further enhance their representations. We also investigated various data sampling strategies and identified the most effective one to mitigate the negative impacts brought by the original imbalance. This resulted in enhanced overall segmentation performance, more balanced outcomes across modalities, and a faster training process. In summary, the key contributions of this work are shown as follows:

- **Modality and Content Prompts:** We introduce two new types of prompts into the prompt encoder part of the MedSAM framework: the modality prompt and the content prompt. By generating effective embedding representations and integrating them with the mask decoder, the model's performance across all imaging modalities can be further enhanced without significantly increasing computational resource costs.
- **Lightweight Architecture with Efficient Data Sampling Strategy:** We explore multiple data sampling strategies to identify the most efficient one, aiming to mitigate the effects of data imbalance and further accelerate the training process so that it converges within one day.
- **Efficient Medical Segmentation Model for Broad Accessibility:** After evaluation on a large and diverse set of imaging data, our model demonstrates that it is feasible to achieve high-quality medical image segmentation without the need for extensive computational resources and long training time. This encourages more researchers to adopt and further explore general-purpose medical segmentation models.

2.2 Related work

2.2.1 Medical image segmentation

Medical image segmentation has seen substantial advancements through the adoption of deep learning methods. The introduction of the U-Net model by [37] marks a significant milestone with its U-shaped architecture, effectively combining convolutional layers with symmetric contracting and expanding paths. Its widespread success inspires a host of variants aimed at boosting performance and tackling specific segmentation targets [32, 38, 39, 40, 41] by incorporating convolutional layers and attention mechanisms into the model design. Among them, nnU-Net [32] stands out as a representative approach, leveraging optimized pre-processing and post-processing to achieve strong medical segmentation performance for each specific medical task. It has been widely adopted in various competitions and real-world applications. For instance, TotalSegmentator [42, 43] offers comprehensive and practical solutions for multi-organ segmentation tasks in CT and MRI modalities.

Moreover, transformer modules have been widely integrated into U-shaped architectures for medical segmentation tasks, as transformers excel in extracting global contextual features via their self-attention mechanisms. Notable examples include UNETR [44] and SwinUNETR [45], which incorporate transformer modules into the encoder part of the U-Net architecture, yielding enhanced segmentation performance. Likewise, some works [46, 47, 48] integrate CNN and transformer modules to leverage the strengths of both architectures and enhance segmentation performance. With the rise of Mamba [49], recent works [50, 51, 52] have explored integrating the Mamba module into the U-Net architecture for further improvement. Despite these advancements, the aforementioned models are generally tailored to specific segmentation tasks and exhibit limited generalizability across various medical imaging modalities.

2.2.2 SAM

SAM [11] is an innovative model that aims to provide a versatile and generalizable solution for segmenting objects in images. There are two key concepts for the success of the SAM model: i) introducing multiple types of prompts, such as bounding boxes, points, and coarse masks, which allows the model to precisely identify and segment the target area; ii) training SAM with a huge amount of data, which enables SAM to adapt to a wide range of segmentation scenarios easily, reflecting its robust zero-shot capability. SAM has gained significant attention, leading to numerous recent works aimed at improving its performance and efficiency. HQ-SAM [53] adopts a minimal adaptation approach by introducing a High-Quality output token and fusing global and local features from the image encoder to obtain high-quality features. In

this way, HQ-SAM performs better on fine-grained segmentation tasks. SEEM [54] introduces more kinds of prompts, including points, boxes, masks, scribbles and text prompts, and learns to deal with them by combining visual and text information in a joint visual-semantic space. This approach enhances segmentation performance and enables zero-shot adaptation to unseen user prompts. Then SAM2 [55] extends the original SAM by enabling both image and video segmentation. By incorporating a memory mechanism, SAM2 can effectively process video data, leading to improved segmentation performance and broader applications. Although SAM models can achieve impressive segmentation performance, it is challenging to train without sufficient computational resources. SAM uses ViT-H [56] as the image encoder and unifies the input image size to 1024×1024 , both of which contribute to substantial GPU memory usage and make training become time-consuming. Therefore, there are also some works focusing on enhancing SAM's efficiency to make it more suitable for widespread real-world use. MobileSAM [57] distills the image encoder from ViT-H to a tiny ViT model, significantly reducing the parameter count. Furthermore, EfficientViT-SAM [58] and RepViT-SAM [59] replace ViT-H with some lightweight ViT variants, achieving better overall performance with significantly fewer parameters.

2.2.3 Prior knowledge in medical image analysis

For a model designed to handle multiple tasks, incorporating prior knowledge, such as features of anatomical structures or modality-specific characteristics, can enhance learning by helping the model recognize internal differences between tasks. DoD-Net [60] employs one-hot embeddings to represent different organs, combining these embeddings with image features. This integrated representation is then fed into the output head, enabling DoDNet to segment tumors across various organs. Uniseg [61] develops a learnable universal prompt that combines with sample-specific features to create prompts for multiple tasks. Task-specific prompts are then selected based on the task ID, allowing Uniseg to perform segmentation across multiple organs and modalities. Then MedPrompt [62] introduces a self-adaptive prompt block designed to learn and incorporate cross-modal information, enhancing the model's ability to translate effectively across different modalities. Hermes model [63] introduces two kinds of learnable prompts, task-specific and modality-specific, which interact with the model in the bottleneck part to guide the model, enhancing segmentation performance across multiple organs and modalities. Prior knowledge has been effectively used in many models for multi-task processing but remains unexplored in the MedSAM framework for enhancing segmentation across various tasks.

2.2.4 SAM for medical segmentation

Inspired by the success of SAM, some works have begun exploring the utilization of SAM for medical segmentation tasks. MedSAM [13] is proposed to adapt SAM

for medical segmentation tasks by training it with a large dataset of medical images. Likewise, SAM-Med2D [35] fine-tunes the SAM model using a large-scale medical dataset and incorporates a variety of comprehensive prompts, including bounding boxes, points, and masks, rather than relying on just one type of prompt. Med-SA [16] extend the SAM structure by introducing adaptors that highly enhance the capabilities for medical applications, enabling it to work with both 2D and 3D medical data. Furthermore, beyond the traditional prompts introduced by SAM, the ScribblePrompt [64] model explores the utilization of scribble prompts, resulting in improved overall performance. Similarly, SAT [65] incorporates text prompts into the SAM model, offering contextual medical knowledge about modalities, anatomies, and body regions. With the release of SAM2, researchers have begun exploring its potential in medical segmentation, particularly its application for 3D medical segmentation. MedSAM-2 [14] extends SAM2 for 2D and 3D medical segmentation by training on a large-scale medical dataset and incorporating a self-sorting memory bank to efficiently select informative embeddings, enhancing overall performance.

Furthermore, lightweight MedSAM models can be obtained by applying the techniques from lightweight SAM models and training them on medical data. Med-efficientSAM [66] distills the knowledge into an EfficientViT [67] image encoder and fine-tunes it with a large-scale medical dataset, while Rep-MedSAM [68] chooses to distill knowledge into a RepViT [69] image encoder. DAFT [70] also adopts the EfficientViT-SAM structure and introduces a data-aware fine-tuning policy inspired by the mixture of experts (MoE) [71] concept to further improve the performance of each modality. Swin-LiteMedSAM [72] introduces a tiny Swin Transformer [73] as the image encoder and builds skip connections between the image encoder and mask decoder. Furthermore, instead of solely using box prompt, box-based points and scribble prompts are automatically generated based on the provided bounding box. Then, LiteMedSAM-Rep [74] uses tiny Swin Transformer as the image encoder and trains the whole model from scratch, subsequently distilling the image encoder into a more lightweight RepViT while keeping the prompt encoder and mask decoder fixed. Although the above methods have made some progress in lightweighting the MedSAM, striking an optimal balance between training efficiency and performance remains a challenge.

2.3 Materials and methods

2.3.1 Datasets

For this study, we used the dataset [15] from the CVPR 2024 competition titled “SEGMENT ANYTHING IN MEDICAL IMAGES ON LAPTOP”² to train and test the

²<https://www.codabench.org/competitions/1847>

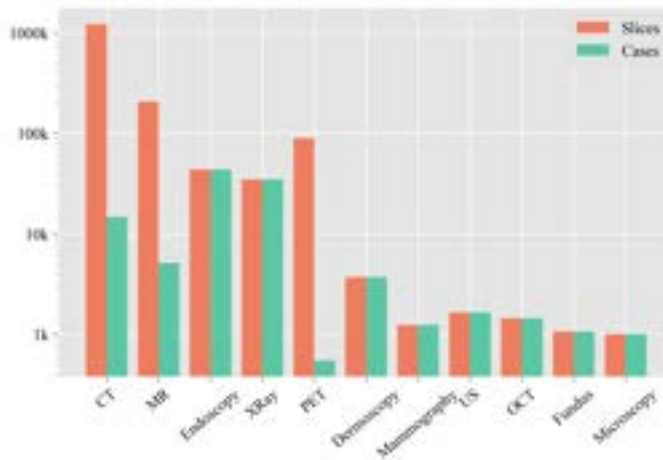


Figure 2.2: Data distribution across imaging modalities in the training set, with the y-axis displayed on a logarithmic scale for enhanced visualization.

model. The training dataset comprises over one million paired 2D images and their corresponding segmentation masks across 11 distinct imaging modalities, including Computed Tomography (CT), Magnetic Resonance Imaging (MRI), Positron Emission Tomography (PET), X-ray, Ultrasound, Mammography, Optical Coherence Tomography (OCT), Endoscopy, Fundus imaging, Dermoscopy, and Microscopy. The distribution of the training dataset across these modalities is illustrated in Figure 2.2. As the testing set was not released after the challenge, we used the competition’s validation set as our testing set. Then this testing set comprises 3,278 samples from 9 imaging modalities, excluding Mammography and OCT.

2.3.2 MCP-MedSAM

Overall, MCP-MedSAM follows the original design of SAM framework, which is composed of three parts: image encoder, prompt encoder and mask decoder, as shown in Figure 2.3. We introduce two additional prompts into the prompt encoder and design a corresponding network to generate effective representations, enriching the output of prompt encoder. Additionally, the mask decoder is modified to better align with the representations of these prompts. Notably, similar to MedSAM, MCP-MedSAM is designed for 2D medical data and processes 3D medical data slice-by-slice.

While MedSAM-2 delivers strong segmentation performance, its training dataset is limited in both volume and modality diversity. MCP-MedSAM, in contrast, aims for faster convergence and broader modality support, requiring a robust pre-trained image encoder. To achieve this, we build on the lite version of MedSAM, which is trained on a larger, more diverse dataset spanning multiple modalities. A pre-

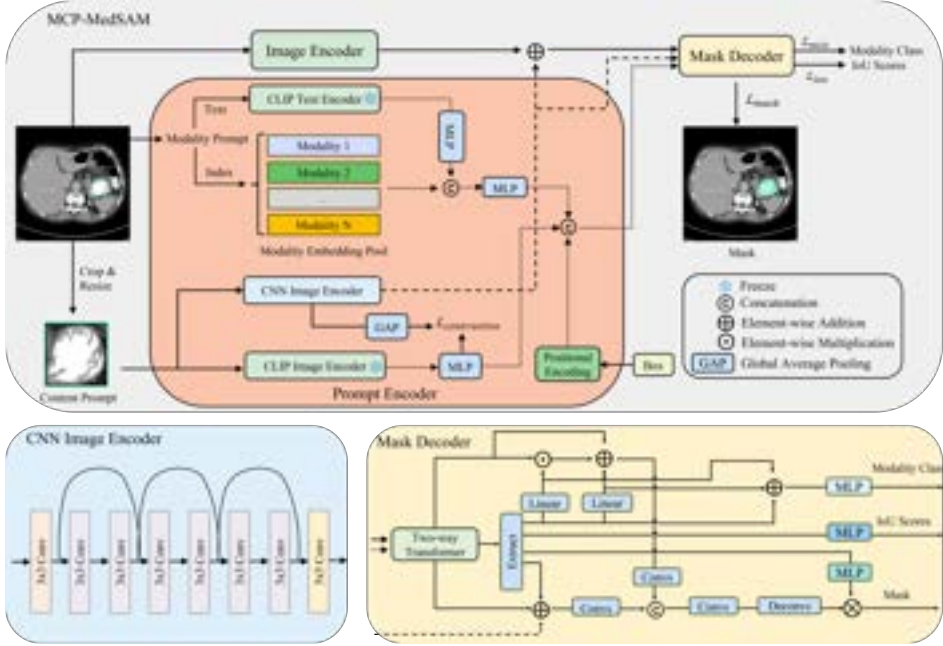


Figure 2.3: The top section provides an overview of MCP-MedSAM, highlighting our newly introduced content prompt and modality prompt in comparison to the SAM framework. Notably, dotted lines indicate the flow of all sparse embedding representations, while dashed ones represent the direction of dense representation. The detailed architectures of the CNN Image Encoder and the modified Mask Decoder are illustrated in the bottom section.

trained tiny ViT¹ image encoder from LiteMedSAM is used to accelerate convergence by providing some prior information. In addition to the traditional box prompt, we incorporate modality and content prompts into the prompt encoder to enhance relevance and robustness. The modality prompt consists of a modality text and an index for extracting the corresponding modality embedding, while the content prompt takes a cropped, resized image from the specified bounding box as input. Furthermore, the embedding representations of these two prompts are integrated into the mask decoder to better fuse modality and content information, resulting in improved segmentation performance. The mask decoder produces three outputs: the target mask, a predicted IoU score, and an additional prediction for the modality class.

2.3.2.1 Modality prompt

Considering the unique characteristics of each modality, integrating modality information into LiteMedSAM is advantageous. Consequently, we introduce the modality prompt to enrich the sparse representation output by the prompt encoder. The modality

prompt consists of two components: a modality text for generating a text embedding and a modality index i for retrieving a modality-specific learnable embedding from the embedding pool $P \in \mathbb{R}^{N \times F}$. Here, N is the number of modalities, F denotes the embedding length, and modality index $i \in \{1, \dots, N\}$ corresponds to a specific modality. In this context, we design each learnable embedding to encapsulate modality-specific information after training, while the text embedding provides a feature representation of the modality from a unique perspective. Given prior information of the modality class, we generate the corresponding text input, $\{Modality\}$ Image, and pass it through the frozen CLIP text encoder to get text embedding. Although the CLIP model incorporates some medical-related prior information through its pre-trained weights, the training set contains a broader range of modalities and includes many unseen segmentation targets, adding complexity to the task. To address this, an MLP is added after the text encoder to tune and to adjust the channel dimensions simultaneously. On the other hand, a corresponding learnable modality-specific embedding is selected from the modality embedding pool based on the modality index. Then using an MLP to combine these two embeddings allows them to complement each other, creating a more complete and comprehensive modality representation that delivers modality-specific information effectively. The final modality embedding is appended to the sparse embedding. Notably, both MLPs in the modality prompt share the same structure, consisting of two linear layers with a GELU activation function. The input channel size is 512, while the output channel size is 256.

2.3.2.2 Content prompt

The prompt encoder typically takes two types of input prompts: a sparse prompt and a dense prompt. The dense prompt serves as an initial coarse mask of the target, generating a dense embedding representation through a series of convolutional layers. When the dense prompt is unavailable, the dense embedding representation is instead modeled as a learnable matrix with weights initialized from a uniform distribution. However, the learnable matrix is challenging to train effectively and lacks initial information about the target, while the coarse mask requires significant labor. Therefore, we introduce a content prompt to effectively capture target information within the specified box and generate a dense representation enriched with target features. Additionally, alongside the dense representation, we also extract a sparse embedding containing content information, aiming to enable more direct interaction with other sparse embeddings. Consequently, the final embedding representation of the content prompt has two components: a sparse one and a dense one, each capturing content information from different perspectives. By leveraging both components, the model can achieve a more comprehensive understanding of the content within the specified box, improving its ability to segment the target with greater accuracy. Given a

bounding box, the content within the box is cropped and resized into a new shape first. The sparse content representation is derived by processing the reshaped input through a frozen CLIP image encoder, followed by an MLP with the same structure as the MLPs in the modality prompt part (two linear layers with GELU activation function, input channel size is 512 and output channel size is 256). This approach ensures that the sparse content embedding captures the comprehensive content information. For the dense content representation, the image is processed through a CNN image encoder based on the ResNet architecture [75], consisting of multiple convolutional layers with skip connections. With an input channel size of 3, all subsequent convolutional layers maintain a consistent output channel size of 256. Given the small input size, this streamlined CNN network can efficiently extract local detailed features and preserve critical content information about the target. Since both the sparse and dense content embedding representations are generated from the same content, they should exhibit strong similarity. To achieve this, the dense content representation is passed through a global average pooling (GAP) layer to produce a sparse embedding. A contrastive loss is then applied between this sparse representation and the corresponding sparse content embedding to align their learning.

2.3.2.3 Mask decoder

In the mask decoder, specific operations are employed to better fuse information from representations of both prompts, enhancing the final performance. After passing the sparse prompts and image embedding through the two-way transformer block, the updated modality embedding is extracted from its corresponding position of the sparse embeddings. In this context, two-way transformer block is a key component of the original SAM [11] mask decoder. It employs self-attention on tokens and bidirectional cross-attention between tokens and image embeddings, facilitating effective information exchange between image embeddings and tokens, which serve as sparse embeddings. These attention mechanisms also ensure the extracted modality embedding integrates both modality-specific and visual information. Then inspired by the thought of FiLM [76], we make the extracted modality embedding pass through two separate linear layers to produce weight and bias embeddings. These embeddings are subsequently combined with the dense matrix output from the two-way transformer through multiplication and addition operations, effectively incorporating modality-specific information. The combined matrix is further processed through two convolutional layers with a skip connection to join in the mask generation process. Furthermore, to better guide the network’s learning of the modality prompt within the prompt encoder, we integrate a classification head into the mask decoder, following the same MLP structure as in the prompt encoder but with different input and output channels (two linear layers with GELU activation function, the input

channel size is 256 and output channel size is the number of modalities, which is 11 in our experiments). This classification head takes the combination of the extracted embedding along with the corresponding weight and bias embeddings as input and predicts the corresponding modality class.

For the dense content representation, inspired by the global-local feature fusion in HQ-SAM [53], we combine it with the dense matrix output from the two-way transformer, which carries global visual information. Furthermore, the sparse content embedding is extracted from the sparse embedding output of the two-way transformer and integrated into this combination as a bias, further enriching the overall information. Finally, similar to the processing of modality information, the resulting matrix is processed through two convolutional layers with a skip connection to refine the features. We concatenate the outputs from both prompt sides and further process them through additional two convolutional layers with a skip connection to fuse the information. The resulting output is then upsampled twice to increase its size and is used to generate the target mask. Notably, the MLPs used for predicting IoU scores and generating segmentation masks all share the same structure, consisting of three linear layers with ReLU activation functions. Both have an input channel size of 256, but their output channel sizes differ: the IoU prediction MLP outputs a single value, while the MLP for mask generation has an output channel size of 32 to align with the feature map.

2.3.2.4 Loss function

In this work, except the traditional mask prediction loss $\mathcal{L}_{\text{mask}}$ and IoU (Intersection over Union) score prediction loss $\mathcal{L}_{\text{iou}}$, we also introduce another two loss components: $\mathcal{L}_{\text{mcls}}$ for the modality classification task and $\mathcal{L}_{\text{contrastive}}$ for approaching the similarity between two kinds of content prompts. Therefore, the overall loss function can be represented as follows:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{mask}} + \lambda_2 \mathcal{L}_{\text{iou}} + \lambda_3 \mathcal{L}_{\text{mcls}} + \lambda_4 \mathcal{L}_{\text{contrastive}}. \quad (2.1)$$

Here, all the λ values are hyperparameters used to tune the importance of each loss component. We set $\lambda_1 = \lambda_2 = 1$ and $\lambda_3 = \lambda_4 = 0.01$, emphasizing the primary focus on segmentation while reflecting the auxiliary role of the classification and contrastive tasks in supporting overall performance.

Mask Prediction Loss $\mathcal{L}_{\text{mask}}$ is the sum of Binary Cross-Entropy (BCE) loss and Dice loss. Given the predicted mask $\hat{M}$ and ground truth mask M , BCE loss evaluates the pixel-wise difference, while Dice loss quantifies the overlap between the two masks.

Then their working principles can be shown as follows:

$$\mathcal{L}_{\text{BCE}} = -\sum_i^N (M_i \log(\hat{M}_i) + (1 - M_i) \log(1 - \hat{M}_i)), \quad (2.2)$$

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_i^N M_i \hat{M}_i}{\sum_i^N M_i^2 + \sum_i^N \hat{M}_i^2}, \quad (2.3)$$

where N is the total number of pixels and i is the pixel index. Then $\hat{M}_i$ and M_i represent the i -th pixel values of the predicted mask and the ground truth mask, respectively.

IoU Loss $\mathcal{L}_{\text{iou}}$ measures the difference between the predicted IoU score from the model and the ground truth IoU score computed from the overlap between the predicted mask and the ground truth mask, aiming to further improve the accuracy of predicted segmentation mask. We employ a mean squared error (MSE) loss to capture this difference, encouraging precise IoU predictions:

$$\mathcal{L}_{\text{iou}} = \frac{1}{N'} \sum_{i=1}^{N'} (s_{\text{iou}}^i - \hat{s}_{\text{iou}}^i)^2, \quad (2.4)$$

where N' is the total number of masks, s_{iou}^i is the ground truth IoU score for the i -th mask, while $\hat{s}_{\text{iou}}^i$ indicates the corresponding predicted IoU score.

Modality Classification Loss To ensure that modality prompt effectively represents modality-specific information, we introduce an auxiliary modality classification task by employing cross entropy loss function, which is defined as follows:

$$\mathcal{L}_{\text{mcls}} = -\sum_{i=1}^C y_i \log(\hat{y}_i), \quad (2.5)$$

where C denotes the total number of modalities, y_i is the label for each modality class i and $\hat{y}_i$ is the predicted probability for each modality class i .

Contrastive Loss To ensure that pairs of sparse and dense content prompts exhibit the highest similarity, we introduce an auxiliary task that utilizes the contrastive loss from the CLIP [36] method. The overall process can be represented as follows:

$$\text{sim}_1 = F_{dc} \cdot F_{sc}^T, \quad (2.6)$$

$$\text{sim}_2 = F_{sc} \cdot F_{dc}^T, \quad (2.7)$$

$$\mathcal{L}_{\text{contrastive}} = (\mathcal{L}_{\text{ce}}(\text{sim}_1, \mathbf{y}) + \mathcal{L}_{\text{ce}}(\text{sim}_2, \mathbf{y})) / 2. \quad (2.8)$$

Here, F_{dc} and F_{sc} are the normalized embeddings of the dense and sparse content prompts, respectively, both having the shape $\mathbb{R}^{B \times C}$, where B is batch size and C denotes the embedding length. sim_1 and sim_2 are two similarity matrices for these two embeddings, each with a shape of $\mathbb{R}^{B \times B}$, where sim_2 is the transpose of sim_1 . $\mathcal{L}_{ce}$ denotes the cross entropy loss function and $\mathbf{y}$ are the collection of labels.

2.3.3 Training strategy

In this section, we detail two core components of our training strategy: pre-trained modules (tiny ViT and CLIP) and data sampling strategy. Both are important for accelerating the model's convergence and obtaining a superior overall performance.

2.3.3.1 Pre-trained modules

As previously mentioned, many methods freeze pre-trained weights from SAM's image encoder, fine-tuning newly introduced learnable components to improve adaptability of new tasks. Using these pre-trained weights ensures a strong baseline ability for the model. However, instead of following this approach, we opt for a lightweight image encoder, tiny ViT, and use pre-trained weights specifically tailored for the medical image. This allows the image encoder of MCP-MedSAM to fully participate in the training process without being frozen, achieving stronger performance. Additionally, we utilize the pre-trained PubMedCLIP³ [77] as the frozen CLIP component of MCP-MedSAM, aiming to leverage its strong zero-shot capability and make it provide some medical domain related prior information as well. This variant of CLIP is fine-tuned for the medical domain using the Radiology Objects in Context (ROCO) dataset [78], including multiple modalities from various human regions.

2.3.3.2 Data sampling strategy

As shown in Figure 2.2, the distribution of different image modalities in our training dataset is highly imbalanced, primarily due to two factors.

The first factor is varying sizes of public datasets: some modalities, such as CT, MR, and X-ray, have much more publicly available data for AI tasks, resulting in a significantly larger number of training samples. The second factor relates to data dimension. Modalities like CT and MR are in 3D format, enabling the extraction of significantly more slices compared to 2D modalities such as Dermoscopy, Fundus, and Microscopy.

In SAM [11] and MedSAM [13], data sampling involves iterating over all image slices. In our training set, this approach leads to a significant imbalance: CT slices account for approximately 76% of the dataset, MR slices nearly 13%, while Microscopy slices represent less than 0.1%, making it the least represented modality. Such severe

³<https://huggingface.co/kaushalya/medclip>

Algorithm 1 Modality-based Data Sampling Strategy

```
1: Input: Total number of modalities  $N$ , number of cases in each modality  $\{C_j\}_{j=1}^N$ ,  
   batch size  $B$   
2: Initialize batch  $\mathcal{B} = \emptyset$   
3: for each epoch do  
4:   for each batch of size  $B$  do  
5:     for each sample in the batch do  
6:       Select modality  $m \sim \mathcal{U}\{0, N-1\}$   
7:       Select case from the chosen modality:  $c \sim \mathcal{U}\{0, C_m-1\}$   
8:       if the selected case is 3D then  
9:         Let the shape of the 3D case be  $[Z, H, W]$   
10:        Choose slice index  $z \sim \mathcal{U}\{0, Z-1\}$   
11:        Extract the slice  $\mathcal{S} = \mathcal{D}_{m,c}[z, :, :] \in \mathbb{R}^{H \times W}$   
12:       else  
13:        Extract the 2D case  $\mathcal{S} = \mathcal{D}_{m,c} \in \mathbb{R}^{H \times W}$   
14:       Select mask  $k \sim \mathcal{U}\{0, K-1\}$ , where  $K$  is the number of masks in  $\mathcal{S}$   
15:        $\mathcal{B} \leftarrow \mathcal{B} \cup \{(\mathcal{S}, k)\}$   
16: Output: Batch  $\mathcal{B}$ 
```

imbalance would negatively impact the model’s performance, while training on a large number of slices also results in extended training time.

To address the limitations of existing data sampling methods, we implement a modality-based data sampling strategy, a variant of stratified sampling. This approach prioritizes achieving balance across all modalities, as determined through comparisons with other commonly used data sampling strategies. The details of this strategy are outlined in Algorithm 1. The key point of this approach is randomly selecting a slice from each data case and ensuring that all modalities are evenly sampled within each training batch. This method alleviates the negative impacts of severe data imbalance, allowing all modalities to be trained with an approximately equal number of slices.

2.4 Experiments and results

2.4.1 Experimental setup

All experiments were performed using Python 3.10 and PyTorch 2.0.0 on a single NVIDIA A100 GPU with 40GB of memory. The AdamW optimizer was used with an initial learning rate of 2×10^{-4} and a weight decay of 1×10^{-3} . During training, the learning rate was reduced by a factor of 0.9 every 5 epochs. The batch size was set to 16, and training was conducted for a total of 25 epochs. In the final epoch, we set a small learning rate to 5×10^{-5} for an additional fine-tuning of the model. Data augmentation included vertical and horizontal flips, each applied with a 50% probability. Additionally, the Wilcoxon signed-rank test is used to assess the statistical

significance of the proposed method in the experiments.

2.4.2 Evaluation metrics

Accuracy Metrics Following [13], we adopted the Dice Similarity Coefficient (DSC) and Normalized Surface Dice (NSD) as evaluation metrics. DSC measures the overlap between two sets, quantifying the similarity between the predicted segmentation and the ground truth. In contrast, NSD evaluates how closely the surfaces of the predicted segmentation align with the ground truth, focusing on surface accuracy. Higher DSC and NSD values correspond to greater segmentation accuracy. To note, DSC and NSD are first averaged within each modality, and then these modality-wise means are averaged to obtain the overall value. Consequently, the standard deviation is also calculated at the modality level.

Efficiency metrics In order to evaluate the computational efficiency of the model, we established two metrics: GPU training time and CPU inference time. GPU training time is primarily used to assess GPU resource consumption, while CPU inference time focuses on evaluating model performance on edge devices without GPU support, such as laptops or standard CPU workstations. CPU inference time is also part of the competition evaluation criteria² for the testing set. The CPU inference time was measured on our local platform with an Intel Xeon(R) W3-2435 @ 3.1 GHz processor and 8GB of memory, where the average time per case was tested in a Docker environment provided by the authors.

2.4.3 Results

2.4.3.1 Comparison with benchmark models

First, we used the lightweight MedSAM model released by the organizers of the "SEGMENT ANYTHING IN MEDICAL IMAGES ON LAPTOP" competition at CVPR 2024 as our baseline. This model was developed by distilling the image encoder of MedSAM [13] and fine-tuning both the encoder and decoder together using the challenge dataset. Then we compared our model with the top-ranking models from the competition leaderboard: (1) MedficientSAM [66]: MedficientSAM utilizes EfficientViT as its image encoder and increases the input size from 256×256 to 512×512 ; (2) DAFT [70]: DAFT also employs EfficientViT, fine-tuning it on different data subsets based on modality to generate multiple modality-specific models; (3) Rep-MedSAM [68]: Rep-MedSAM employs RepViT as its image encoder and, after distillation, trains the entire model using the dataset; (4) Swin-LiteMedSAM [72]: Swin-LiteMedSAM utilizes the tiny Swin Transformer as its image encoder and introduces point and scribble prompts, which are automatically generated based on bounding boxes. (5) LiteMedSAM-Rep [74]: LiteMedSAM-Rep's prompt encoder

and mask decoder are initially trained from scratch alongside a tiny Swin Transformer as the image encoder. Then they distill the knowledge from the tiny Swin Transformer to RepViT. Overall, these models mainly focus on modifying the image encoder by distilling knowledge from a large ViT into a lightweight transformer encoder. Table 2.1 summarizes the segmentation performance of all models, reported in terms of DSC (%) and NSD (%). MCP-MedSAM achieved the best DSC and NSD performance compared to other benchmark models on the testing set. Notably, the segmentation predictions of these benchmark models were reproduced using Docker images provided by the authors, available on Docker Hub⁴. Then Figure 2.4 presents visualizations of the models’ predictions, providing additional details. Overall, our proposed MCP-MedSAM produced segmentation masks that closely resembled the ground truth, whereas some other models exhibited either a higher degree of over-segmentation or of under-segmentation.

Table 2.2 demonstrates GPU training time (in hours) and CPU inference time (in seconds) across different methods. MCP-MedSAM achieved the shortest GPU training time (23.8 hours), significantly outperforming all other models. GPU training times were sourced from the respective papers, while the baseline model’s training and inference times were not disclosed by the challenge organizers. To note, DAFT has three training stages, but it does not specify the time required for its third stage. Therefore, we estimated the total training time to exceed the combined duration of the first two stages. Additionally, LiteMedSAM-Rep and Rep-MedSAM utilized different types of GPUs (NVIDIA RTX 4090 and NVIDIA V100). However, based on Lambda GPU benchmark analysis⁵, we find that the A100 40GB is slightly faster than the RTX 4090 (by approximately 1.2 $\times$) and significantly faster than the V100 (by about 3.6 $\times$). Hence, we estimate that the equivalent training time on an A100 would be approximately 1333 hours for LiteMedSAM-Rep and 52 hours for Rep-MedSAM. As MCP-MedSAM requires only one day (23.8 hours) for training, and LiteMedSAM-Rep and Rep-MedSAM would take an estimated 56 and 2.2 days respectively under the same type of GPU, we can still conclude that MCP-MedSAM requires the least training time. For CPU inference time, most methods achieved approximately 1 second. MCP-MedSAM was the slowest among the compared methods, whereas DAFT was the fastest.

2.4.3.2 Ablation study

In this section, we present a comprehensive ablation study of the key modules in our approach. It is mainly composed of four parts: prompt encoder components of MCP-MedSAM, pre-trained components of MCP-MedSAM, data sampling strategy and

⁴<https://hub.docker.com/>

⁵<https://lambda.ai/gpu-benchmarks>

Table 2.1: Accuracy comparison with state-of-the-art methods on the challenge leaderboard, with the best result for each metric highlighted in bold. The † after each metric value indicates a significant difference ($p < 0.05$) compared to the proposed method.

Models	DSC (%)	NSD (%)
Baseline	$83.81 \pm 15.31^\dagger$	$83.26 \pm 22.67^\dagger$
LiteMedSAM-Rep [74]	$84.51 \pm 10.11^\dagger$	$85.03 \pm 17.13^\dagger$
Rep-MedSAM [68]	$86.19 \pm 7.67^\dagger$	$87.97 \pm 11.85^\dagger$
Swin-LiteMedSAM [72]	$86.78 \pm 8.63^\dagger$	$88.44 \pm 12.79^\dagger$
MedficientSAM [66]	$86.20 \pm 8.00^\dagger$	$87.65 \pm 11.61^\dagger$
DAFT [70]	$87.18 \pm 8.29^\dagger$	$88.32 \pm 13.41^\dagger$
MCP-MedSAM (proposed)	87.50 ± 6.91	89.40 ± 10.37

Table 2.2: Efficiency comparison with state-of-the-art methods on the challenge leaderboard, with the best result for each metric highlighted in bold. Notably, LiteMedSAM-Rep is trained on an NVIDIA RTX 4090, Rep-MedSAM on an NVIDIA V100, and the other models on an NVIDIA A100.

Models	GPU Training Time (hours)	CPU Inference Time (seconds)
Baseline	-	-
LiteMedSAM-Rep [74]	1600.0	0.9
Rep-MedSAM [68]	188.0	1.3
Swin-LiteMedSAM [72]	106.8	2.6
MedficientSAM [66]	118.5	0.7
DAFT [70]	> 42.9	0.4
MCP-MedSAM (proposed)	23.8	4.6

training GPU type.

Prompt Encoder Components of MCP-MedSAM Within the prompt encoder, both the modality and content prompt processing networks are composed of two key components. To assess the impact of each, we trained the model under the modality sampling strategy, omitting each component separately. Additionally, we evaluated the model using only one type of introduced prompt, as well as without any introduced prompts, to provide a comprehensive view. To note, when the CNN image encoder branch was excluded, we used a learnable matrix with randomly drawn weights from a uniform distribution as the outputting dense embedding representation. The detailed results are shown in Table 2.3. Removing a component from either introduced prompt processing network resulted in a decline in overall performance and might be even worse than removing the entire prompt processing network. Then using only the complete modality or content prompt processing network achieved comparable

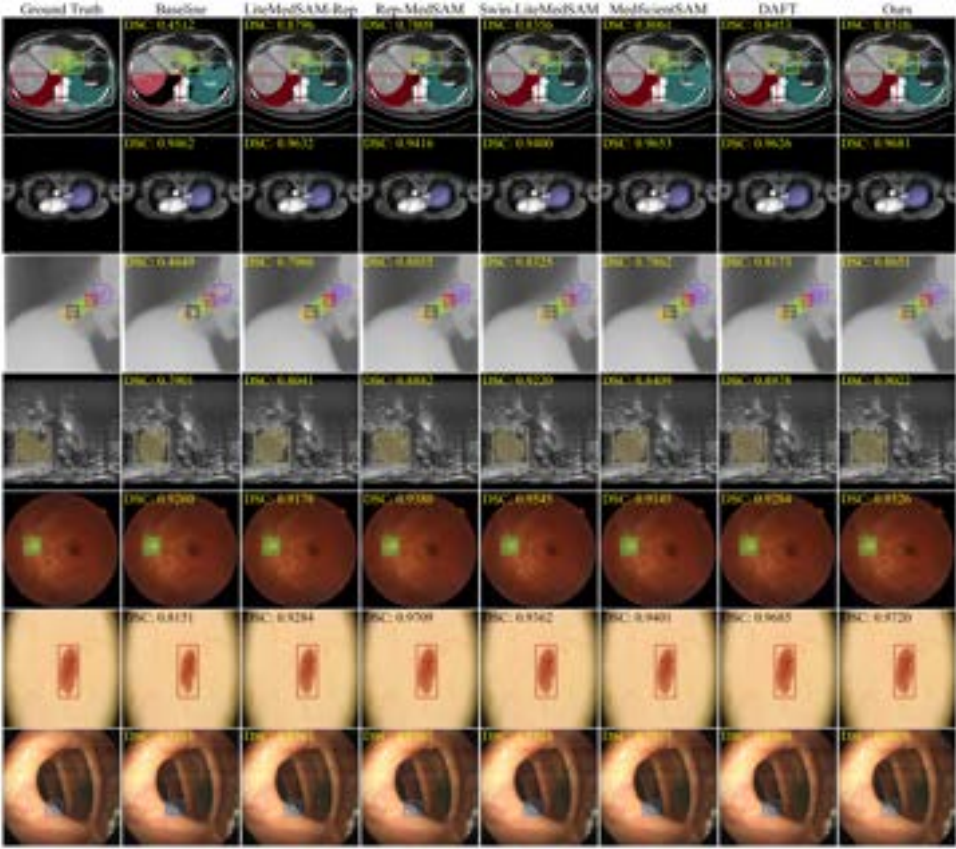


Figure 2.4: Visualization of multiple modalities yielded by our proposed method and the other benchmark models. The DSC value of each sample is displayed in the upper left corner.

performance, both of which outperformed the absence of both prompts. Figure 2.5 further represents the performance of each introduced prompt with some visualizations. Each prompt produced different segmentation results based on its unique perspective, with varying types of errors. Giving both prompts resulted in the best segmentation performance.

Pre-trained Components of MCP-MedSAM As previously mentioned, we incorporated pre-trained weights into the image encoder and CLIP component of MCP-MedSAM to improve performance. To support our approach and gain deeper insights into their effects, we conducted several related ablation studies. For the tiny ViT, we tested three scenarios: without pretraining, with pre-trained weights from natural images, and with pre-trained weights from medical images. The CLIP was tested with two

options: with pre-trained weights from natural images, and with pre-trained weights from medical images. As shown in Table 2.4, both CLIP and tiny ViT had a better overall performance when using medical domain pre-trained weights compared to those initialized with natural domain pre-trained weights. Furthermore, when tiny ViT was trained from scratch without any pre-trained weights, it exhibited the lowest performance among all the options.

Data Sampling Strategy We analyzed three different data sampling strategies: (1) Slice Sampling, used in [13], randomly selecting a slice from the whole training dataset; (2) Case Sampling, randomly selecting a slice from each training case; (3) Modality Sampling (see Algorithm 1), introducing control over modality balance in each training batch, building on the case sampling approach. The detailed results are shown in Table 2.5. Among them, slice sampling achieved the highest scores for CT and MR but lagged behind the other two strategies on other modalities. While case sampling and modality sampling showed stronger performance on specific modalities, modality sampling delivered the best overall results with more balanced performance across all modalities.

Training GPU Type To further evaluate the applicability of MCP-MedSAM on smaller GPUs, we conducted an experiment on a mid-range GPU (NVIDIA RTX 6000, 24GB) by reducing the batch size from 16 to 8. The final results achieved a DSC of 86.87 ± 7.53 and an NSD of 88.34 ± 12.07 , with a total training time of 54.6 hours. Overall, compared to training on an A100 GPU, segmentation performance decreased and training time increased significantly. This is mainly due to the reduced batch size and lower GPU performance. A smaller batch size leads to more iterations per epoch, resulting in longer training time. More importantly, it can also negatively affect model performance by introducing higher variance in gradient estimation, which makes training less stable and slower convergence.

2.5 Discussion and conclusion

In accuracy comparisons with the benchmark models (Table 2.1), MCP-MedSAM achieved the best results and showed a significant statistical difference compared to the other methods. In general, it can be attributed to several key factors: 1) the introduction of two types of prompts offers some valuable cues for target segmentation by integrating information from different perspectives. As shown in Table 2.3, using either prompt improved overall segmentation accuracy: the modality prompt processing network captures inter-modality differences and unique target characteristics, while the content prompt processing network focuses on features within the bounding box to directly extract information about the segmentation

Table 2.3: Ablation study of the component of each prompt processing network in the prompt encoder part of MCP-MedSAM model. The checkmark means including the component in the model. And the best result for each evaluation metrics is shown in bold. The $\dagger$ after each metric value indicates a significant difference ($p < 0.05$) compared to the proposed method.

Modality Prompt		Content Prompt		DSC (%)	NSD (%)
Text CLIP	Modality Embedding	Image CLIP	CNN Encoder		
				$86.36 \pm 8.39^\dagger$	$87.64 \pm 13.23^\dagger$
	✓	✓	✓	$86.82 \pm 7.97^\dagger$	$88.35 \pm 12.72^\dagger$
✓		✓	✓	$86.57 \pm 7.21^\dagger$	$88.39 \pm 10.91^\dagger$
✓	✓			$87.07 \pm 7.19^\dagger$	$88.78 \pm 11.07^\dagger$
✓	✓		✓	$86.92 \pm 7.83^\dagger$	$88.47 \pm 12.31^\dagger$
✓	✓	✓		$86.92 \pm 7.59^\dagger$	$88.55 \pm 12.02^\dagger$
		✓	✓	$87.13 \pm 6.89^\dagger$	$88.76 \pm 11.12^\dagger$
✓	✓	✓	✓	87.50 ± 6.91	89.40 ± 10.37

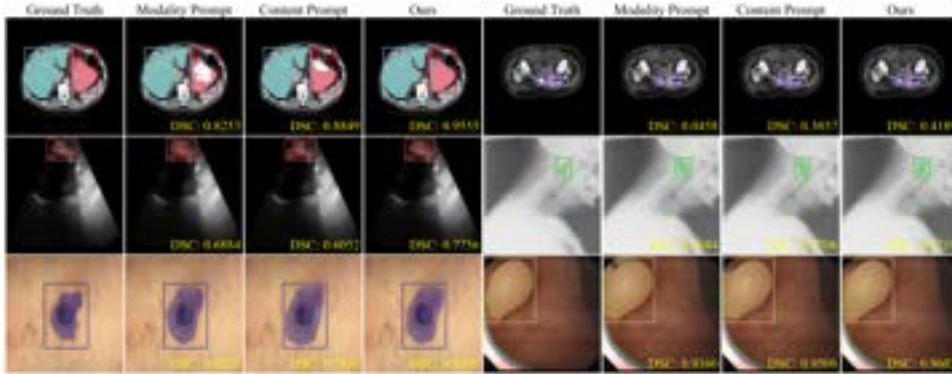


Figure 2.5: Visualizations generated from multiple imaging modalities using three prompting strategies: modality-only, content-only, and combined prompts. The DSC value of each sample is displayed in the bottom right corner.

target. The comparable performance achieved with either prompt also indicates the similar importance of modality information and content information contained in the bounding box. Moreover, the components within each prompt processing network complement each other, enriching the overall representation. In contrast, relying on a single prompt processing component may interfere with the processing of the other prompt, ultimately reducing overall performance; 2) the incorporation of pre-trained components supplies the model with prior knowledge, enhancing its final performance. In Table 2.4, it is obvious that medical-related initial weights lead to optimal results by incorporating valuable prior information; 3) the modality-based data sampling strategy mitigates the negative effects of data imbalance, leading to a more balanced overall performance compared to the two other strategies. While this slightly lowers

Table 2.4: Ablation study of pre-trained components with different weights (natural and medical images) on the MCP-MedSAM model, with the best result for each evaluation metrics highlighted in bold. The † after each metric value indicates a significant difference ($p < 0.05$) compared to the proposed method.

Method	DSC (%)	NSD (%)
Tiny ViT (not pre-trained)	85.53 ± 8.67 [†]	87.11 ± 13.35 [†]
Tiny ViT (pre-trained on natural images)	86.95 ± 8.03 [†]	88.61 ± 12.36 [†]
Tiny ViT (pre-trained on medical images, proposed)	87.50 ± 6.91	89.40 ± 10.37
CLIP (pre-trained on natural images)	86.99 ± 7.53 [†]	88.44 ± 12.20 [†]
CLIP (pre-trained on medical images, proposed)	87.50 ± 6.91	89.40 ± 10.37

Table 2.5: Performance comparison across different data sampling strategies, with the performance of each modality detailed. And the best performance for each modality and overall performance is shown in bold. The † after each metric value indicates a significant difference ($p < 0.05$) compared to the modality sampling strategy.

Modality	Slice Sampling		Case Sampling		Modality Sampling	
	DSC (%)	NSD (%)	DSC (%)	NSD (%)	DSC (%)	NSD (%)
CT	91.00 ± 9.69	93.85 ± 9.70	90.31 ± 8.10	93.55 ± 8.35	90.02 ± 7.98	93.43 ± 8.14
MR	87.69 ± 10.70	91.68 ± 12.41	85.73 ± 11.07	89.79 ± 12.06	85.53 ± 11.14	89.81 ± 12.59
PET	66.21 ± 11.38	49.40 ± 30.40	68.06 ± 9.36	52.05 ± 29.13	73.38 ± 7.04	61.68 ± 25.88
US	82.50 ± 10.52	87.16 ± 7.20	83.97 ± 11.63	88.71 ± 8.35	84.77 ± 9.62	89.61 ± 6.55
X-ray	83.44 ± 7.19	88.24 ± 7.36	86.33 ± 5.91	91.00 ± 6.36	85.83 ± 5.93	90.57 ± 6.27
Dermoscopy	93.08 ± 5.21	94.61 ± 4.36	94.58 ± 4.22	96.07 ± 3.23	94.84 ± 4.54	96.32 ± 3.56
Endoscopy	93.19 ± 5.12	96.08 ± 3.70	96.25 ± 4.11	98.45 ± 2.19	95.17 ± 6.67	97.66 ± 5.08
Fundus	94.57 ± 1.70	96.27 ± 1.52	95.61 ± 1.41	97.21 ± 1.18	95.77 ± 1.39	97.35 ± 1.21
Microscopy	76.42 ± 15.67	83.07 ± 12.80	82.53 ± 15.79	88.50 ± 12.74	82.17 ± 15.20	88.17 ± 12.28
Average	85.34 ± 8.83 [†]	86.71 ± 13.84 [†]	87.04 ± 7.97 [†]	88.37 ± 12.87 [†]	87.50 ± 6.91	89.40 ± 10.37

the performance of common modalities like CT and MRI, it significantly improves the performance of underrepresented modalities with much fewer training samples, such as PET, as shown in Table 2.5. Additionally, our proposed method has the lowest standard deviation values, also highlighting the effectiveness of the modality-based data sampling strategy in balancing the performance of multiple modalities.

The visualizations also offered valuable insights for assessing the performance of MCP-MedSAM. In Figure 2.4, the CT and X-ray samples show multiple overlapping segmentation targets and some of them are small in size, both of which increase segmentation difficulty. In contrast, the other displayed modalities have fewer and larger targets, resulting in lower overall segmentation difficulty. However, noticeable over-segmentation occurred in Endoscopy for many benchmark models, likely due to background features resembling those of the target. They indicate that small target sizes and complex backgrounds could negatively impact segmentation performance, while the proposed MCP-MedSAM represents the best ability to mitigate these effects, demonstrating its robustness and confirming the usefulness of the

previously mentioned key factors. Overall, the visualization results align with the quantitative findings in Table 2.1. Likewise, Figure 2.5 more clearly demonstrates the benefits of incorporating both introduced prompts into the prompt encoder, as their combination further enriches the feature representation, leading to enhanced performance.

In efficiency comparisons with the benchmark models (Table 2.2), MCP-MedSAM required the least training time, while the other models took significantly longer time to finish training. For the inference time, the main goal of MCP-MedSAM is to achieve superior segmentation performance without requiring long training time and significant GPU resource consumption, so improving inference speed is not our primary focus. Furthermore, incorporating additional components such as the CLIP component and CNN image encoder will inevitably increase inference time. Several of the benchmark models have adopted different strategies to reduce inference time while maintaining performance, for example, DAFT [70] replaced PyTorch with the OpenVINO Runtime⁶. Similarly, MedficientSAM [66] further optimized inference by integrating OpenVINO and leveraging C++ for pre-processing and post-processing optimizations. It could be interesting to adopt such strategies for MCP-MedSAM as well, which will be considered in our future work. Nonetheless, for most clinical scenarios, inference times in the range of seconds are very acceptable, while they can be further reduced by running the models on a GPU.

MCP-MedSAM features a lightweight structure compared to MedSAM [13]. While a direct performance comparison is not feasible due to differences in training and testing datasets, the reported performance metrics, such as DSC (ranging from 0.85 to 0.90), indicate comparable performance levels. This highlights MCP-MedSAM's ability to achieve effective medical image segmentation despite its lightweight design. These findings underscore the potential to deliver high-quality segmentation performance without extensive GPU resources, encouraging the development of lightweight and accessible models for advanced medical image analysis. However, MCP-MedSAM adopts the overall design of MedSAM and segments 3D data slice by slice in a 2D manner. This approach is not only time-consuming but also negatively impacts the segmentation performance of 3D samples, as it prevents the model from learning the correlations between slices. With the proposal of the SAM2 framework [55], which achieves superior performance in segmenting both images and videos, integrating SAM2's working principles with the MCP-MedSAM structure has the potential to enhance segmentation performance for 3D modalities, while also reducing overall inference time. Furthermore, although our results clearly demonstrate quantitative improvement in segmentation performance, further investigation is still needed to explore how these enhancements translate into practical clinical value. For example, in

⁶<https://docs.openvino.ai/2024/openvino-workflow/running-inference.html>

radiotherapy (RT) planning, the segmentation of targets such as tumors is critical, as even small errors can significantly impact dose distribution and treatment effectiveness. Therefore, DSC improvements in tumor segmentation are clinically meaningful. However, for certain organs-at-risk (OARs) that are located far from the tumor region—such as the esophagus in some head and neck cancer cases, [79] showed a low correlation between DSC and dose errors. In such cases, a marginal gain in DSC may not translate into a noticeable clinical difference. Some future studies can further investigate the real-world impacts of these segmentation improvements across various clinical scenarios.

In this work, we proposed a lightweight medical segment anything model called MCP-MedSAM, designed to achieve strong overall performance without long training time and large GPU resource consumption. By integrating pre-trained components, the model training process is accelerated, leading to improved performance. Then the introduction of the modality prompt and the content prompts offers valuable diverse information, improving upon the lightweight MedSAM design. Furthermore, a modality-based data sampling strategy ensures that each modality is trained equally, leading to a more balanced overall performance. In conclusion, MCP-MedSAM achieves better overall segmentation performance against top-ranking methods in the challenge², demonstrating its effectiveness and potential.

3

On Factors that Influence Deep Learning-Based Dose Prediction of Head and Neck Tumors

This chapter was adapted from:

Ruochen Gao, Prerak Mody, Chinmay Rao, Frank Dankers, and Marius Staring, "On Factors that Influence Deep Learning-Based Dose Prediction of Head and Neck Tumors", *Physics in Medicine & Biology*, vol. 70, no. 11, p. 115006, 2025.

Abstract

Objective. This study investigates key factors influencing deep learning-based dose prediction models for head and neck cancer radiation therapy (RT). The goal is to evaluate model accuracy, robustness, and computational efficiency, and to identify key components necessary for optimal performance.

Approach. We systematically analyze the impact of input and dose grid resolution, input type, loss function, model architecture, and noise on model performance. Two datasets are used: a public dataset (OpenKBP) and an in-house clinical dataset (LUMC). Model performance is primarily evaluated using two metrics: dose score and dose-volume histogram (DVH) score.

Main results. High-resolution inputs improve prediction accuracy (dose score and DVH score) by 8.6–13.5% compared to low resolution. Using a combination of CT, planning target volumes (PTVs), and organs-at-risk (OARs) as input significantly enhances accuracy, with improvements of 57.4–86.8% over using CT alone. Integrating mean absolute error (MAE) loss with value-based and criteria-based DVH loss functions further boosts DVH score by 7.2–7.5% compared to MAE loss alone. In the robustness analysis, most models show minimal degradation under Poisson noise (0–0.3 Gy) but are more susceptible to adversarial noise (0.2–7.8 Gy). Notably, certain models, such as SwinUNETR, demonstrate superior robustness against adversarial perturbations.

Significance. These findings highlight the importance of optimizing deep learning models and provide valuable guidance for achieving more accurate and reliable radiotherapy dose prediction.

3.1 Introduction

Radiation therapy (RT) is one of the primary treatment modalities for head and neck cancers, aiming to deliver high doses of radiation to the planning target volume (PTV) while minimizing exposure to surrounding organs at risk (OARs). Due to the irregular shape of PTV, the multiple dose levels of PTV (such as primary tumors and lymph nodes) and the close proximity of OARs, head and neck cancer is considered one of the most complex clinical treatment areas [80].

Modern RT techniques, such as Intensity-Modulated Radiotherapy (IMRT) and Volumetric Modulated Arc Therapy (VMAT), rely on an inverse planning process. This process, which involves iterative adjustment of treatment objectives for the tumor and OARs, can be time-consuming and heavily dependent on the planner's expertise, leading to significant variability in plan quality [20].

To enhance the consistency, quality, and efficiency of RT plans, Knowledge-Based Planning (KBP) has been introduced to automate the process of RT plan generation. The KBP approach leverages knowledge from prior high-quality, clinically accepted plans to predict optimal RT plans for new patients. Traditional KBP models use anatomical and geometric features to predict the Dose-Volume Histogram (DVH) and the dose distribution [81, 82, 83]. However, with the rise of deep learning in recent years, deep learning-based dose prediction methods no longer require manually designed features as input. Instead, they can learn directly from raw image data, marking a new stage in the development of KBP [84]. Typically, the dose distribution predicted by deep learning-based methods is then mimicked to create a clinically deliverable plan [85].

Segmentation models based on the encoder-decoder architecture have become dominant in medical imaging segmentation tasks [86, 87, 32, 45, 88]. These models take inputs, such as CT or MRI images, and output segmentation masks with matching dimensions, a feature that also makes them suitable for dose prediction tasks. This compatibility has led many researchers to adopt the encoder-decoder structure for dose prediction. For example, [89] introduced DoseNet, a fully convolutional volumetric network, to predict the dose distribution for prostate cancer. [23] proposed a hierarchically densely connected U-Net (HDUNet), which combines U-Net and DenseNet to predict the dose distribution for the head and neck. In 2020, the American Association of Physicists in Medicine (AAPM) launched the Open Knowledge-Based Planning Challenge (OpenKBP), significantly advancing the field of radiation dose prediction by providing a standardized open source dataset for head and neck cancer (OpenKBP dataset) along with standardized evaluation metrics [22]. This initiative has spurred the development of various innovative methods, broadly categorized into two main approaches: (1) U-Net-based methods and (2) Transformer-based methods.

In the first category, numerous studies have leveraged U-Net and its variants to predict dose distributions. [90] proposed a three-dimensional (3D) densely connected U-Net with dilated convolutions to capture complex spatial relationships in the data. [91] presented a 3D cascaded U-Net model (C3D) with a knowledge distillation technique, achieving first place in the OpenKBP Challenge. Additionally, [26] proposed a beam-wise dose decomposition 3D cascaded network, while [92] developed MemU-net, which integrates MemNet memory blocks within a U-Net framework to improve network efficiency and prediction accuracy. Recently, [93] introduced the LENAS framework, an approach that incorporates Neural Architecture Search (NAS) and knowledge distillation to optimize the search for effective components in U-Net. In the second category, transformer architectures have recently emerged as a promising alternative in head and neck dose prediction, capitalizing on their ability to capture long-range dependencies. [94] introduced the TrDosePred framework, pioneering the use of Transformer architectures for this task. Following this, [24] proposed DOSE-PYFER, a cascaded Transformer-based model specifically designed to predict the dose distribution. However, most of these studies focus mainly on proposing a new model structure to improve accuracy.

In addition to innovations in model architecture, only a few studies have explored other factors that influence the accuracy of dose prediction. For example, [95] studied the impact of the loss function, suggesting that incorporating a domain-specific DVH loss can further enhance the model performance. [96] investigated the impact of model input on performance, finding that CT and PTV are sufficient to predict dose distribution. However, these investigations are often isolated and based on private datasets, which limits the generalizability of their findings. Consequently, a comprehensive analysis of the factors that influence deep learning-based dose prediction across various datasets is still needed.

To facilitate clinical translation, research should go beyond accuracy to rigorously evaluate both robustness and computational efficiency. Robustness is crucial for managing various forms of noise, including inherent image noise and adversarial disturbances. In particular, adversarial noise, potentially introduced by malicious attacks, such as ransomware, poses a significant threat to hospital and patient safety [97]. However, computational efficiency, which encompasses hardware requirements and runtime, is equally critical to ensure that models can be feasibly deployed within clinical settings. Although accuracy often takes precedence in most studies, there is a noticeable gap in assessing how models handle noise and perform efficiently. Bridging this gap is essential to develop dose prediction models that are not only accurate but also robust and practically viable in a clinical context.

Figure 3.1 provides an overview of the deep learning-based dose prediction pipeline and highlights the key factors analyzed in this study. In this paper, we make two

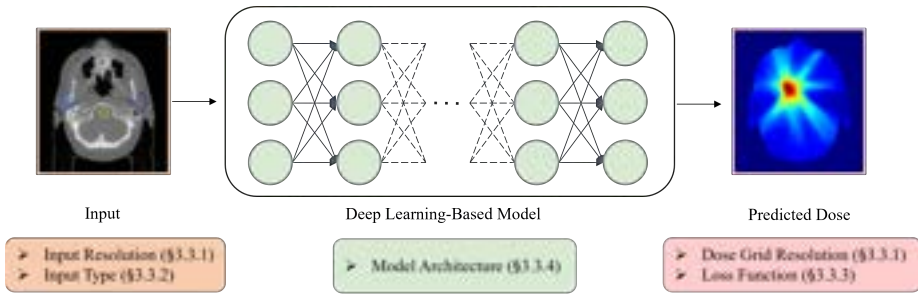


Figure 3.1: Pipeline of deep learning-based dose prediction and focus of this paper.

primary contributions:

- **Comprehensive Analysis of Influential Factors:** We conduct a systematic investigation into various factors that influence deep learning-based dose prediction for head and neck tumors. These factors include input resolution, dose grid resolution, input type, loss function, and model structure. By analyzing these factors using both a public dataset and an in-house clinical dataset, we provide a holistic understanding of how each factor impacts the performance of dose prediction.
- **Evaluation of Model Robustness and Efficiency:** In addition to focusing on the accuracy of the prediction, we evaluate the robustness of different model structures against Poisson noise and adversarial noise. Furthermore, we assess the computational efficiency of these models, including GPU memory consumption and runtime. This dual focus on robustness and efficiency addresses practical concerns that are critical for clinical applications.

3.2 Materials and methods

3.2.1 Patient dataset

In this study, we used two datasets. The first is the OpenKBP dataset, which contains data from 340 head and neck cancer patients: 200 for training, 40 for validation, and 100 for testing [22]. The second is an in-house dataset (referred to as LUMC) of 104 patients treated for oropharyngeal and hypopharyngeal cancer at Leiden University Medical Center between 2017 and 2024. The study was approved by the Medical Ethics Committee of Leiden, The Hague, and Delft (G21.142, October 15, 2021). Patient consent was waived due to the retrospective nature of the study.

The two datasets are very similar in composition. Each patient's data includes a planning CT, PTV contours, OAR contours, and the corresponding 3D dose distribution. There are several differences. For OpenKBP, the dose distribution was generated from

Table 3.1: Comparison between the OpenKBP and LUMC dataset

Dataset	Plan Type	PTVs	OARs	Dose Grid Resolution
OpenKBP	IMRT	PTV ₇₀ , PTV ₆₃ , PTV ₅₆	Brainstem, Spinal cord, Right parotid, Left parotid, Esophagus, Larynx, and Mandible	(3.9, 3.9, 2.5)
LUMC	VMAT	PTV ₇₀ , PTV _{54.25}	Brainstem, Spinal cord, Right parotid, Left parotid, Esophagus, and Larynx	(2, 2, 2)

a 9-beam equidistant coplanar IMRT setup with 6 MV fields. For LUMC, the dose distribution was generated from dual-arc full rotation VMAT beams. The PTVs in OpenKBP include three dose levels: PTV₅₆, PTV₆₃, and PTV₇₀, representing 56 Gy, 63 Gy, and 70 Gy, respectively. In contrast, LUMC’s PTVs include two dose levels: PTV_{54.25} and PTV₇₀, representing 54.25 Gy and 70 Gy. In addition, the OpenKBP dataset has a lower dose grid resolution compared to the LUMC dataset. For detailed comparisons, see table 3.1.

3.2.2 Data preprocessing and training strategy

We used the same data preprocessing and training strategies for both the OpenKBP and LUMC datasets. First, during data pre-processing, we followed [91] to clip the intensity of the CT images to the range $[-1024, 1500]$ Hounsfield Units (HU). We then normalized the values by dividing by 1000 HU. For the dose distribution, we performed normalization by dividing by 70 Gy. For dataset splitting, we adhered to the original settings for the OpenKBP dataset. For the LUMC dataset, we used three-fold cross-validation.

During training, we used online data augmentation to avoid overfitting, including random translation, random flipping, and random rotation along the cranio-caudal axis. We used the AdamW optimizer with an initial learning rate of 3×10^{-4} . All models presented in this study were trained for a maximum of 1000 epochs, with early stopping employed to prevent overfitting.

3.2.3 Evaluation metrics

3.2.3.1 Accuracy metric

Following the OpenKBP-2020 AAPM Grand Challenge, we evaluate the accuracy of the model using the Dose score and the DVH score [22]. The Dose score (in Gy) is measured by the mean absolute error (MAE) between the actual and predicted doses. The DVH score (in Gy), specific to radiation therapy, includes criteria for both PTVs and OARs. For PTVs, the DVH score covers three criteria: $D_{1\%}$, $D_{95\%}$, and $D_{99\%}$, indicating the minimum doses received by 1%, 95%, and 99% of the volume, respectively. For OARs, the DVH score involves calculating D_{mean} (the mean dose received by volume) and $D_{0.1cc}$ (the near maximum dose received by 0.1 cc of volume). The Dose score and the DVH score are defined as follows:

$$\text{Dose score} = \frac{1}{N} \sum_n |\hat{D}_n - D_n|, \quad (3.1)$$

where $\hat{D}_n$ and D_n denote the predicted dose and the ground truth dose for the n -th voxel, respectively, and N is the total number of voxels.

$$\begin{aligned} \text{DVH score} = \frac{1}{3P + 2O} & \left(\sum_{p=1}^P (|\hat{D}_{1\%}^p - D_{1\%}^p| + |\hat{D}_{95\%}^p - D_{95\%}^p| + |\hat{D}_{99\%}^p - D_{99\%}^p|) \right. \\ & \left. + \sum_{o=1}^O (|\hat{D}_{\text{mean}}^o - D_{\text{mean}}^o| + |\hat{D}_{0.1cc}^o - D_{0.1cc}^o|) \right) \end{aligned} \quad (3.2)$$

where P is the total number of PTVs and O is the total number of OARs.

In addition to using the aggregate Dose Score and DVH Score as the main accuracy metrics in this study, we also use individual clinical dose evaluation metrics for comparison with the clinical plan, including $V_{95\%}$ (the percentage of a volume that receives at least 95% of the prescribed dose), D_{mean} and $D_{0.03cc}$ (the near maximum dose received by 0.03 cc of volume). It is important to note that the OpenKBP dataset does not include a clinically approved plan; therefore, clinical dose evaluation metrics are only calculated for the LUMC dataset.

3.2.3.2 Robustness metric

To evaluate the robustness of the model, we measure the change in accuracy of the model before and after introducing noise. To note, the noise is added during the model inference stage without retraining the model. Robustness is quantified using $\Delta\text{Dose score}$ and $\Delta\text{DVH score}$, which represent the differences in the accuracy metrics between noise-free conditions and those under noisy conditions. Higher values of $\Delta\text{Dose score}$ and $\Delta\text{DVH score}$ indicate poorer robustness, reflecting higher sensitivity to noise. They are defined as follows:

$$\Delta\text{Dose score} = \text{Dose score}^* - \text{Dose score}, \quad (3.3)$$

$$\Delta\text{DVH score} = \text{DVH score}^* - \text{DVH score}. \quad (3.4)$$

Here, the asterisk (*) denotes the scores calculated under noisy conditions, while the unmarked scores represent those obtained under noise-free (normal) conditions.

3.2.3.3 Computational efficiency metric

To evaluate the computational efficiency of the model, we prioritize two key metrics: GPU memory usage and GPU runtime, rather than focusing on the number of model parameters and floating point operations (FLOPs). This choice is made because GPU memory usage and runtime are more relevant for clinical deployment scenarios, as they accurately represent the hardware resources needed and the actual

computation time. Lower GPU memory usage and shorter runtime indicate a more computationally efficient model. To note, our evaluation focuses specifically on the model inference, without including data loading or preprocessing, to better reflect the model’s computational efficiency. All tests were performed on an NVIDIA A100 40GB GPU, and we report the average usage of GPU memory and runtime to predict a 3D dose distribution.

3.2.4 Input and dose grid resolution

Typically, the input resolution is set to match the dose grid resolution. This means that we determine the input resolution by defining the dose grid resolution. However, due to the large size of the 3D dose distribution, many studies employ interpolation algorithms to down-sample the dose distribution to use as ground truth [22]. As a result, the model predicts a low-resolution dose distribution, which should be upsampled to the original resolution for clinical use. Although this approach reduces hardware requirements and speeds up training, it introduces the risk of potential interpolation-related errors.

To evaluate the influence of dose grid resolution on dose prediction, we use a cubic-spline interpolation algorithm to resample the dose distribution in different resolutions. Since the resolution of the OpenKBP dataset is low by default, we conducted this experiment using only the LUMC dataset, which is of higher resolution. We evaluated dose grid resolutions at three levels: $2 \times 2 \times 2 \text{ mm}^3$ (high and original resolution), $3 \times 3 \times 3 \text{ mm}^3$ (medium resolution), and $4 \times 4 \times 4 \text{ mm}^3$ (low resolution).

3.2.5 Input type

Selecting the right input is essential for deep learning-based dose prediction models. Before developing an RT plan, we typically have anatomical information such as the patient’s planning CT, PTV contours, and OARs contours. This information identifies critical areas to treat and high-risk organs to avoid. Using these data as input enables the neural network to learn complex spatial relationships and dose distribution patterns.

To assess how different types of input affect dose prediction accuracy, we chose four types of input: (1) CT only, (2) PTVs and OARs, (3) CT combined with PTVs, and (4) CT combined with PTVs and OARs.

3.2.6 Loss function

The loss function of neural networks plays a crucial role during training because it quantifies the difference between the predicted output and the ground truth. By minimizing the loss function, the model continuously adjusts its parameters, thereby enhancing its prediction accuracy. Most prior work uses the mean squared error (MSE) or MAE loss [89, 23, 91, 92, 24]. [95] proposed making the DVH a differentiable target

and introduced a value-based DVH loss function. However, this approach requires a high computational overhead. [26] proposed two more efficient DVH-based loss functions, including a value-based DVH loss function ($\mathcal{L}_{vDVH}$) and a criteria-based DVH loss function ($\mathcal{L}_{cDVH}$), defined as follows:

$$\mathcal{L}_{vDVH} = \sum_{s=1}^{P+O} \frac{1}{N_s} \sum_{n=1}^{N_s} |R(\hat{D} \cdot W_s)_n - R(D \cdot W_s)_n|, \quad (3.5)$$

where N_s is the number of voxels in the s -th ROI (W_s), and $R(\cdot)$ denotes the sorting operation,

$$\mathcal{L}_{cDVH} = \frac{1}{3P+2O} \left(\sum_{p=1}^P (|\hat{D}_{1\%}^p - D_{1\%}^p| + |\hat{D}_{95\%}^p - D_{95\%}^p| + |\hat{D}_{99\%}^p - D_{99\%}^p|) + \sum_{o=1}^O (|\hat{D}_{\text{mean}}^o - D_{\text{mean}}^o| + |\hat{D}_{0.1cc}^o - D_{0.1cc}^o|) \right), \quad (3.6)$$

where these DVH criteria are calculated by the sorted dose values. We interpret $\mathcal{L}_{vDVH}$ as penalizing the discrepancy between the predicted and ground-truth DVH curves, while $\mathcal{L}_{cDVH}$ focuses on penalizing differences specifically at critical points of the DVH curve.

To evaluate the influence of different loss functions on the dose prediction task, we conducted three sets of experiments: (1) using $\mathcal{L}_{MAE}$ alone, (2) using $\mathcal{L}_{MAE} + \mathcal{L}_{vDVH}$, and (3) using $\mathcal{L}_{MAE} + \mathcal{L}_{vDVH} + \mathcal{L}_{cDVH}$. For (2), the loss weight parameters are set to 1 and 1, and for (3), they are set to 1, 0.5, and 0.5.

3.2.7 Model structure

For model structure selection, we focus on 3D neural networks proposed in the existing literature, as 2D neural networks often overlook spatial information between slices. There are several representative works in the literature based on UNet for dose prediction, such as DoseNet [89], HDUNet [23], C3D (the champion of the OpenKBP challenge, [91]), and U-NAS [93]. In recent years, transformer-based architectures have also been investigated, such as SwinUNETR [45] and DOSE-PYFER [24], which we also include. We selected these networks because of their superior performance in dose prediction tasks, making them strong candidates for this study. It is important to note that C3D and DOSE-PYFER have cascaded structures, while U-NAS is available in both single and cascaded setups.

3.2.8 Noise

To evaluate the robustness of the model structure, we introduce noise, including Poisson noise and adversarial noise, into the input CT image.

3.2.8.1 Poisson noise

Poisson noise frequently appears in CT images due to the nature of the photon detection process [98]. In CT imaging, the emitted X-ray photons are detected after passing through the body. However, the number of detected photons follows a Poisson distribution, which leads to statistical fluctuations in the image. The probability mass function (PMF) of a Poisson distribution is given by $P(k; \lambda) = \lambda^k e^{-\lambda} / k!$, where λ represents the average rate (i.e., the mean number of detected photons), and k is a discrete random variable representing the actual number of photons detected. In this study, we set the value of $\lambda = 20$ based on visual inspection.

3.2.8.2 Adversarial noise

Research indicates that deep learning models are susceptible to adversarial noise [99]. By adding a small, carefully designed perturbation to the original image, which may not be perceptible to humans, the model can produce outputs with a certain degree of error, leading to a significant decline in its accuracy. In this study, we focus on two commonly used methods to generate adversarial noise.

Projected Gradient Descent (PGD) [100] is an iterative method that generates an adversarial example by taking multiple small steps. For a single step, given input data x with its true dose distribution D , the adversarial example $\bar{x}$ is generated by perturbing the input in the direction that maximizes the loss with respect to D . The perturbation is updated by projecting it onto a specified ϵ -ball (a small region around the original data controlled by the parameter ϵ) while ensuring it maximizes the loss function:

$$\bar{x}^{(t+1)} = \text{Clip}_{x,\epsilon} \left(\bar{x}^{(t)} + \alpha \cdot \text{sign}(\nabla_x J(\theta, \bar{x}^{(t)}, D)) \right), \quad (3.7)$$

where t is the iteration index, α is the step size, $J(\theta, \bar{x}^{(t)}, D)$ is the loss function, and $\text{Clip}_{x,\epsilon}$ projects $\bar{x}$ back into the ϵ -neighborhood of x .

Momentum Iterative FGSM (Mi-FGSM) [101] improves on FGSM [102] by introducing momentum in gradient updates, helping to escape local optima more effectively. The update procedure becomes:

$$g^{(t+1)} = \mu \cdot g^{(t)} + \frac{\nabla_x J(\theta, \bar{x}^{(t)}, D)}{\|\nabla_x J(\theta, \bar{x}^{(t)}, D)\|_1}, \quad (3.8)$$

$$\bar{x}^{(t+1)} = \text{Clip}_{x,\epsilon} \left(\bar{x}^{(t)} + \alpha \cdot \text{sign}(g^{(t+1)}) \right), \quad (3.9)$$

where $g^{(t)}$ represents the accumulated gradient momentum up to iteration t , μ is the momentum parameter that controls the influence of previous gradients, and $\|\cdot\|_1$ denotes the L1 norm, which sums the absolute values of the gradient components. In this study, we set $\epsilon = 16$ HU based on visual inspection.

Table 3.2: Accuracy comparison of different input and dose grid resolution

Resolution	LUMC	
	Dose score (Gy) ↓	DVH score (Gy) ↓
$4 \times 4 \times 4 \text{ mm}^3$ (low resolution)	1.39 ± 0.28	1.85 ± 0.45
$3 \times 3 \times 3 \text{ mm}^3$ (medium resolution)	1.37 ± 0.26	1.67 ± 0.49
$2 \times 2 \times 2 \text{ mm}^3$ (high and original resolution)	1.27 ± 0.26	1.60 ± 0.55

Table 3.3: Accuracy comparison of different input types

Network input	OpenKBP		LUMC	
	Dose score (Gy) ↓	DVH score (Gy) ↓	Dose score (Gy) ↓	DVH score (Gy) ↓
CT	5.91 ± 2.20	8.82 ± 3.80	3.10 ± 0.87	12.09 ± 3.84
PTVs + OARs	2.68 ± 1.06	1.77 ± 1.23	1.31 ± 0.26	1.69 ± 0.65
CT + PTVs	2.73 ± 1.15	1.86 ± 1.14	1.28 ± 0.25	1.63 ± 0.49
CT + PTVs + OARs	2.52 ± 1.00	1.52 ± 1.13	1.27 ± 0.26	1.60 ± 0.55

Table 3.4: Accuracy comparison of different loss functions

Loss	OpenKBP		LUMC	
	Dose score (Gy) ↓	DVH score (Gy) ↓	Dose score (Gy) ↓	DVH score (Gy) ↓
$\mathcal{L}_{MAE}$	2.52 ± 1.00	1.52 ± 1.13	1.27 ± 0.26	1.60 ± 0.55
$\mathcal{L}_{MAE} + \mathcal{L}_{vDVH}$	2.60 ± 1.08	1.48 ± 1.11	1.30 ± 0.26	1.59 ± 0.55
$\mathcal{L}_{MAE} + \mathcal{L}_{vDVH} + \mathcal{L}_{cDVH}$	2.59 ± 1.09	1.41 ± 1.10	1.29 ± 0.27	1.48 ± 0.50

3.3 Results

3.3.1 Input and dose grid resolution

For the experiment evaluating the impact of different input and dose grid resolutions on dose prediction, we used the C3D model, the winning solution of the OpenKBP Challenge. The model input consisted of CT, PTVs, and OARs, and we employed the MAE loss function. Table 3.2 reports the accuracy on the LUMC dataset. The results indicate that using high and original resolution input ($2 \times 2 \times 2 \text{ mm}^3$) yielded the lowest Dose score and DVH score. Therefore, we selected this resolution for subsequent experiments on the LUMC dataset.

3.3.2 Input type

For the experiment that evaluated the impact of different input types on dose prediction, we used the C3D model and the MAE loss function. Table 3.3 reports the accuracy on the OpenKBP and LUMC datasets. The results show that the incorporation of CT, PTVs, and OARs resulted in the lowest Dose score and DVH score for both datasets. In contrast, using CT alone resulted in the worst scores.

Table 3.5: Accuracy comparison of different model structures

Model Structure		OpenKBP		LUMC	
		Dose score (Gy) ↓	DVH score (Gy) ↓	Dose score (Gy) ↓	DVH score (Gy) ↓
Single	DoseNet [89]	3.06 ± 1.22	1.67 ± 1.17	1.49 ± 0.27	1.56 ± 0.55
	HDUNet [23]	2.70 ± 1.05	1.46 ± 1.10	1.34 ± 0.26	1.57 ± 0.59
	SwinUNETR [45]	2.86 ± 1.16	1.50 ± 1.25	1.40 ± 0.26	1.54 ± 0.54
	U-NAS [93]	2.72 ± 1.17	1.45 ± 1.22	1.32 ± 0.26	1.51 ± 0.53
Cascade	C3D [91]	2.59 ± 1.09	1.41 ± 1.10	1.29 ± 0.27	1.48 ± 0.50
	DOSE-PYFER [24]	2.75 ± 1.18	1.49 ± 1.23	1.32 ± 0.24	1.54 ± 0.53
	U-NAS [93]	2.65 ± 1.05	1.37 ± 1.12	1.31 ± 0.27	1.45 ± 0.51

3.3.3 Loss function

For the experiment that assessed the impact of different loss functions on dose prediction, we used the C3D model with input consisting of CT, PTVs and OARs. As shown in table 3.4, for the OpenKBP dataset, the $\mathcal{L}_{MAE}$ loss achieved the lowest Dose score, while the $\mathcal{L}_{MAE} + \mathcal{L}_{vDVH} + \mathcal{L}_{cDVH}$ loss achieved the lowest DVH score along with a competitive Dose score. A similar trend was observed in the LUMC dataset. This pattern suggests that while the $\mathcal{L}_{MAE}$ loss alone provided the best Dose score, $\mathcal{L}_{MAE} + \mathcal{L}_{vDVH} + \mathcal{L}_{cDVH}$ offered the best DVH score and a more balanced overall improvement (that is, better Dose and DVH scores compared to $\mathcal{L}_{MAE} + \mathcal{L}_{vDVH}$). Given the high clinical relevance of the DVH score, we selected $\mathcal{L}_{MAE} + \mathcal{L}_{vDVH} + \mathcal{L}_{cDVH}$ for subsequent experiments.

3.3.4 Model structure

In this section, we used the input consisting of CT, PTVs, and OARs, and the loss function was $\mathcal{L}_{MAE} + \mathcal{L}_{vDVH} + \mathcal{L}_{cDVH}$. Table 3.5 shows that in the single model setup, U-NAS performed well on both datasets. In the cascade model setup, C3D achieved the best dose scores in both datasets, while U-NAS achieved the best DVH scores. In the LUMC dataset, the accuracy differences between all models are not great, being less than or equal to 0.2 Gy. However, for some of the clinical dose evaluation metrics shown in figures 3.2 and 3.3, there remain significant differences between predictions and the clinical plan (mainly with respect to $V_{95\%}$ of PTV_{54.25}, D_{mean} of PTV₇₀, $D_{0.03cc}$ of Spinal Cord, and D_{mean} of the contralateral submandibular gland).

Under Poisson noise (table 3.6), all models demonstrated strong robustness, with most models having Δ Dose and Δ DVH scores below 0.1 Gy. Figure 3.4 also illustrates this point: even in extreme conditions with $\lambda = 20$, although the quality of the CT images had severely degraded, the dose prediction remained almost unchanged.

The robustness of different model structures was further evaluated under adversarial noise by PGD and Mi-FGSM attacks (table 3.7, table 3.8). The tables reveal that SwinUNETR consistently demonstrated superior resilience across both the OpenKBP

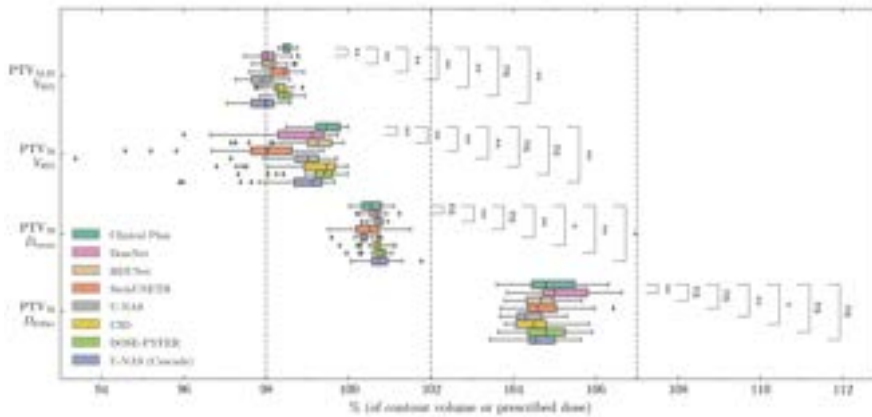


Figure 3.2: Boxplots with clinical dose evaluation metrics for targets on the LUMC dataset. For $PTV_{54.25}$, the clinical goal is $V_{95\%} \geq 98\%$. For PTV_{70} , the clinical goals are $V_{95\%} \geq 98\%$, $D_{mean} \leq 102\%$ (of the prescribed dose), and $D_{0.03cc} \leq 107\%$ (of the prescribed dose). Statistical significance was tested using a two-tailed Wilcoxon signed-rank test. **: $p \leq 0.01$, *: $0.01 < p \leq 0.05$, ns = not significant.

Table 3.6: Robustness comparison of different model structures against Poisson noise ($\lambda = 20$)

Model Structure		OpenKBP		LUMC	
		Δ Dose score (Gy) ↓	Δ DVH score (Gy) ↓	Δ Dose score (Gy) ↓	Δ DVH score (Gy) ↓
Single	DoseNet [89]	0.01 ± 0.02	0.01 ± 0.10	0.03 ± 0.01	0.02 ± 0.05
	HDUNet [23]	0.28 ± 0.27	0.10 ± 0.32	0.08 ± 0.06	0.03 ± 0.21
	SwinUNETR [45]	0.04 ± 0.10	0.09 ± 0.18	0.12 ± 0.05	0.16 ± 0.17
	U-NAS [93]	0.00 ± 0.03	0.00 ± 0.03	0.00 ± 0.01	0.01 ± 0.03
	C3D [91]	0.00 ± 0.04	0.00 ± 0.06	0.02 ± 0.02	0.00 ± 0.08
Cascade	DOSE-PYFER [24]	0.00 ± 0.07	0.00 ± 0.07	0.01 ± 0.01	0.00 ± 0.06
	U-NAS [93]	0.00 ± 0.04	0.00 ± 0.04	0.01 ± 0.01	0.00 ± 0.02

and LUMC datasets when subjected to these adversarial noises. Notably, U-NAS (cascade), which had performed well under Poisson noise, struggled considerably under adversarial noise, especially on the OpenKBP dataset. Figure 3.5 visually aligns with these findings. As the attack strength increases, both U-NAS (cascade) and SwinUNETR show increasing degradation from the original dose predictions. However, SwinUNETR maintains greater stability, with less pronounced deviations compared to U-NAS (cascade), particularly as the attack strength becomes stronger.

Table 3.9 compares the computational efficiency of various model structures. DoseNet was the most computationally efficient among the single models, having the lowest GPU memory usage and runtime. Among the cascade models, U-NAS was the most efficient compared to others. SwinUNETR, among all models, demanded the most resources and had the longest runtimes. The results also indicated that the LUMC dataset, which contains higher resolution images, led to higher GPU memory usage

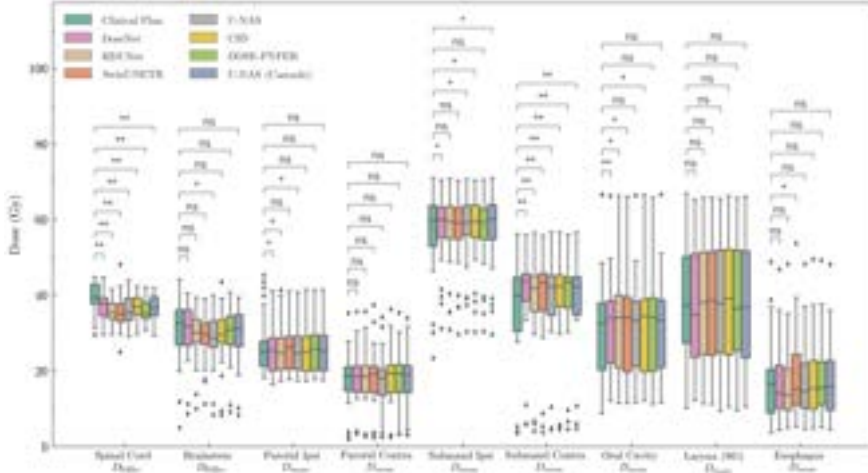


Figure 3.3: Boxplots with clinical dose evaluation metrics for OARs on the LUMC dataset. Ipsi = ipsilateral, Contra = contralateral, Submand = submandibular gland, SG = supraglottis. Statistical significance was tested using a two-tailed Wilcoxon signed-rank test. **: $p \leq 0.01$, *: $0.01 < p \leq 0.05$, ns = not significant.

Table 3.7: Robustness comparison of different model structures against PGD attack ($\epsilon = 16$ HU)

Model Structure		OpenKBP		LUMC	
		Δ Dose score (Gy) ↓	Δ DVH score (Gy) ↓	Δ Dose score (Gy) ↓	Δ DVH score (Gy) ↓
Single	DoseNet [89]	3.63 ± 1.85	6.41 ± 1.87	0.66 ± 0.23	0.64 ± 0.42
	HDUNet [23]	0.20 ± 0.30	0.23 ± 0.43	0.22 ± 0.14	0.18 ± 0.26
	SwinUNETR [45]	0.06 ± 0.19	0.17 ± 0.37	0.18 ± 0.12	0.01 ± 0.17
	U-NAS [93]	0.35 ± 0.43	0.31 ± 0.54	0.67 ± 0.20	0.46 ± 0.29
Cascade	C3D [91]	0.14 ± 0.27	0.24 ± 0.40	0.08 ± 0.07	0.13 ± 0.20
	DOSE-PYFER [24]	0.29 ± 0.41	0.55 ± 0.65	0.11 ± 0.09	0.12 ± 0.21
	U-NAS [93]	0.35 ± 0.39	0.31 ± 0.49	0.30 ± 0.14	0.31 ± 0.26

and longer runtimes compared to the OpenKBP dataset. Note that all models achieved GPU runtimes of less than 1 second in both datasets.

3.4 Discussion

In this paper, we explore the factors that affect the performance of deep learning-based dose prediction using the publicly available OpenKBP dataset and in-house LUMC dataset. From a dataset perspective, the OpenKBP dataset has certain limitations. The data originates from The Cancer Imaging Archive (TCIA), has been re-annotated and processed; the resolution of the CT image and dose grid is lower than clinical standard. In addition, it does not include a clinically approved RT plan. However, the LUMC dataset compensates for these shortcomings well. Furthermore, the IMRT-based

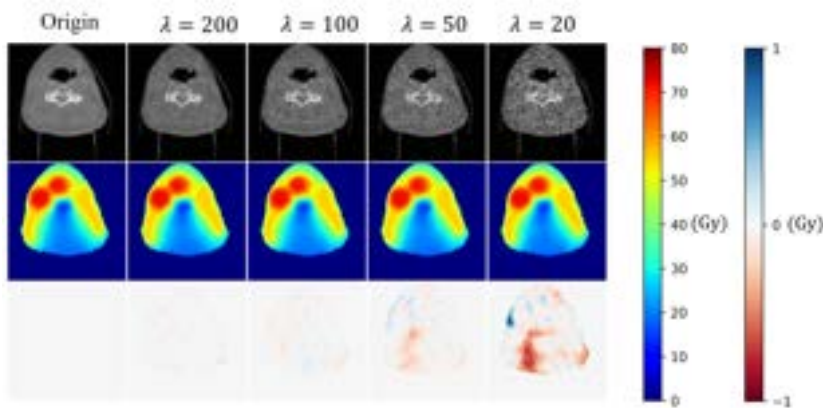


Figure 3.4: Impact of different Poisson noise levels (λ) on U-NAS (cascade) dose prediction, using a randomly selected slice from the LUMC dataset. The first row shows CT images with increasing noise levels. The second and third rows depict U-NAS (cascade) dose predictions and the difference map from the original dose prediction (noise-free), respectively. The color bars represent dose values (0 to 80 Gy) and the difference map from the original prediction (-1 to 1 Gy).

Table 3.8: Robustness comparison of different model structures against Mi-FGSM attack ($\epsilon = 16$ HU)

Model Structure		OpenKBP		LUMC	
		Δ Dose score (Gy) ↓	Δ DVH score (Gy) ↓	Δ Dose score (Gy) ↓	Δ DVH score (Gy) ↓
Single	DoseNet [89]	4.42 ± 1.72	7.77 ± 1.87	0.78 ± 0.23	0.50 ± 0.26
	HDUNet [23]	0.63 ± 0.24	0.90 ± 0.35	0.48 ± 0.10	0.43 ± 0.20
	SwinUNETR [45]	0.24 ± 0.13	0.52 ± 0.24	0.20 ± 0.07	0.22 ± 0.18
	U-NAS [93]	0.93 ± 0.56	0.93 ± 0.51	0.70 ± 0.27	0.41 ± 0.38
	C3D [91]	0.51 ± 0.52	0.63 ± 0.61	0.23 ± 0.07	0.28 ± 0.31
Cascade	DOSE-PYFER [24]	0.63 ± 0.53	1.11 ± 0.69	0.31 ± 0.17	0.40 ± 0.37
	U-NAS [93]	0.99 ± 0.73	1.28 ± 0.66	0.32 ± 0.09	0.31 ± 0.20

OpenKBP and VMAT-based LUMC datasets well represent current treatment planning techniques in the field of radiation therapy. Our findings indicate that most conclusions are applicable to both datasets, suggesting that the results discussed in this paper have considerable generalizability.

To ensure a fair comparison, all models were either re-implemented based on their original publications or adapted from publicly available GitHub repositories (e.g., C3D, DOSE-PYFER, U-NAS). For models without released code, we re-implemented the architectures using the MONAI¹ framework and closely followed the original publication. Although there may be implementation differences, we standardized the training settings across all models as described in Chapter 3.2.2 (e.g., same optimizer,

¹<https://www.monai.io>

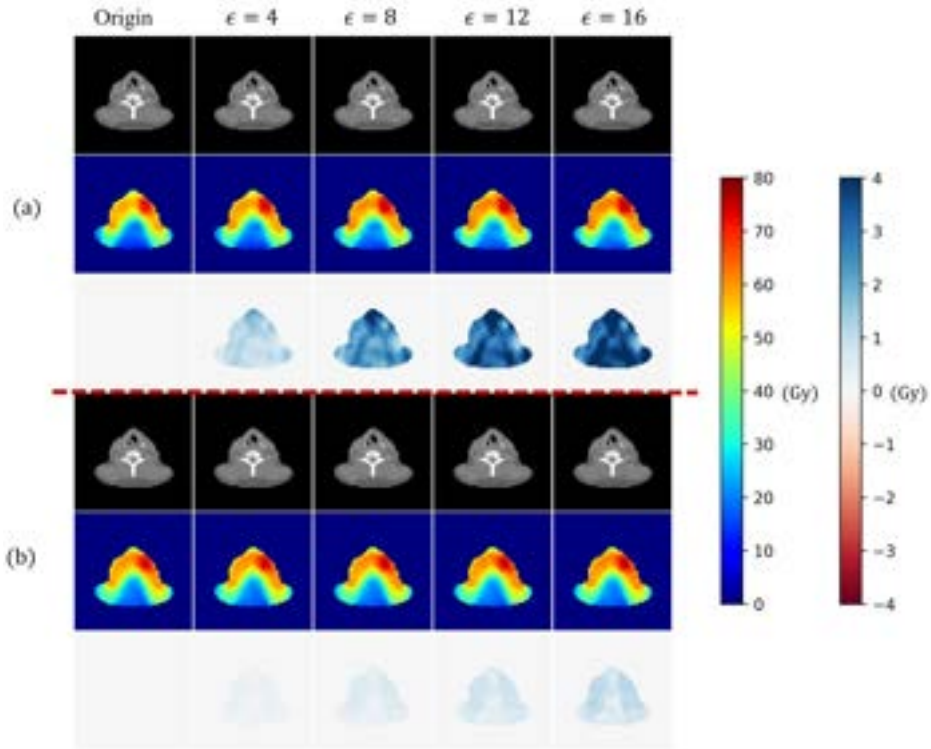


Figure 3.5: Impact of different attack strengths (ϵ) of the MI-FGSM attack on U-NAS (cascade) and SwinUNETR, using a randomly selected slice from the OpenKBP dataset. Panel (a) showcases the results on U-NAS (cascade): the first row displays the original and perturbed CT images, the second and third rows depict the corresponding dose predictions and the difference map from the original dose prediction (noise-free), respectively. Panel (b) presents similar sequence results for SwinUNETR. The color bars represent dose values (0 to 80 Gy) and the difference map from the original prediction (-4 to 4 Gy).

Table 3.9: Computational efficiency comparison of different model structures

Model Structure		OpenKBP		LUMC	
		GPU memory (MB)	GPU runtime (ms)	GPU memory (MB)	GPU runtime (ms)
Single	DoseNet [89]	297	11	708	31
	HDUNet [23]	2589	109	9437	179
	SwinUNETR [45]	3593	126	11359	440
	U-NAS [93]	1412	58	4778	142
Cascade	C3D [91]	2254	77	8036	241
	DOSE-PYFER [24]	2309	186	6798	512
	U-NAS [93]	1550	115	5150	356

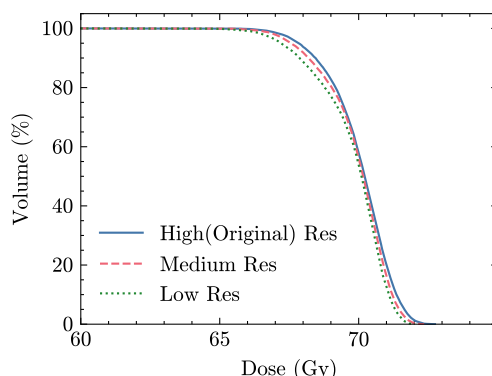


Figure 3.6: Comparison of the DVH curves of PTV₇₀ under high (original), medium, and low resolutions.

learning rate, and maximum training epoch). These settings were chosen because we found them to be robust across different model architectures and generally allowed the models to reach performance levels comparable to those reported in the original papers. We acknowledge that further hyperparameter tuning could improve individual model performance. However, our goal was to benchmark under a unified and reproducible training setup. All code and configurations used in this study are publicly available at <https://github.com/RuochenGao/HaN-DosePrediction>.

As shown in figure 3.6, the differences observed in the dose fall-off region of the DVH plot are due to interpolation effects. The primary goal of RT planning is to deliver a high radiation dose to the tumor while minimizing exposure to surrounding healthy tissues, creating a sharp dose gradient visible as a steep decline in the DVH curve. However, interpolation algorithms can smooth out this gradient between high-dose and low-dose regions, introducing deviations from the actual dose distribution. Therefore, we recommend using high-resolution input and dose grid to preserve the integrity of the gradient.

From table 3.3, we can see that the model performs poorly when using only CT

images. This is because CT scans do not provide the neural network with explicit information about the tumor’s location, shape, and size, and the neural network must therefore learn to extract this information from the CT data directly, without the use of supervision. This significantly increases the difficulty of the overall learning task for dose prediction, as tumors in the head and neck region exhibit irregular shapes and sizes. In contrast, when using only PTVs + OARs, we observe that the model performed well. This indicates that the spatial and anatomical relationships between targets and OARs contain the most critical information for the dose prediction task. While CT (HU values) play a key role in traditional dose calculation engines by enabling photon attenuation modeling, deep learning models appear to rely more heavily on explicit geometric representations provided by PTV and OAR contours. However, the combination of CT + PTVs + OARs achieved the highest accuracy, and therefore we still recommend using this combination.

The selection of a loss function is closely related to the desired output. In RT planning, clinicians are more concerned with ensuring adequate dose coverage for PTVs and minimizing the mean and maximum dose to OARs, which is related to the DVH score in our study. Previous work has demonstrated the benefits of incorporating a value-based DVH loss [95]. As demonstrated in table 3.4, we found that combining MAE, value-based DVH loss, and criteria-based DVH loss functions improves the DVH score even further. Moreover, the criteria-based DVH loss can be tailored to meet specific clinical requirements in different scenarios. Thus, we recommend using the combined loss function $\mathcal{L}_{MAE} + \mathcal{L}_{vDVH} + \mathcal{L}_{cDVH}$ to achieve optimal results.

From table 3.5, we observe that models with a cascade architecture often outperformed single models, achieving lower Dose and DVH scores, indicating improved accuracy in dose prediction. This suggests that the progressive refinement inherent in cascade architectures enables more precise adjustments during the prediction process, resulting in better accuracy in capturing complex patterns within CT scans, combined with PTVs and OARs. However, as shown in table 3.9, the GPU runtimes for cascade models were generally longer than those for single models, indicating a trade-off in computational efficiency. Since the inference time remained below 1 second, this extended runtime may not pose a significant issue. We believe that for the complex task of dose prediction, a cascade architecture may be necessary to achieve optimal performance. Furthermore, it should be noted that U-NAS (cascade) achieved the best results on the DVH score. However, as shown in figure 3.2, U-NAS (cascade) shows significant differences ($p \leq 0.01$) compared to the clinical plan in terms of $V_{95\%}$ for PTV_{54.25} and PTV₇₀. In contrast, DOSE-PYFER performs better on these metrics, showing no significant differences from the clinical plan ($p > 0.05$). This discrepancy arises because the DVH score is an aggregated metric composed of several DVH metrics. As such, it does not necessarily reflect good performance in all individual DVH metrics.

Furthermore, U-NAS focuses on finding the optimal neural network structure based on the current loss function, which does not include metrics such as $V_{95\%}$.

In the experiments on model robustness, we found that for Poisson noise, which commonly occurs in CT scans, all models exhibited strong robustness (table 3.6). As shown in figure 3.4, where the quality of CT image had been significantly degraded at $\lambda = 20$, the dose prediction showed almost no change. However, when facing adversarial noise, where imperceptible noise is added to the original CT image (shown in figure 3.5), the accuracy of the model can be affected (table 3.7, table 3.8). This is mainly because adversarial noise is generated based on the model's gradient, maximizing the model's prediction error. However, we observed that SwinUNETR demonstrated stronger robustness against adversarial noise. This may be due to SwinUNETR divides the input data into non-overlapping patches and processes these patches individually. This patch-based approach can act as a form of regularization, reducing the impact of adversarial noise by ensuring that localized perturbations in each patch do not significantly affect the overall representation. However, SwinUNETR does not perform as well as other models, such as C3D and U-NAS, in terms of the accuracy metric. Furthermore, it has the highest GPU memory consumption among the evaluated models. Therefore, for practical clinical deployment, it is essential to choose an appropriate model structure that balances accuracy, robustness, and computational efficiency.

One aspect not explored in this study is the incorporation of prior knowledge information such as beam configuration. As the beam setup is fixed in our dataset, the influence of beam configuration was not investigated in this study. However, recent studies, such as [25], have shown that including beam-related information (e.g., beam angles and beam plates) can improve dose prediction accuracy, particularly in more diverse clinical scenarios. Although this was beyond the scope of the current work, incorporating this prior knowledge is a promising direction for future research to improve model generalizability.

Another limitation is that we add noise to CT images to verify the robustness of the model structure. However, in clinical practice, the uncertainty introduced by inter-physician variability in the contouring of tumors and OARs [10] represents another critical element affecting model robustness. Future work will focus on investigating the impact of this uncertainty on dose prediction. We suggest that in dose prediction tasks, it is essential not only to focus on the accuracy of the model but also to consider the robustness of the model, including its ability to counter noise and handle uncertainty. These aspects have significant practical value in clinical applications.

Finally, our study focuses on photon-based radiotherapy (IMRT and VMAT), while proton therapy is also currently used for head and neck cancer treatment. Proton therapy introduces additional challenges in dose prediction due to its higher sensitivity

to variations in CT HU values and its sharper dose gradients. While our findings suggest that photon-based deep learning models benefit in a relatively small amount from CT as an input (table 3.3), this input may be more relevant for proton therapy dose prediction. In the future, a dedicated evaluation using proton therapy datasets will be needed to determine if our findings can be generalized to proton radiotherapy.

3.5 Conclusion

This study presents a comprehensive analysis of key factors that influence deep learning-based dose prediction models for head and neck cancer radiotherapy. By systematically examining input and dose grid resolution, input type, loss function, and model architecture using both public and in-house clinical datasets, we demonstrate their significant effects on model accuracy, robustness, and computational efficiency. Our findings show that high-resolution inputs, specifically CT images with PTVs and OARs, combined with a hybrid loss function that incorporates MAE and value-based and criteria-based DVH components, substantially improve prediction accuracy. Robustness testing reveals that while most models exhibit greater resistance to Poisson noise than adversarial noise, certain models, such as SwinUNETR, demonstrate superior robustness against adversarial perturbations. These insights provide valuable guidance for optimizing deep learning-based dose prediction models, contributing to more precise and reliable radiotherapy planning.

4

Clinical DVH Metrics as a Loss Function for 3D Dose Prediction in Head and Neck Radiotherapy

This chapter was adapted from:

Ruochen Gao, Marius Staring, and Frank Dankers, "Clinical DVH metrics as a loss function for 3D dose prediction in head and neck radiotherapy", accepted for publication in the International Journal of Computer Assisted Radiology and Surgery (IJCARS), CARS 2026 Special Issue.

Abstract

Purpose Deep-learning-based three-dimensional (3D) dose prediction has become an important component of automated radiotherapy workflows. However, most existing models are trained using voxel-wise regression losses, which are poorly aligned with clinical plan evaluation criteria that rely on dose-volume histogram (DVH)-derived metrics. This study aims to develop a clinically guided loss formulation that directly optimizes clinically used DVH metrics while remaining computationally efficient for head and neck (H&N) dose prediction.

Methods We propose a clinical DVH metric loss (CDM loss) that jointly incorporates differentiable formulations of common *D-metrics* and differentiable surrogates of *V-metrics*. The loss is driven by a JSON-based clinical plan evaluation template to ensure alignment with clinical planning criteria. In addition, we introduce a lossless bit-mask encoding scheme to efficiently represent a large number of overlapping ROIs, substantially improving training efficiency. The method was evaluated on a retrospective cohort of 174 H&N patients treated with VMAT, using a temporal split for training ($n = 137$) and testing ($n = 37$).

Results Compared with MAE- and DVH-based losses, the proposed CDM loss substantially improved clinical target coverage, and satisfied all predefined clinical constraints. Using a standard 3D U-Net, the PTV Score was reduced from 1.544 (MAE) to 0.491 (MAE+CDM), while consistently satisfying all PTV clinical constraints. OAR sparing remained comparable or improved. Bit-mask ROI encoding reduced average training epoch time from 241 s to 43 s (82.2% reduction) and lowered peak GPU memory usage. Notably, with the CDM loss, a standard 3D U-Net achieved performance comparable to or better than more complex state-of-the-art architectures.

Conclusion Directly optimizing clinically used DVH metrics enables 3D dose predictions that are better aligned with clinical treatment planning criteria than conventional only voxel-wise or DVH-curve supervision. The proposed CDM loss, combined with efficient ROI bit-mask encoding, provides a practical and scalable framework for H&N dose prediction.

4.1 Introduction

Radiotherapy (RT) is widely used in cancer treatment to deliver sufficient radiation to eradicate tumor cells while protecting surrounding healthy tissues. Conventional treatment planning directly optimizes beam parameters within the treatment planning system (TPS), often requiring multiple rounds of manual refinements and substantial planner expertise. This process is highly time-consuming, especially for anatomically complex sites. To improve efficiency and consistency, modern workflows increasingly introduce dose prediction as an intermediate step, providing an anatomy-informed estimate of a clinically realistic three-dimensional (3D) dose distribution. This predicted dose can then be used to guide subsequent TPS optimization through a dose mimicking process, in which a deliverable treatment plan is optimized to closely reproduce the predicted 3D dose distribution [85].

Among all treatment sites, head and neck (H&N) radiotherapy is particularly challenging due to its intricate anatomy, a large number of critical organs-at-risk (OARs), and the need for steep dose gradients between planning target volumes (PTVs) and nearby healthy organs [9]. These complexities make high-quality optimization particularly demanding and contribute to considerable variability in plan quality across clinicians and institutions [21]. Therefore, developing accurate and automated 3D dose prediction models tailored for H&N treatment is essential to reduce planning workload and promote standardization in clinical practice.

Early dose prediction approaches relied on atlas-based and knowledge-based planning, which generated 3D dose estimates from prior clinical plans and handcrafted anatomical features [103]. Although these methods improve planning consistency, they are limited in capturing the complex nonlinear relationship between anatomy and dose. Recently, deep learning (DL) methods have shown strong potential for learning complex mappings between patient anatomy and clinical dose patterns. Most DL-based methods use 3D convolutional or transformer-based U-shaped architectures [91, 93, 24, 28], taking CT and region-of-interest (ROI) masks as input to generate 3D voxel-wise dose predictions, as shown in Fig. 4.1. These approaches generally achieve high voxel-wise accuracy because they are trained with regression losses such as mean absolute error (MAE) or mean squared error (MSE), which encourage global similarity to clinical dose distributions. However, voxel-wise accuracy is not the primary criterion used in clinical plan evaluation. Instead, clinicians focus on achieving adequate planning target volume (PTV) coverage and sufficient organ-at-risk (OAR) sparing, which are typically assessed using dose-volume histogram (DVH)-derived metrics. Because standard regression losses do not directly optimize these clinical objectives, DL-based models may achieve high numerical accuracy, yet still perform sub-optimally from a clinical standpoint.



Figure 4.1: Workflow of a typical deep-learning-based H&N dose prediction method. The planning CT and masks of the PTVs and OARs are provided as inputs to a U-shaped neural network to generate a 3D voxel-wise dose distribution. In H&N cases, the PTVs are typically assigned two or three dose levels, while the number of OARs is relatively large compared to other treatment sites.

Table 4.1: Definitions of common D - and V -metrics in H&N RT.

Parameter	Definition
$D_{x\%}$	Minimum dose received by $x\%$ of the ROI volume.
$D_{x\ cc}$	Minimum dose received by the hottest $x\ cc$ of the ROI.
$D_{\max}$	Maximum dose within the ROI.
$D_{\min}$	Minimum dose within the ROI.
D_{mean}	Mean dose within the ROI.
$V_{x\%}$	Volume percentage of an ROI receiving at least $x\%$ of the prescription dose.
$V_x\ \text{Gy}$	Volume percentage of an ROI receiving at least $x\ \text{Gy}$.

To bridge this gap, several studies have incorporated differentiable DVH losses, which allow comparison of predicted and ground-truth DVH curves during training [95, 104, 26, 27]. In clinical practice, however, plan evaluation is typically based on a collection of dose metrics rather than full DVH curves. For H&N RT, these metrics commonly include both D -metrics (e.g., $D_{x\%}$, the minimum dose received by $x\%$ of an ROI) and V -metrics (e.g., $V_{x\%}$, the volume of an ROI receiving at least $x\%$ of the prescription dose), as summarized in Table 4.1. While D -metrics are differentiable and can be incorporated into training [26, 27], V -metrics contain threshold operations that are inherently non-differentiable, posing challenges for the gradient-based training framework. Therefore, effective methods for directly optimizing V -metrics remain limited, and a comprehensive framework that jointly supports both D - and V -metrics is still needed.

Another practical challenge arises from the large number of OARs in H&N anatomy. Unlike DL-based medical imaging segmentation models, which often train on small 3D patches, dose-prediction networks typically require the full 3D volume as input to guarantee spatial dose continuity. Current methods generally implement these ROIs as one input channel per ROI. As a result, when many OARs are included, the

number of input channels grows rapidly, leading to substantial preprocessing overhead, higher CPU—to—GPU data transfer cost, increased GPU memory usage, and longer training time. Although merging all ROIs into a single indexed mask could reduce input dimensionality, this approach faces two major issues: (1) many OARs overlap spatially (e.g., the brain and brainstem overlap), making direct merging impossible; and (2) the numerical labels introduce unintended ordinal bias (e.g., a label “9” may be treated differently from a label “1” purely because it is larger), even though ROI labels should carry no numeric significance. This unintended numerical meaning can mislead the network during training and negatively affect learning [105]. As a result, prior studies often limit the input to a subset of OARs [22, 93, 24, 28, 26, 27], which is restrictive because clinical evaluation typically involves a much broader set of OARs. This challenge highlights the need for models that can efficiently incorporate all clinically relevant ROIs while maintaining computational efficiency.

To address these limitations, we propose a clinical DVH metric loss (CDM loss), together with an efficient bit—mask ROI encoding strategy for 3D H&N dose prediction. Our framework enables direct optimization of the full set of clinically defined dose evaluation metrics while maintaining computational efficiency. Our main contributions are summarized as follows:

- We introduce a clinical DVH metric loss (CDM loss) that combines differentiable formulations for *D-metrics* with differentiable approximations for *V-metrics*. By directly optimizing the metrics used in clinical plan evaluation, we show that even a standard 3D U-Net can achieve clinically satisfactory dose predictions.
- We develop an efficient lossless bit—mask encoding for ROI representation that reduces CPU preprocessing, CPU—to—GPU data transfer, and GPU memory usage through lightweight, on-demand decoding. This strategy substantially accelerates training, while supporting the full set of OARs required for clinical evaluation.
- We organize the clinical evaluation dose metrics used by CDM loss function through a simple JSON-based template, as summarized in Table 4.2. This template specifies the ROIs and their associated dose metrics in a clear, structured, and easily updated format, ensuring that the training objectives remain aligned with clinical evaluation criteria used in a given institute.

4.2 Materials

4.2.1 Dataset

In this study, data were collected retrospectively from 174 H&N cancer patients treated at Leiden University Medical Center from February 2021 to July 2025. The study

Table 4.2: The H&N RT clinical plan evaluation template used in our institution. The PTVs are denoted by their prescribed dose levels (e.g., PTV_{54.25} for 54.25 Gy and PTV₇₀ for 70 Gy). ROIs with “L/R” denote paired organs and are evaluated independently for the left and right sides. Each ROI is evaluated using one or more dose metrics. “Aim” represents the desired planning goal, while “Constraint” denotes a mandatory criterion. The template is implemented via a JSON file.

ROI	Metric	Aim	Constraint
PTV _{54.25}	V _{95%}		≥ 98%
PTV _{54.25}	D _{mean}	≤ 102%	
PTV ₇₀	V _{95%}		≥ 98%
PTV ₇₀	D _{0.03cc}	≤ 107%	≤ 110%
PTV ₇₀	D _{mean}	≤ 102%	
SpinalCord	D _{0.03cc}		≤ 50 Gy
SpinalCord + 3 mm	D _{0.03cc}		≤ 52 Gy
Brainstem_Surf	D _{0.03cc}		≤ 60 Gy
Brainstem_Core	D _{0.03cc}		≤ 54 Gy
Brain	D _{0.03cc}	≤ 65 Gy	
Brain	D _{2%}	≤ 70 Gy	
Cochlea_(L/R)	D _{mean}	≤ 45 Gy	
Parotid_(L/R)	D _{mean}	≤ 28 Gy	
Gland_Submand_(L/R)	D _{mean}	≤ 35 Gy	
Oral_Cavity	D _{mean}	≤ 28 Gy	
Musc_Constrict_S	D _{mean}	≤ 40 Gy	
Musc_Constrict_M	D _{mean}	≤ 40 Gy	
Musc_Constrict_I	D _{mean}	≤ 40 Gy	
Cricopharyngeus	D _{mean}	≤ 40 Gy	
Larynx_SG	D _{mean}	≤ 40 Gy	
Glottic_Area	D _{mean}	≤ 40 Gy	
Bone_Mandible	D _{2%}	≤ 70 Gy	
Bone_Mandible-PTV	D _{2%}	≤ 50 Gy	
Eye_(L/R)	D _{0.03cc}	≤ 35 Gy	
Lens_(L/R)	D _{0.03cc}	≤ 6 Gy	
Pituitary	D _{mean}	≤ 20 Gy	
OpticNrv_(L/R)	D _{0.03cc}	≤ 55 Gy	

was approved by the Medical Ethics Committee of Leiden, The Hague, and Delft (approval number G21.142, October 15, 2021). The cohort covered a range of tumor sites, including the oropharynx, hypopharynx, nasopharynx, larynx, and oral cavity. All patients were treated on Elekta Synergy linear accelerators (Elekta, Stockholm, Sweden) using 6 MV dual-arc, full-rotation volumetric modulated arc therapy (VMAT) and all RT plans were generated in RayStation™ (RaySearch Laboratories, Stockholm,

Sweden).

To better reflect real-world clinical application, in which a model trained on historical data is used to make predictions for future patients, the dataset was divided chronologically. Patients treated between February 2021 and July 2024 ($n = 137$) were assigned to the training cohort, while those treated between August 2024 and July 2025 ($n = 37$) formed the test cohort, enabling temporal (out-of-time) validation on later cases.

The original CT images had an average resolution of $1.2 \times 1.2 \times 2 \text{ mm}^3$, whereas the dose grid resolution was fixed at $2 \times 2 \times 2 \text{ mm}^3$. To ensure uniform spatial resolution, all CT scans were resampled to $2 \times 2 \times 2 \text{ mm}^3$. Cropping or padding were performed around the treatment isocenter, defined as the geometric center of the planned radiotherapy beams, to obtain a uniform volume size of (160, 192, 256). The same preprocessing pipeline was applied to all ROIs.

For intensity normalization, CT values were clipped to the range $[-1024, 1024]$ and then normalized to $[0, 1]$ using min–max normalization. Dose values were normalized by dividing by 70 (Gy), the highest prescription dose in our cohort.

4.2.2 Evaluation metrics

To quantitatively evaluate model performance, three complementary evaluation metrics are defined: PTV Score, OAR Score, and Dose Score. The PTV Score and OAR Score are directly derived from the clinical plan evaluation template used in our institution (Table 4.2), which specifies the clinical dose evaluation metrics for each ROI. These two scores are reported separately because they represent different clinical objectives and optimization priorities.

Let $\mathcal{P}$ denote the set of all PTVs, and $\mathcal{O}$ denote the set of all OARs. For each ROI $r \in \mathcal{P} \cup \mathcal{O}$, a corresponding set of clinical evaluation dose metrics $\mathcal{M}_r$ is defined according to Table 4.2. For example, $\mathcal{M}_r = \{V_{95\%}, D_{\text{mean}}, D_{0.03\text{cc}}\}$ for PTV₇₀, and $\mathcal{M}_r = \{D_{0.03\text{cc}}, D_{2\%}\}$ for Brain. For each metric $m \in \mathcal{M}_r$, let $M_{r,m}^{\text{pred}}$ and $M_{r,m}^{\text{gt}}$ denote the predicted and ground truth (clinical) values, respectively.

The PTV Score is defined as:

$$\text{PTV Score} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \frac{1}{|\mathcal{M}_p|} \sum_{m \in \mathcal{M}_p} \left| M_{p,m}^{\text{pred}} - M_{p,m}^{\text{gt}} \right|, \quad (4.1)$$

where all PTV metrics are normalized to the prescribed dose and expressed in percentage (%). Similarly, the OAR Score is defined as:

$$\text{OAR Score} = \frac{1}{|\mathcal{O}|} \sum_{o \in \mathcal{O}} \frac{1}{|\mathcal{M}_o|} \sum_{m \in \mathcal{M}_o} \left| M_{o,m}^{\text{pred}} - M_{o,m}^{\text{gt}} \right|, \quad (4.2)$$

where all OAR metrics are measured in absolute dose (Gy).



Figure 4.2: (a) The 3D dose distribution within an ROI is flattened into a one-dimensional dose array $\{d_i\}_{i=1}^N$. After a descending sorting operation, we get $d_1^{\downarrow} \geq d_2^{\downarrow} \geq \dots \geq d_N^{\downarrow}$. A D -metric, such as $D_{x\%}$, is obtained by retrieving the dose value $d_k^{\downarrow}$ at the index k that corresponds to the top $x\%$ portion of the ROI volume in the sorted dose array. (b) Using the same flattened dose array $\{d_i\}_{i=1}^N$, a V -metric such as $V_{x\%}$ is computed by applying a dose threshold equal to $x\%$ of the prescription dose, summing the number of voxels that exceed this threshold, and computing the corresponding volume percentage. For example, $V_{95\%}$ of PTV_{70} denotes the fraction of the PTV_{70} volume receiving at least 95% of the 70 Gy dose level.

The Dose Score is also reported to quantify the overall voxel-wise difference between the predicted and ground truth (clinical) 3D dose distributions. Let Ω denote the set of all voxels within the dose grid, and $D^{\text{pred}}(v)$ and $D^{\text{gt}}(v)$ be the predicted and clinical dose at voxel $v \in \Omega$. The Dose Score is defined as:

$$\text{Dose Score} = \frac{1}{|\Omega|} \sum_{v \in \Omega} \left| D^{\text{pred}}(v) - D^{\text{gt}}(v) \right|, \quad (4.3)$$

reported in Gy. Although this metric has limited direct clinical relevance, it is widely adopted in the literature and facilitates consistent comparison with prior studies.

4.3 Methods

4.3.1 CDM loss formulation

Based on the clinical plan evaluation criteria summarized in Table 4.2, we formulate the proposed CDM loss that directly optimizes the specified dose metrics. It incorporates both D - and V -metrics, allowing end-to-end training guided by clinical evaluation

criteria. The following sections describe the differentiable formulation of *D-metrics* and the surrogate-based differentiable approximation of *V-metrics*.

4.3.1.1 Differentiable formulation of *D-metrics*

As shown in Fig. 4.2(a), given a set of voxel doses $\{d_i\}_{i=1}^N$ within an ROI of size N , sorted in descending order as $d_1^\downarrow \geq d_2^\downarrow \geq \dots \geq d_N^\downarrow$, the minimum dose received by the hottest $x\%$ of the ROI volume, is denoted as:

$$D_{x\%} = d_k^\downarrow, \text{ where } k = \left\lceil \frac{x}{100} N \right\rceil. \quad (4.4)$$

Thus, $D_{x\%}$ corresponds to the k -th largest dose value, which can be found at index k after sorting. Similarly, the minimum dose received by the hottest x cc of the ROI, denoted as $D_{x \text{ cc}}$, is computed in an analogous manner:

$$D_{x \text{ cc}} = d_k^\downarrow, \text{ where } k = \left\lceil \frac{x}{V_{\text{voxel}}} \right\rceil, \quad (4.5)$$

where V_{voxel} denotes the volume of a single voxel (in cc). Both $D_{x\%}$ and $D_{x \text{ cc}}$ can therefore be interpreted as quantile-based metrics, differing only in whether the cutoff is defined by a fractional or absolute volume.

In addition to these quantile metrics, maximum and minimum dose within an ROI can be expressed naturally in the same framework:

$$D_{\max} = d_1^\downarrow, \quad D_{\min} = d_N^\downarrow. \quad (4.6)$$

Although sorting indices are non-differentiable, deep learning frameworks such as PyTorch¹ allow gradients to propagate through the sorted values via gather/scatter operations. Following prior work [26, 27], we use a top- k algorithm that computes the required quantiles without explicitly sorting all voxels, offering both differentiability and computational efficiency.

Finally, the mean dose $D_{\text{mean}} = \frac{1}{N} \sum_{i=1}^N d_i$ is inherently differentiable.

4.3.1.2 Differentiable surrogate of *V-metrics*

V-metrics, including $V_{x\%}$ and $V_{x \text{ Gy}}$, describe the fraction of an ROI receiving at least a specified dose level. As illustrated in Fig. 4.2(b), $V_{x\%}$ denotes the fraction of the ROI receiving at least $x\%$ of the prescription dose. Given the prescription dose D_{presc} , the corresponding threshold is $T = x D_{\text{presc}} / 100$ Gy. The computation of $V_{x\%}$ is equivalent to counting the number of voxels with dose $d_i \geq T$, normalized by the total number of voxels N :

$$V_{x\%} = \frac{1}{N} \sum_{i=1}^N H(d_i - T), \quad (4.7)$$

¹<https://pytorch.org/>

where $H(\cdot)$ is the non-differentiable Heaviside step function. The same principle applies to $V_{x\text{ Gy}}$, differing only in that the threshold is specified in absolute dose.

To enable gradient-based training, we approximate H with a logistic sigmoid $\sigma_\alpha(t) = 1/(1 + e^{-\alpha t})$, with slope parameter $\alpha > 0$:

$$V_{x\%}^{\text{approx}} = \frac{1}{N} \sum_{i=1}^N \sigma_\alpha(d_i - T). \quad (4.8)$$

A larger α causes σ_α to more closely approximate the hard threshold, but could also increase the risk of gradient saturation: when α is too large, most voxels have $|\alpha(d_i - T)| \gg 1$, placing them in the flat tails of the sigmoid where $\sigma'_\alpha(t) \approx 0$. Consequently, only voxels very close to T contribute meaningful gradients, which may destabilize training or slow convergence. Therefore, rather than arbitrarily increasing α , we seek the smallest value that ensures sufficient approximation accuracy while mitigating the risk of gradient saturation.

4.3.1.3 Selection of α via a bound on the approximation error

Let $z_i = d_i - T$. We first quantify how closely the smooth surrogate $\sigma_\alpha(z)$ matches the hard indicator $H(z)$ at the voxel level. The mean pointwise approximation error is defined as:

$$\delta(\alpha) = \frac{1}{N} \sum_{i=1}^N |H(z_i) - \sigma_\alpha(z_i)|. \quad (4.9)$$

Since

$$V_{x\%} = \frac{1}{N} \sum_{i=1}^N H(z_i), \quad V_{x\%}^{\text{approx}}(\alpha) = \frac{1}{N} \sum_{i=1}^N \sigma_\alpha(z_i), \quad (4.10)$$

the triangle inequality implies that the $V_{x\%}$ approximation error is always bounded by

$$|V_{x\%} - V_{x\%}^{\text{approx}}(\alpha)| = \left| \frac{1}{N} \sum_{i=1}^N (H(z_i) - \sigma_\alpha(z_i)) \right| \leq \delta(\alpha). \quad (4.11)$$

We now derive an upper bound for $\delta(\alpha)$. Using the identity

$$|H(z) - \sigma_\alpha(z)| = \begin{cases} 1 - \sigma_\alpha(z) = \frac{1}{1 + e^{\alpha z}}, & z \geq 0, \\ \sigma_\alpha(z) = \frac{1}{1 + e^{-\alpha z}}, & z < 0, \end{cases} \quad (4.12)$$

we obtain the symmetric form

$$|H(z) - \sigma_\alpha(z)| = \frac{1}{1 + e^{\alpha|z|}} \leq e^{-\alpha|z|}. \quad (4.13)$$

A voxel-wise application yields

$$\delta(\alpha) \leq \frac{1}{N} \sum_{i=1}^N e^{-\alpha|z_i|}. \quad (4.14)$$

To obtain an interpretable and clinically meaningful bound, let q_m denote the fraction of voxels whose dose values lie within a margin m around the threshold:

$$q_m = \frac{1}{N} \#\{i : |z_i| \leq m\}. \quad (4.15)$$

For voxels inside this margin, the worst-case error is $1/2$ (since $\sigma_\alpha(0) = 1/2$). For voxels outside the margin, the error is bounded by $e^{-\alpha m}$. Thus both $\delta(\alpha)$ and the approximation error of $V_{x\%}$ in Eq. (4.11) satisfy

$$|V_{x\%} - V_{x\%}^{\text{approx}}(\alpha)| \leq \delta(\alpha) \leq q_m\left(\frac{1}{2}\right) + (1 - q_m)e^{-\alpha m}. \quad (4.16)$$

To ensure that the total approximation error remains below a user-specified tolerance ε (%), it suffices that

$$q_m\left(\frac{1}{2}\right) + (1 - q_m)e^{-\alpha m} \leq \varepsilon. \quad (4.17)$$

Solving for α gives the required lower bound

$$\alpha \geq \frac{1}{m} \ln\left(\frac{1 - q_m}{\varepsilon - \frac{1}{2}q_m}\right). \quad (4.18)$$

Eq. (4.18) provides a lower bound on the slope parameter, ensuring a balance between approximation accuracy and avoidance of gradient saturation.

In summary, for each V -metric we first choose a tolerance ε that serves as an upper bound on the error tolerated by the user. We then define a margin m around the dose threshold and compute q_m , the fraction of voxels whose dose values fall within this margin, from the dose distributions of the training set. Finally, we take the smallest α satisfying Eq. (4.18) and use it in the loss calculation.

4.3.1.4 Overall loss formulation

Using the differentiable formulation of D -metrics and the differentiable surrogate of V -metrics described above, we compute each metric in the template (Table 4.2) for both the predicted and ground truth dose in the same way. The loss is then defined as:

$$\mathcal{L}_{\text{CDM}} = \sum_{r \in \mathcal{R}} \sum_{k \in \mathcal{M}_r} w_{r,k} |M_{r,k}^{\text{pred}} - M_{r,k}^{\text{gt}}|, \quad (4.19)$$

where $M_{r,k}^{\text{pred}}$ and $M_{r,k}^{\text{gt}}$ denote the predicted and ground truth metric k for ROI r , and $w_{r,k}$ denotes a weight. To preserve voxel-wise accuracy of the predicted dose distribution, we also use the MAE loss:

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|\Omega|} \sum_{v \in \Omega} |D^{\text{pred}}(v) - D^{\text{gt}}(v)|. \quad (4.20)$$

The final objective combines both components:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{MAE}} + \lambda_2 \mathcal{L}_{\text{CDM}}, \quad (4.21)$$

where λ_1 and λ_2 balance voxel-wise consistency and alignment of clinical objectives.

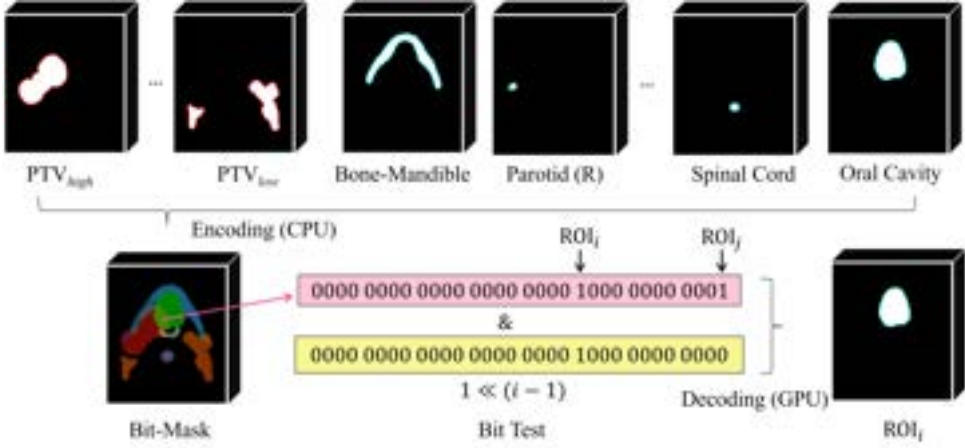


Figure 4.3: Multiple individual ROI masks are losslessly encoded into a single integer bit-mask, where each bit position represents a specific ROI (illustrated in pink). Because the raw bit-mask values span a wide numerical range, direct visualization is not intuitive. Therefore, the bit-mask is visualized in pseudo-color, with colors assigned solely based on their unique values for display purposes. During decoding, the binary mask for ROI_i is obtained by testing whether the $(i-1)$ -th bit of the bit-mask is active, implemented as a bitwise AND operation on GPU between the bit-mask and $(1 \ll (i-1))$, where the left-shifted single-bit mask $(1 \ll (i-1))$ is illustrated in yellow. A nonzero result indicates voxel membership in ROI_i .

4.3.2 Bit-mask encoding

The CDM loss in Chapter 4.3.1 requires access to all ROIs listed in Table 4.2. In H&N RT, the number of ROIs is typically large (often exceeding twenty), and many of them overlap spatially. In conventional dose prediction pipelines, each ROI is stored as an independent 3D binary mask channel using a one-hot encoding. When full 3D volumes are used as network input, this representation leads to substantial overhead in CPU preprocessing and CPU-to-GPU data transfer.

To alleviate these inefficiencies, we introduce a lossless bit-mask encoding as a compact intermediate representation for ROI handling, as illustrated in Fig. 4.3. Importantly, the bit-mask is not used as the direct input to the network. Instead, it serves as an efficient representation where standard binary ROI masks are decoded when needed, either to form the network input or to compute ROI-specific loss terms.

Formally, let $S_i(\mathbf{r}) \in \{0, 1\}$ denote voxel membership in ROI_i . Assigning one bit position to each ROI yields the lossless encoding

$$B(\mathbf{r}) = \sum_{i=1}^{N_{ROI}} S_i(\mathbf{r}) 2^{i-1}, \quad N_{ROI} \leq B_{max}, \quad (4.22)$$

where B_{max} denotes the bit width of the integer type. In this work, a 32-bit unsigned

integer (uint32) is used, supporting up to 32 ROIs. Overlapping anatomical structures are represented by activating multiple bits within the same voxel, preserving all ROI relationships in a single channel.

A key advantage of this representation arises during data preprocessing. With a conventional one-hot encoding, geometric augmentations such as rotations, flips, and affine transformations must be applied independently to each ROI channel, causing preprocessing cost to scale linearly with the number of ROIs. In contrast, when ROIs are stored in a single bit-mask volume, each augmentation operation is applied once to the bit-mask, implicitly transforming all ROIs simultaneously. This substantially reduces CPU preprocessing time while maintaining exact ROI geometry.

The bit-mask representation also reduces CPU-to-GPU data transfer. Instead of transferring the CT volume together with more than twenty one-hot ROI channels, only the CT volume and a single compact bit-mask are transferred per iteration. This reduces I/O overhead and improves training throughput, especially when using full-resolution 3D volumes.

Once on the GPU, the bit-mask B is decoded into binary ROI masks as required. When forming the network input, the decoded ROI masks $\hat{S} \in \mathbb{R}^{N_{\text{ROI}} \times D \times H \times W}$ are concatenated with the CT volume to obtain

$$I_{\text{in}} \in \mathbb{R}^{(1+N_{\text{ROI}}) \times D \times H \times W}. \quad (4.23)$$

Thus, the network always operates on standard one-hot ROI channels, identical to those used in conventional pipelines. The bit-mask itself serves only as an intermediate encoding and does not alter the inputs of the network.

The decoding is implemented via lightweight, element-wise bit-test, as shown in Fig. 4.3. In particular, the binary mask for ROI _{i} is obtained via

$$S_i(\mathbf{r}) = \text{bool}(B(\mathbf{r}) \& (1 \ll (i - 1))), \quad (4.24)$$

where $\text{bool}(\cdot)$ maps nonzero values to 1, “&” denotes the bitwise AND, and “ $\ll$ ” denotes the bit left-shift operation. These operations are highly efficient on modern GPUs and introduce negligible computational overhead.

During loss computation, only the bit-mask B remains resident in GPU memory, and the binary mask for each ROI _{i} is decoded on demand at the moment its corresponding metric $M_{r,k}^{\text{pred}}$ or $M_{r,k}^{\text{gt}}$ is evaluated. The decoded mask is used to gather voxel doses for that ROI and is then immediately released. This on-demand decoding strategy avoids storing all N_{ROI} ROI channels simultaneously and thereby reduces peak GPU memory usage.

Table 4.3: Comparison of model performance under different loss-function configurations. Lower scores indicate better performance.

Loss function	PTV Score (%) ↓	OAR Score (Gy) ↓	Dose Score (Gy) ↓
MAE	1.544 ± 1.188	2.103 ± 0.704	1.300 ± 0.229
MAE + DVH	0.992 ± 0.466	2.162 ± 0.518	1.385 ± 0.217
MAE + DVH + CDM	0.567 ± 0.300	1.933 ± 0.481	1.389 ± 0.228
MAE + CDM (proposed)	0.491 ± 0.250	1.999 ± 0.497	1.370 ± 0.245

4.4 Experiments and results

4.4.1 Experiment setup

The ROIs used in our experiments are listed in Table 4.2. The template contains two PTV structures (PTV_{54.25} and PTV₇₀) and 28 OARs, resulting in a total of $N_{\text{ROI}} = 30$ structures. Because this number fits within a 32-bit range, all ROI masks are encoded using a 32-bit unsigned integer (uint32) following the bit-mask representation described in Chapter 4.3.2.

For the CDM loss, all PTV-related $V_{95\%}$ metrics were included in the differentiable surrogate of V -metrics. To ensure controlled approximation accuracy, we adopt a margin of $m = 0.5$ Gy (i.e., a 1 Gy window) and a tolerance of $\varepsilon = 1\%$. Using the lower-bound condition in Eq. (4.18), we compute the minimum α values that satisfy the prescribed tolerance. Thus, we set $\alpha = 209$ for PTV_{54.25} and $\alpha = 176$ for PTV₇₀. To reflect the higher clinical priority of PTVs, all PTV-related metric weights are set to $w(r, k) = 1$, whereas all OAR-related metric weights are set to $w(r, k) = 0.1$. The balancing coefficients are set to $\lambda_1 = 1$ and $\lambda_2 = 0.5$ for the final loss $\mathcal{L}_{\text{total}}$.

Unless otherwise stated, all experiments were conducted using a 3D U-Net baseline implemented in the MONAI² framework. The architecture follows the standard encoder-decoder design with skip connections [106]. Compared with the conventional 3D U-Net, the major architectural modification is the use of 3D Instance Normalization instead of Batch Normalization, which has been shown to provide more stable training for small-batch 3D medical imaging DL tasks [107].

All models were trained for 1000 epochs using the AdamW optimizer with an initial learning rate of 3×10^{-4} and a cosine annealing schedule. Mixed-precision training (bf16-mixed) was used throughout to reduce memory consumption and improve computational efficiency. All experiments were performed on an NVIDIA RTX 6000 Ada GPU.

4.4.2 Results

4.4.2.1 Loss function comparison

We first compared different loss-function configurations using the 3D U-Net baseline to evaluate the impact of the proposed CDM loss. Four loss-function configurations were investigated: MAE, MAE + DVH, MAE + DVH + CDM, and MAE + CDM (our proposed setting). Here, the DVH loss refers to the DVH value loss introduced in [26], which provides a more efficient way to penalize DVH-curve discrepancies compared with the formulation in [95]. In all experiments, the MAE loss was weighted as 1, while both the DVH and CDM components were assigned weights of 0.5. Table 4.3 reports the model performance under different loss-function configurations. The results show that introducing the CDM loss led to a substantial improvement in the PTV Score, achieving the lowest value when combined with MAE. In contrast, both MAE and MAE + DVH alone failed to effectively reduce the PTV Score. The CDM loss also provided moderate benefits for the OAR Score, suggesting its positive influence on balancing target coverage and organ-at-risk sparing. Although the MAE-only configuration yielded the lowest Dose Score, it also had a very high PTV Score.

We further examined whether the predicted dose distributions satisfied the predefined planning constraints. As illustrated in Fig. 4.4, the MAE + CDM configuration was the only setting that met both the $V_{95\%}$ ($\geq 98\%$) and $D_{0.03cc}$ ($\leq 110\%$) constraints across all test cases. No statistically significant differences from the clinical plans were observed for these metrics. In contrast, the other loss configurations showed instances of insufficient PTV coverage or excessive dose, as shown in Fig. 4.5. For OARs, all loss configurations satisfied the relevant clinical limits, but only the MAE + CDM model showed no statistically significant differences from the clinical plans for all metrics.

4.4.2.2 Ablation study

To assess the influence of α used in the differentiable V -metric approximation, we performed an ablation study on the parameter α (Table 4.4). Smaller values (e.g., 100) resulted in higher PTV Scores. The theoretically derived values (209, 176) for $PTV_{54.25}$ and PTV_{70} achieved the best PTV Score while maintaining stable OAR and Dose Scores. Increasing α beyond this range (300–500) did not yield further improvement and in some cases slightly degraded PTV performance.

To further assess the computational impact of the proposed bit-mask ROI encoding, an ablation study was performed using the same 3D U-Net baseline and identical training settings. The comparison is reported in Table 4.5. When using conventional one-hot ROI channels, training required a peak GPU memory of 20.50 GB and an average epoch duration of 241 s. With bit-mask encoding, the peak memory was

²<https://github.com/Project-MONAI/MONAI>

Table 4.4: Ablation study on the parameter α . Lower scores indicate better performance.

α (PTV _{54.25})	α (PTV ₇₀)	PTV Score (%) ↓	OAR Score (Gy) ↓	Dose Score (Gy) ↓
100	100	0.564 ± 0.287	1.919 ± 0.502	1.349 ± 0.228
209	176	0.491 ± 0.250	1.999 ± 0.497	1.370 ± 0.245
300	300	0.500 ± 0.276	1.961 ± 0.491	1.346 ± 0.226
400	400	0.563 ± 0.298	1.988 ± 0.475	1.352 ± 0.214
500	500	0.563 ± 0.308	1.993 ± 0.452	1.327 ± 0.204

Table 4.5: Ablation experiment on the 3D U-Net baseline comparing computational efficiency with and without the bit-mask ROI encoding.

Method (3D U-Net baseline)	Peak GPU Memory (GB)	Epoch Duration (s)
Without bit-mask	20.50	241
With bit-mask (proposed)	19.57	43

Table 4.6: Comparison of dose prediction performance across different network architectures using the MAE + CDM loss configuration. Lower scores indicate better performance.

Model	PTV Score (%) ↓	OAR Score (Gy) ↓	Dose Score (Gy) ↓
SwinUNETR [45]	0.699 ± 0.292	2.035 ± 0.566	1.461 ± 0.253
C3D [91]	0.497 ± 0.314	1.945 ± 0.473	1.289 ± 0.192
MedNeXt [108]	0.496 ± 0.251	2.154 ± 0.571	1.395 ± 0.204
Pyfer [24]	0.491 ± 0.287	2.019 ± 0.517	1.290 ± 0.207
3D U-Net (baseline) [106]	0.491 ± 0.250	1.999 ± 0.497	1.370 ± 0.245

19.57 GB (4.5% reduction) and the epoch duration was 43 s (82.2% reduction). Peak GPU memory is reported because memory usage during training is dynamic, and the maximum allocation determines whether the model fits within available GPU memory.

4.4.2.3 Model comparison

Building upon the best-performing loss setting (MAE + CDM), finally we compared the 3D U-Net baseline with several state-of-the-art architectures (Table 4.6), including transformer-based models (SwinUNETR [45]), large-kernel ConvNeXt-style convolutional networks (MedNeXt [108]), and cascade-style frameworks (C3D [91], Pyfer [24]). Notably, the 3D U-Net achieved a PTV Score of 0.491, matching the best-performing model (Pyfer) and outperforming more complex architectures such as SwinUNETR and MedNeXt. The OAR Score of 1.999 was also competitive, ranking second among all models.

4.5 Discussion

Overall, our results demonstrate that directly optimizing clinically used dose metrics is substantially more effective than voxel-wise or DVH-curve-based supervision. As shown in Table 4.3, MAE alone achieved the lowest voxel-wise Dose Score, but failed to ensure adequate target coverage. Previously proposed DVH-based losses improved DVH curve similarity [95, 104]. However, they did not reliably enforce clinically mandatory constraints, such as $V_{95\%}$ for PTVs. This limitation is evident in Fig. 4.4 and Fig. 4.5. Both MAE and MAE + DVH exhibit cases of insufficient target coverage or excessive hot spots. In contrast, the proposed CDM loss directly optimizes the dose metrics used in clinical plan evaluation. This task-aligned supervision consistently satisfied all PTV constraints in all patients, without compromising OAR sparing. These results indicate that the proposed CDM loss improves clinical relevance and outperforms loss formulations based on voxel-wise or DVH-curve supervision.

A key component of the proposed framework is the differentiable surrogate for V -metrics, which uses a sigmoid slope parameter α . The analytically derived lower bound provides a principled and reproducible way to select α without requiring exhaustive tuning. The ablation study (Table 4.4) shows that a small α leads to overly smooth threshold approximations. As a result, the model struggled to distinguish doses above and below the clinical prescription dose level. In contrast, very large α values created overly sharp transitions. This limits the number of voxels that contribute meaningful gradients and makes it harder for the model to correct small deviations near the prescription threshold. Between these extremes, the analytically derived values (209, 176) in this study provided stable performance. Additionally, the results also show that slightly larger values such as $\alpha = 300$ achieve similar PTV, OAR, and Dose Scores. Accordingly, the chosen values should be interpreted as a conservative and principled choice, rather than an attempt to identify an optimal setting.

An important advantage of the proposed CDM loss is its flexible, template-driven design. All optimized metrics, ROIs, and their relative priorities are specified through a simple JSON-based clinical evaluation template. This allows the loss to be configured for different planning protocols without modifying the network architecture or training pipeline. Although this study focuses on a single-institution H&N dataset, the template-based formulation in principle can be adapted to other datasets by updating the clinical criteria template. Compared with fixed, hard-coded loss formulations, this approach provides a practical and transparent way to align training objectives with clinically used evaluation standards.

Beyond loss formulation, this work addresses a practical but often overlooked bottleneck in H&N dose prediction: the computational overhead caused by a large number of ROIs. The proposed bit-mask encoding strategy replaces more than twenty

one-hot ROI channels with a single integer-valued mask and decodes ROI membership on the GPU. This substantially reduces training time, with the average epoch duration decreasing from 241 s to 43 s (Table 4.5), accelerating training by more than 5 $\times$. In contrast to conventional pipelines [22, 91, 24, 28] that often restrict supervision to a subset of structures due to computational constraints, the proposed encoding enables more comprehensive and clinically faithful supervision. This supports more informative learning of spatial dose relationships across PTVs and OARs.

An additional observation from the model comparison experiments is that architectural complexity becomes a secondary factor when the training objective is well aligned with clinical goals. Under the MAE + CDM loss, a standard 3D U-Net achieved performance comparable to, or better than, transformer-based [45] and cascade-style architectures [91, 24] (Table 4.6). This suggests that, for 3D dose prediction, improving *what the model is optimized for* may be more impactful than increasing architectural complexity. From a clinical deployment perspective, this is an encouraging result, as simpler architectures are typically easier to train, validate, and maintain.

Several limitations of this study should be acknowledged. First, all data were obtained from a single institution with a unified planning protocol. The generalizability of the approach to other institutions therefore remains to be validated. Second, while the predicted dose distributions satisfied all clinical constraints, this study did not evaluate downstream integration into deliverable treatment planning [85, 109]. Future work will focus on multi-institutional validation and on assessing the impact of CDM-guided dose prediction on dose mimicking and final plan quality.

In summary, this work presents a flexible and clinically aligned framework for 3D head and neck dose prediction that combines a clinical DVH metric loss with an efficient, lossless ROI encoding strategy. By directly optimizing clinically relevant evaluation metrics, the proposed approach satisfied all predefined clinical constraints and improves the clinical acceptability of the predicted dose distributions. As such, the framework provides a practical foundation for AI-assisted radiotherapy workflows and may facilitate automated planning by improving plan consistency and reducing manual effort.

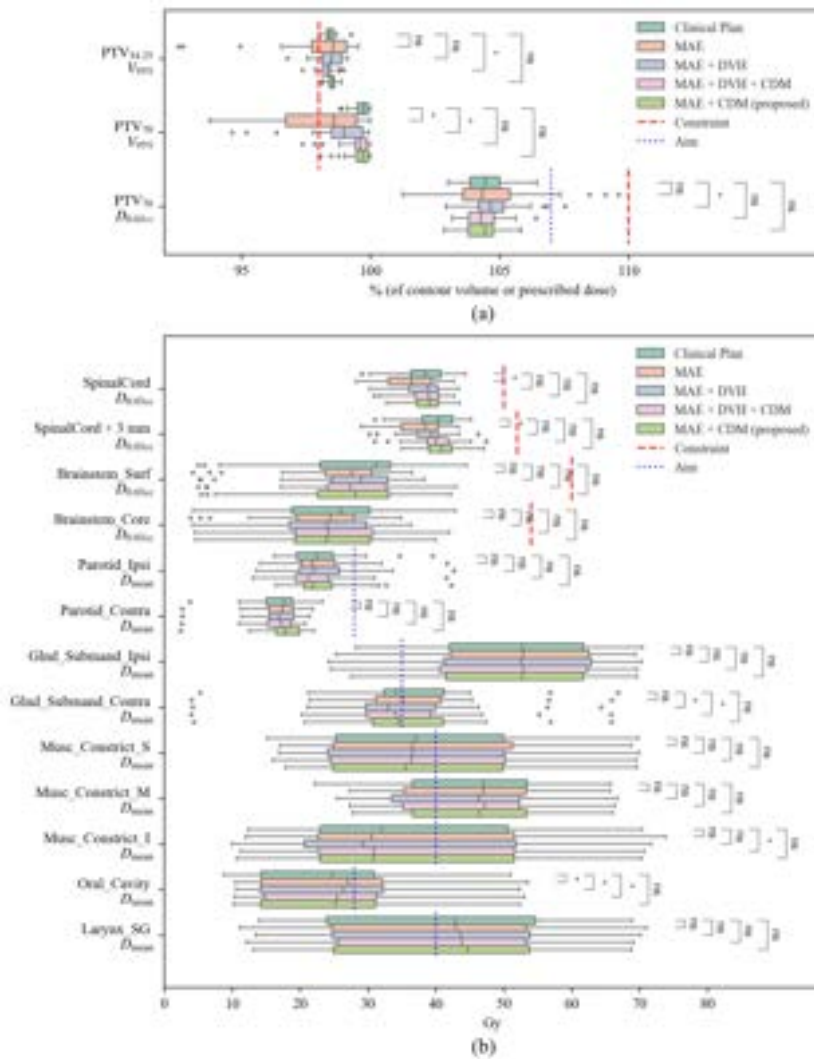


Figure 4.4: Boxplots of dose evaluation metrics for the clinically most relevant ROIs for different loss function configurations. (a) Boxplots for PTVs. (b) Boxplots for OARs. The corresponding planning aims and dose constraints are also shown (aims are clinically regularly violated due to proximity of the tumor). For $V_{95\%}$, the constraint is defined as $\geq$ the corresponding threshold, whereas for D_{mean} and $D_{0.03\text{cc}}$, both the constraint and the aim are defined as $\leq$ the corresponding threshold. Parotid and submandibular glands were categorized as ipsilateral (Ipsi) or contralateral (Contra) according to received dose. Statistical significance was tested using a two-tailed Wilcoxon signed-rank test. *: $p \leq 0.05$, ns = not significant.

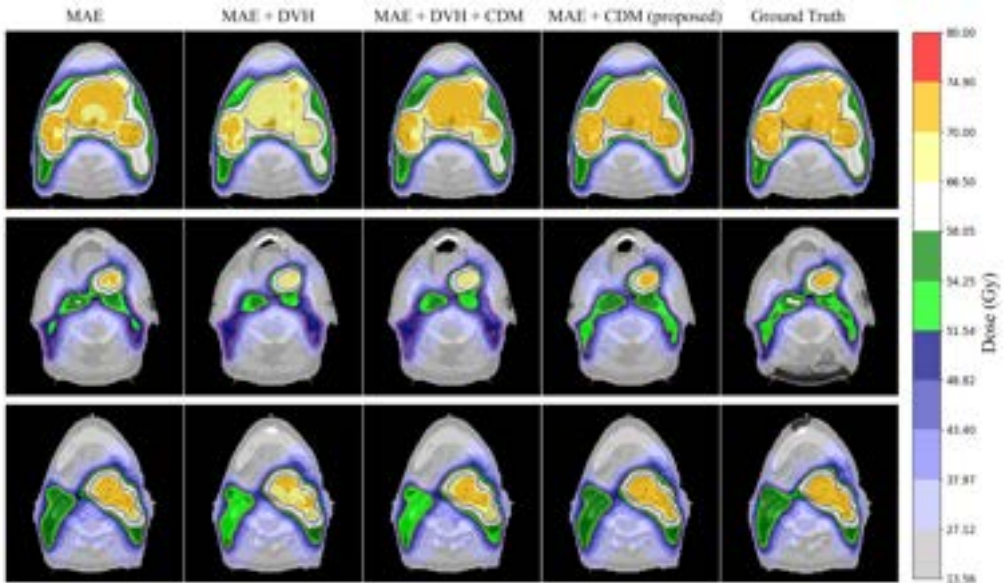


Figure 4.5: Axial visualization of dose distributions under different loss function configurations. The red contour denotes the PTV_{54.25}, while the blue contour represents the PTV₇₀. A colormap routinely used in clinical practice is employed to enhance visual interpretability.

5

Automated Head and Neck Radiotherapy Planning from Deep Learning-Based Dose Prediction via Dose Mimicking

This chapter was adapted from:

Ruochen Gao, Noud van Ruremonde, Alice Onderwater, Martin de Jong, Marius Staring, and Frank Dankers, "Automated Head and Neck Radiotherapy Planning from Deep Learning-Based Dose Prediction via Dose Mimicking", submitted, 2026.

Abstract

Background: Deep learning-based dose prediction is promising for automated radiotherapy planning, but predicted dose distributions are not directly deliverable because they do not specify machine delivery parameters. Dose mimicking enables the conversion of predicted dose distributions into clinically deliverable plans.

Purpose: To implement a fully automated planning pipeline that integrates deep learning-based dose prediction with dose mimicking for head and neck cancer (HNC) radiotherapy.

Methods: A retrospective cohort of 174 patients with HNC treated with dual-arc VMAT was included. A hierarchical densely connected U-Net was trained on 137 patients to predict three-dimensional dose distributions from planning CT images, target volumes, and organs-at-risk (OARs), using a hybrid loss that combined mean absolute error and clinical dose metric loss. A separate cohort of 7 patients was used exclusively for dose mimicking parameter fine-tuning. For the final temporally independent test cohort of 30 patients, predicted dose distributions were imported into RayStation v2025 and converted into deliverable VMAT plans using a fully scripted dose mimicking workflow. Automated plans were compared with clinical plans and predicted dose distributions using dose–volume histogram (DVH) metrics, normal tissue complication probability (NTCP), and plan complexity metrics.

Results: Automated plans achieved comparable target coverage, with $V95\% \geq 98\%$ achieved for PTV_{70} and $PTV_{54.25}$ in 96.7% and 86.7% of plans, respectively. For OARs, no significant differences were observed for most evaluated organs. No significant differences in NTCP estimates for xerostomia or dysphagia were observed between the clinical and automated plans. Automated plans showed lower MU/cGy, small aperture score, and circumference-to-area ratio, but higher MLC leaf travel.

Conclusions: The proposed pipeline enabled fully automated generation of clinically deliverable HNC treatment plans, with comparable DVH and NTCP performance to clinical plans and lower plan complexity in several modulation-related metrics.

5.1 Introduction

Radiotherapy treatment planning is a complex and time-consuming process that requires iterative optimization to balance target coverage and sparing of surrounding organs-at-risk (OARs). Plan quality depends strongly on planner experience, leading to inter-planner variability [20]. Automated planning strategies are therefore of considerable clinical interest because they may improve efficiency, consistency, and robustness.

Deep learning-based dose prediction has emerged as a promising approach toward this goal. By learning the relationship between patient anatomy, target volumes, OARs, and delivered dose distributions, deep learning algorithms can generate patient-specific three-dimensional (3D) dose predictions. Prior studies have shown good agreement with clinical plans in terms of dose-volume histogram (DVH) metrics across multiple disease sites, supporting their use within automated planning pipelines [23, 22, 91, 27, 28].

However, a key limitation is that a predicted dose distribution is inherently not a deliverable treatment plan. A treatment plan is defined by machine delivery parameters, including beam configurations, gantry and couch angles, multi-leaf collimator (MLC) leaf positions, and monitor units (MUs). In the treatment planning system (TPS), the dose distribution is computed using a dose calculation algorithm based on these parameters [110]. In contrast, deep learning models directly predict dose without specifying the underlying delivery parameters. Consequently, the predicted dose alone cannot be used for treatment delivery. Converting a predicted dose into a deliverable plan can be formulated as an inverse problem: determining a set of physically achievable machine parameters whose computed dose reproduces the predicted distribution [85].

Several approaches have been proposed to bridge the gap between predicted dose distributions and deliverable treatment plans, primarily based on either traditional inverse optimization [30, 111, 112] or dose mimicking [113]. Among these, the combination of dose prediction and dose mimicking is particularly attractive. Dose prediction provides a patient-specific estimate of the desired dose distribution before TPS optimization, thereby reducing reliance on iterative trial-and-error tuning of planning objectives. Dose mimicking can then be used to reproduce the predicted dose with physically deliverable machine parameters, potentially offering a faster alternative to conventional inverse optimization workflows.

Existing TPS-based studies have mainly focused on treatment sites with relatively less complex planning scenarios, such as prostate, lung, and breast cancers [113, 30, 111, 112]. However, head and neck radiotherapy represents one of the most challenging treatment sites for planning due to the numerous OARs in close proximity

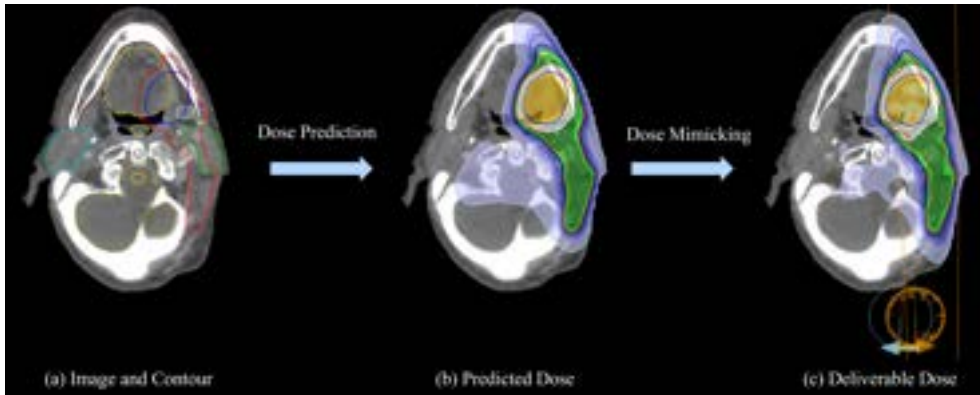


Figure 5.1: Overview of the proposed automated radiotherapy planning pipeline.

to targets, steep dose gradients, multiple prescription dose levels, and substantial anatomical variability [80]. In addition, head and neck cancer (HNC) includes multiple primary tumor subsites, such as the oropharynx, hypopharynx, larynx, and oral cavity, each with distinct geometric and dosimetric characteristics. This heterogeneity poses challenges for developing robust and generalizable automated planning approaches. Despite the clinical importance of commercial TPS integration, few studies have implemented fully automated planning workflows within a commercial TPS for HNC.

The agreement between automated and clinical dose distributions is typically evaluated using DVH metrics [30, 111]. However, other clinically relevant aspects, such as normal tissue complication probability (NTCP) and plan complexity, have received less attention. NTCP models provide an estimate of potential treatment-related toxicity, while plan complexity is closely related to deliverability and treatment efficiency. Evaluating these factors is important for assessing the clinical suitability of automated planning approaches beyond dose accuracy alone.

In this work, we present an automated planning pipeline that integrates deep learning-based dose prediction with dose mimicking in a commercial TPS (RayStation, RaySearch Laboratories, Stockholm, Sweden) to generate deliverable photon treatment plans for HNC. The overall workflow is illustrated in Figure 5.1. The model was trained on an institutional dataset using a hybrid loss function that combines voxel-wise and DVH-based components. The proposed workflow establishes a fully automated, closed-loop process from dose prediction to a deliverable treatment plan and was evaluated on a temporally independent test cohort covering multiple tumor subsites. In addition to dose-based evaluation, we assessed NTCP and plan complexity to ensure a comprehensive, clinically relevant evaluation.

5.2 Materials and methods

5.2.1 Patient cohort

This retrospective study included 174 patients with HNC who were treated at Leiden University Medical Center (LUMC) between February 2021 and July 2025. The study was approved by the Medical Ethics Committee of Leiden–The Hague–Delft (approval number G21.142, October 15, 2021). The cohort comprised tumors originating from the oropharynx, hypopharynx, nasopharynx, larynx, and oral cavity, and included both unilateral and bilateral treatment cases. Patients undergoing reirradiation were excluded.

All patients were treated with external beam photon radiotherapy on Elekta Synergy linear accelerators (Elekta, Stockholm, Sweden) using dual-arc volumetric modulated arc therapy (VMAT) with 6 MV photons. Clinical treatment plans were generated in RayStation following institutional protocols.

To mimic the intended clinical deployment scenario, the 174 patients were split chronologically according to their roles in the automated planning pipeline. For the dose prediction model, 137 patients (February 2021–June 2024) were used for training, and 37 patients (July 2024–July 2025) were used for testing. The 37 dose prediction test patients were further divided for the dose mimicking stage: 7 patients (July 2024–August 2024) were used exclusively for parameter fine-tuning, and the remaining 30 patients (September 2024–July 2025) constituted the temporally independent final test cohort. The fixed dose mimicking parameters were applied to the final test cohort without further adjustment.

5.2.2 Deep learning dose prediction

A deep learning model was trained to predict patient-specific 3D dose distributions from planning CT images and structure information. A hierarchical densely connected U-Net (HD U-Net) was used for voxel-wise dose prediction [23]. Model inputs consisted of the planning CT, binary masks of planning target volumes (PTVs) corresponding to two prescription dose levels (70 Gy and 54.25 Gy), and masks of 28 OARs. The model output was a predicted 3D dose distribution.

The model was trained on 137 patients using a hybrid loss function that combines mean absolute error (MAE) with a novel clinical dose metric (CDM) loss. The MAE term enforces voxel-wise agreement between predicted and reference doses, while the CDM loss incorporates clinically relevant DVH metrics into the training process. The CDM loss, proposed in Chapter 4.3.1, uses differentiable approximations of DVH metrics, particularly V-metrics, to enable gradient-based optimization. This formulation allows the model to directly optimize clinically relevant endpoints, thereby improving both spatial dose accuracy and DVH consistency compared with state-of-

the-art methods. The trained model was then applied to the test cohort to generate predicted dose distributions.

5.2.3 Dose mimicking and plan generation

To generate a clinically deliverable VMAT treatment plan, the predicted dose distribution was imported into the TPS and used as the reference dose for dose mimicking. Dose mimicking was performed using RayStation’s built-in optimization algorithm in a research build of RayStation v2025 SP1 and consisted of two phases. In the first phase, the optimization focused on reproducing the reference dose distribution, with particular emphasis on OAR sparing objectives. In the second phase, the optimization was refined to improve target coverage while maintaining similarity to the reference dose.

For each patient, a treatment plan was initialized using a predefined beam setup template and the anatomical structures required for optimization. During mimicking, MLC leaf positions and segment weights (i.e., monitor units assigned to each control point) were iteratively adjusted to reproduce the reference dose distribution. The gantry angle sampling and arc geometry were predefined and kept fixed (full dual-arc at 2° gantry spacing). Delivery parameters such as dose rate and gantry speed were not explicitly optimized; instead, they were implicitly determined by the system to satisfy machine delivery constraints.

The mimicking optimization objectives and their corresponding weights were initially determined through manual fine-tuning on the validation cohort. Once established, the same parameter set was fixed and applied to all test patients without further adjustment. The described workflow, including predicted dose import, plan initialization, and dose mimicking optimization, was fully automated using script-based execution in the TPS and did not require user interaction.

5.2.4 Evaluation metrics

To evaluate the performance of the proposed approach, the automated plans were compared with the corresponding clinical plans using several categories of evaluation metrics.

5.2.4.1 DVH metrics

DVH metrics were used to assess plan quality in terms of target coverage and OAR sparing. $V_{95\%}$, D_{mean} , and $D_{0.03\text{cc}}$ were used to assess target coverage, dose homogeneity, and hot spots within the target volumes. For evaluating OARs, D_{mean} , $D_{0.03\text{cc}}$ and $D_{2\%}$ were used according to clinical relevance. The institutional dose evaluation criteria applied in this study are summarized in Table 5.1.

Table 5.1: Institutional DVH-based dose evaluation criteria for head and neck radiotherapy planning. ROIs with “L/R” denote paired organs and were evaluated separately for the left and right sides. Each ROI is evaluated using one or more dose metrics. “Aim” represents the desired planning goal, while “Constraint” denotes a mandatory criterion.

ROI	Metric	Aim	Constraint
PTV ₇₀	$V_{95\%}$		$\geq 98\%$
PTV ₇₀	D_{mean}	$\leq 102\%$	
PTV ₇₀	$D_{0.03\text{cc}}$	$\leq 107\%$	$\leq 110\%$
PTV _{54.25}	$V_{95\%}$		$\geq 98\%$
PTV _{54.25}	D_{mean}	$\leq 102\%$	
SpinalCord	$D_{0.03\text{cc}}$		$\leq 50\text{ Gy}$
SpinalCord + 3 mm	$D_{0.03\text{cc}}$		$\leq 52\text{ Gy}$
Brainstem_Surf	$D_{0.03\text{cc}}$		$\leq 60\text{ Gy}$
Brainstem_Core	$D_{0.03\text{cc}}$		$\leq 54\text{ Gy}$
Brain	$D_{0.03\text{cc}}$	$\leq 65\text{ Gy}$	
Brain	$D_{2\%}$	$\leq 70\text{ Gy}$	
Cochlea_(L/R)	D_{mean}	$\leq 45\text{ Gy}$	
Parotid_(L/R)	D_{mean}	$\leq 28\text{ Gy}$	
GlnD_Submand_(L/R)	D_{mean}	$\leq 35\text{ Gy}$	
Oral_Cavity	D_{mean}	$\leq 28\text{ Gy}$	
Musc_Constrict_S	D_{mean}	$\leq 40\text{ Gy}$	
Musc_Constrict_M	D_{mean}	$\leq 40\text{ Gy}$	
Musc_Constrict_I	D_{mean}	$\leq 40\text{ Gy}$	
Cricopharyngeus	D_{mean}	$\leq 40\text{ Gy}$	
Larynx_SG	D_{mean}	$\leq 40\text{ Gy}$	
Glottic_Area	D_{mean}	$\leq 40\text{ Gy}$	
Bone_Mandible	$D_{2\%}$	$\leq 70\text{ Gy}$	
Bone_Mandible-PTV	$D_{2\%}$	$\leq 50\text{ Gy}$	
Eye_(L/R)	$D_{0.03\text{cc}}$	$\leq 35\text{ Gy}$	
Lens_(L/R)	$D_{0.03\text{cc}}$	$\leq 6\text{ Gy}$	
Pituitary	D_{mean}	$\leq 20\text{ Gy}$	
OpticNrv_(L/R)	$D_{0.03\text{cc}}$	$\leq 55\text{ Gy}$	

5.2.4.2 NTCP metrics

NTCP models were used to assess the clinical impact of each plan by quantifying the predicted risk of toxicity. They estimate the probability of radiation-induced toxicity based on OAR dose distributions and patient-specific clinical factors. For HNC, the evaluated endpoints included grade ≥ 2 and grade ≥ 3 xerostomia, as well as grade ≥ 2 and grade ≥ 3 dysphagia, both at six months after treatment. NTCP values for

xerostomia and dysphagia were calculated using published NTCP models [114, 115]:

$$\text{NTCP} = \frac{1}{1 + e^{-S}}. \quad (5.1)$$

For xerostomia, the linear predictor S incorporates the mean doses to salivary glands, as well as baseline xerostomia:

$$S = \beta_0 + \beta_1 \left(\sqrt{D_{\text{mean}}^{\text{Parotid-L}}} + \sqrt{D_{\text{mean}}^{\text{Parotid-R}}} \right) + \beta_2 D_{\text{mean}}^{\text{Glands-Submand}} + \beta_{\text{baseline}}, \quad (5.2)$$

where Glands_Submand is the joint contour of left and right submandibular glands. All regression coefficients depend on the outcome grade (≥ 2 or ≥ 3) and treatment setting (primary or postoperative). In addition, β_{baseline} is specified according to baseline xerostomia severity.

For dysphagia, the linear predictor S incorporates the mean doses to swallowing-related structures, including the oral cavity and pharyngeal constrictor muscles, as well as baseline dysphagia and tumor location:

$$S = \beta_0 + \beta_1 D_{\text{mean}}^{\text{Oral-Cavity}} + \beta_2 D_{\text{mean}}^{\text{Musc-Constrict-S}} + \beta_3 D_{\text{mean}}^{\text{Musc-Constrict-M}} + \beta_4 D_{\text{mean}}^{\text{Musc-Constrict-I}} + \beta_{\text{baseline}} + \beta_{\text{tumor}}, \quad (5.3)$$

where all regression coefficients depend on the outcome grade (≥ 2 or ≥ 3) and treatment setting (primary or postoperative). In addition, β_{baseline} is specified according to baseline dysphagia grade, and β_{tumor} according to tumor location (oral cavity, pharynx, or larynx).

5.2.4.3 Plan complexity metrics

Plan complexity was evaluated to assess the deliverability and efficiency of the automatically generated treatment plans. Complexity was quantified using monitor units normalized by prescription dose (MU/cGy), small aperture score (SAS), circumference-to-area ratio (CoA), and MLC leaf travel. These metrics were selected to characterize complementary components of plan complexity, including global modulation intensity, aperture size, aperture shape complexity, and mechanical delivery demands, while maintaining interpretability and clinical relevance.

MU/cGy was defined as the total number of monitor units divided by the prescription dose (cGy), reflecting the overall modulation level of the plan:

$$\text{MU/cGy} = \frac{\text{Total MU}}{\text{Prescription dose (cGy)}}. \quad (5.4)$$

Higher MU/cGy values indicate increased modulation and thus greater plan complexity.

The small aperture score (SAS) quantifies the MU-weighted proportion of leaf pairs with openings smaller than a predefined threshold d_{th} across all control points:

$$SAS = \frac{\sum_{i=1}^{N_{CP}} w_i \sum_{j=1}^{N_{leaf}} \mathbb{I}(d_{i,j} < d_{th})}{N_{leaf} \sum_{i=1}^{N_{CP}} w_i}, \quad (5.5)$$

where $d_{i,j}$ is the distance between opposing MLC leaves for leaf pair j at control point i , w_i is the MU delivered at control point i , and $\mathbb{I}(\cdot)$ is the indicator function. In this study, d_{th} was set to 10 mm, and leaf pairs were considered open only when $d_{i,j} > 6$ mm. Higher SAS values indicate increased delivery complexity due to a greater proportion of small apertures.

The circumference-to-area ratio (CoA) was used to quantify the geometric complexity of MLC apertures. For each control point, CoA was defined as the ratio of the aperture perimeter to its area, $CoA_i = P_i / A_i$, where P_i and A_i denote the perimeter and area of the aperture at control point i , respectively. The overall CoA for each plan was calculated as a MU-weighted average across all control points:

$$CoA = \frac{\sum_{i=1}^{N_{CP}} w_i \cdot CoA_i}{\sum_{i=1}^{N_{CP}} w_i}. \quad (5.6)$$

CoA has units of mm^{-1} , and higher values indicate increased aperture shape complexity due to a larger perimeter relative to the irradiated area.

MLC leaf travel quantifies the cumulative displacement of all MLC leaves between consecutive control points:

$$\text{Leaf travel} = \sum_{i=1}^{N_{CP}-1} \sum_{j=1}^{N_{leaf}} |x_{i+1,j} - x_{i,j}|, \quad (5.7)$$

where $x_{i,j}$ denotes the position of leaf j at control point i . Higher leaf travel indicates greater mechanical complexity during treatment delivery.

For all metrics, higher values correspond to increased modulation and delivery complexity, whereas lower values indicate more efficient and potentially more robust treatment plans.

5.2.5 Statistical analysis

All statistical analyses in this study were performed on the final independent test cohort. A two-tailed Wilcoxon signed-rank test was used for paired comparisons of evaluation metrics between clinical plans, predicted dose distributions, and automated plans, as applicable. A p -value of less than 0.05 was considered statistically significant.

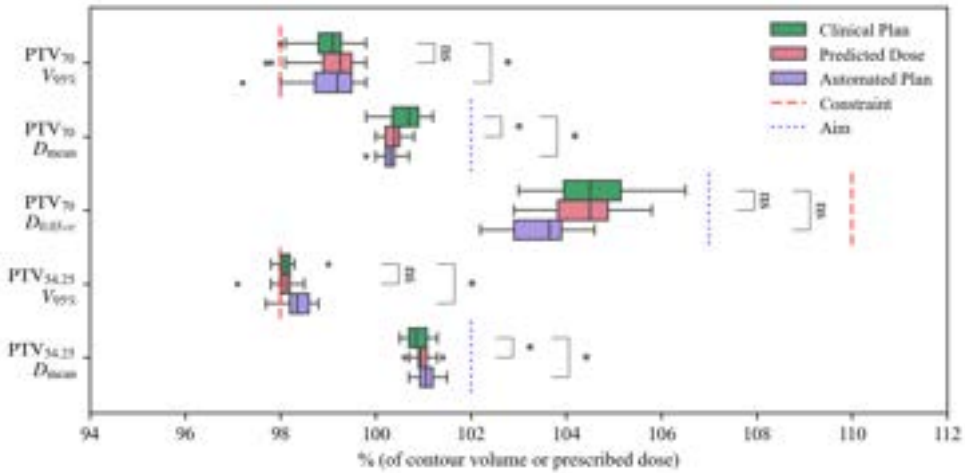


Figure 5.2: Comparison of PTV DVH metrics between clinical plans, predicted dose, and automated plans. For $V_{95\%}$, the constraint is defined as $\geq$ the corresponding threshold, whereas for D_{mean} and $D_{0.03\text{cc}}$, both the constraint and the aim are defined as $\leq$ the corresponding threshold. The detailed threshold values are provided in Table 5.1. *: $p < 0.05$, ns = not significant.

5.3 Results

5.3.1 DVH metrics comparison

As shown in Figure 5.2, the automated plans achieved target coverage comparable to clinical plans. For PTV_{70} , all automated plans met the clinical constraint of $\geq 98\%$, except for one outlier with a $V_{95\%}$ of 97.2%, which corresponded to a patient with an unusually large tumor volume. For $\text{PTV}_{54.25}$, the automated plans also showed a slightly higher $V_{95\%}$, with 26 of 30 plans (86.7%) satisfying the clinical constraint of $V_{95\%} \geq 98\%$ and a lowest observed value of 97.7%. The automated plans showed a lower $\text{PTV}_{70} D_{\text{mean}}$ and a slightly higher $\text{PTV}_{54.25} D_{\text{mean}}$, while no significant difference was observed for $\text{PTV}_{70} D_{0.03\text{cc}}$. Representative dose distributions are presented in Figure 5.3.

For OAR DVH metrics (Figure 5.4), no statistically significant differences were observed for most organs, such as the spinal cord, brainstem, and parotid glands. Statistically significant differences were observed for the ipsilateral submandibular gland, oral cavity, and inferior constrictor muscle; however, the absolute dose differences appeared limited.

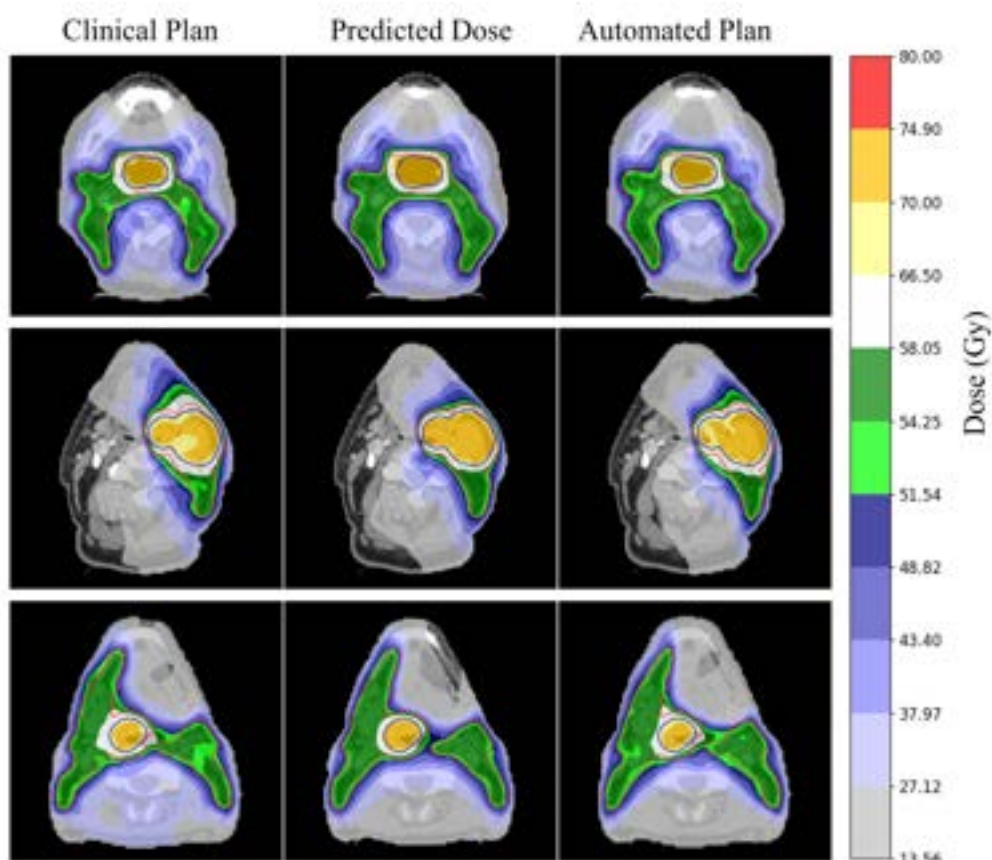


Figure 5.3: Axial visualization of dose distributions of clinical plans, predicted dose, and automated plans. The red contour denotes the $PTV_{54.25}$, while the blue contour represents the PTV_{70} . A colormap routinely used in clinical practice is employed to enhance visual interpretability.

5.3.2 NTCP comparison

No statistically significant differences were observed among clinical plans, predicted dose distributions, and automated plans across the evaluated NTCP endpoints, which included grade ≥ 2 and grade ≥ 3 xerostomia and dysphagia (Figure 5.5). For all endpoints, the distributions were largely overlapping among the three plan types. This suggests that dose mimicking preserved the predicted toxicity profile while generating deliverable treatment plans.

5.3.3 Plan complexity comparison

The automated plans exhibited significantly reduced modulation complexity compared to clinical plans (Figure 5.6), as reflected by lower MU/cGy, SAS, and CoA. Conversely,

MLC leaf travel was significantly increased, indicating greater mechanical movement during delivery. Overall, these results suggest a reduction in modulation complexity with a trade-off in increased leaf motion.

5.4 Discussion and conclusion

This work presents an automated radiotherapy planning pipeline that integrates clinically guided deep learning-based dose prediction with TPS-based dose mimicking for HNC. By converting predicted dose distributions into clinically deliverable plans, the proposed approach addresses a key limitation of dose prediction methods: predicted dose distributions do not define machine delivery parameters and therefore cannot be used directly for treatment delivery.

A key feature of the workflow is its script-based execution within the TPS, enabling end-to-end plan generation from predicted dose to a deliverable treatment plan without user interaction. Once the required inputs are available, predicted dose import, plan initialization, and dose mimicking can be performed automatically using a fixed procedure. This design may improve the reproducibility and consistency of automated plan generation in clinical practice, which is particularly relevant given the known inter-planner variability in radiotherapy treatment planning [20].

An important finding of this study is that a single set of dose mimicking objectives and weights could be applied across a heterogeneous cohort of HNC patients, with differences in primary tumor location, elective and boosted lymph node targets, and treatment laterality (unilateral versus bilateral plans). Despite these differences in target geometry and planning complexity, the workflow produced plans with comparable DVH performance and similar NTCP estimates without site-specific tuning, suggesting reasonable robustness within this cohort. However, this standardization comes with reduced flexibility. A general mimicking setting supports reproducibility and automation, but it does not explicitly accommodate patient-specific clinical preferences or individualized planning priorities. Therefore, the current workflow may be less suitable for cases that require highly personalized trade-offs. In such cases, the automatically generated plans may still serve as useful starting points for subsequent planner-guided refinement.

The changes in plan complexity were not consistent across all metrics (Figure 5.6). Automated plans showed lower MU/cGy, SAS, and CoA, suggesting reduced beam modulation, fewer small apertures, and less irregular aperture shapes. This may be favorable, as highly complex VMAT plans have been associated with greater uncertainty in dose calculation and delivery and, in some settings, poorer patient-specific QA performance [116]. However, automated plans also showed higher MLC leaf travel, indicating increased mechanical motion during delivery. Overall, these findings suggest that the automated plans were less complex than the clinical plans across most

evaluated metrics, although plan complexity remains multidimensional and cannot be fully summarized by a single measure.

Several limitations should be acknowledged. First, this was a single-institution study with a relatively small temporally independent test cohort ($n = 30$), and multi-institutional validation on larger datasets is needed to assess generalizability more rigorously. Second, the NTCP analysis should be interpreted with caution. Although NTCP provides a clinically relevant estimate of toxicity risk, it remains a model-based surrogate rather than a direct measure of observed clinical outcome [115]. Moreover, the applied NTCP models are largely driven by OAR mean dose, which may limit their sensitivity to subtle dose differences between plans.

In summary, we presented a fully automated planning pipeline that converts predicted dose distributions from a deep learning model into clinically deliverable treatment plans for head and neck radiotherapy using dose mimicking in a commercial TPS. The proposed workflow generated plans with comparable DVH quality and predicted toxicity outcomes to clinical plans, while showing lower values for several modulation-related complexity metrics despite increased MLC leaf travel. These findings support the feasibility of integrating deep learning dose prediction with dose mimicking for automated treatment planning in head and neck radiotherapy.

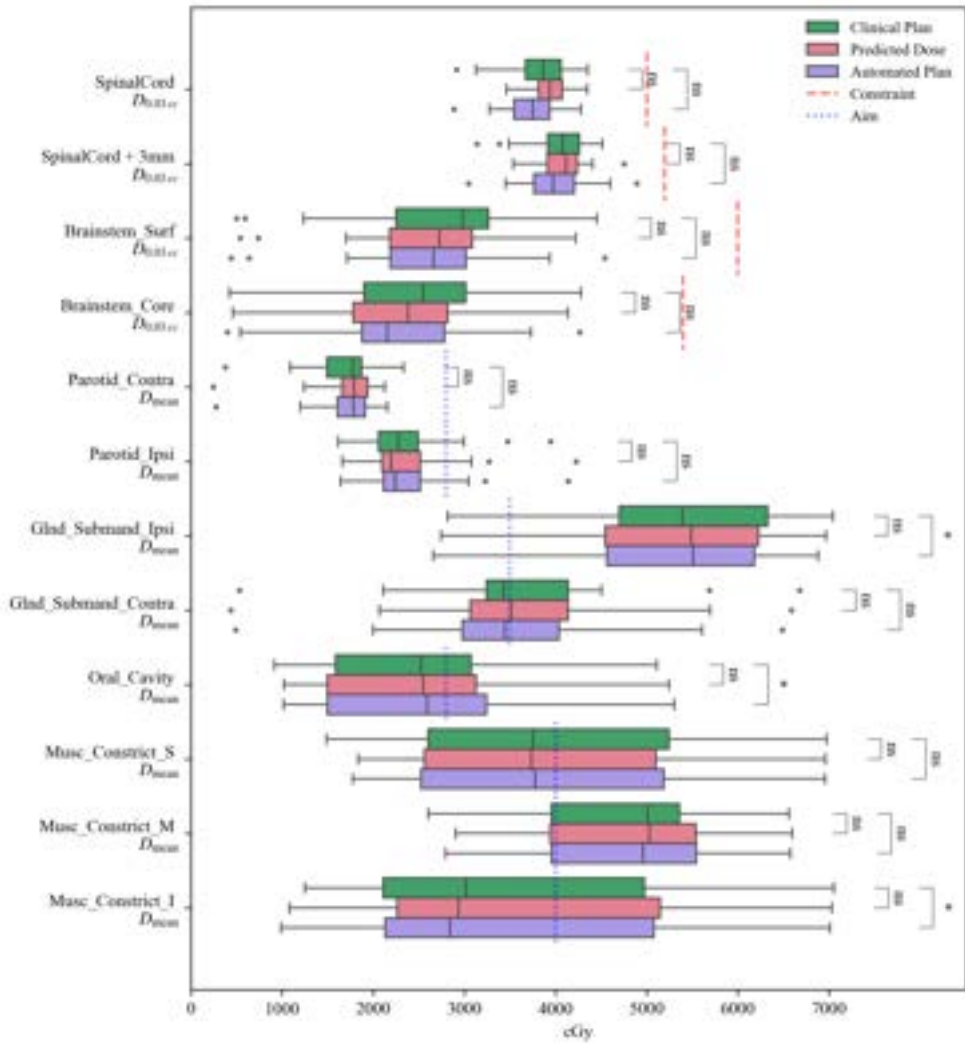


Figure 5.4: Comparison of OAR DVH metrics between clinical plans, predicted dose, and automated plans. Parotid and submandibular glands were categorized as ipsilateral (Ipsi) or contralateral (Contra) according to the mean dose received. For D_{mean} and $D_{0.03cc}$, both the constraint and the aim are defined as $\leq$ the corresponding threshold. The detailed threshold values are provided in Table 5.1. *: $p < 0.05$, ns = not significant.

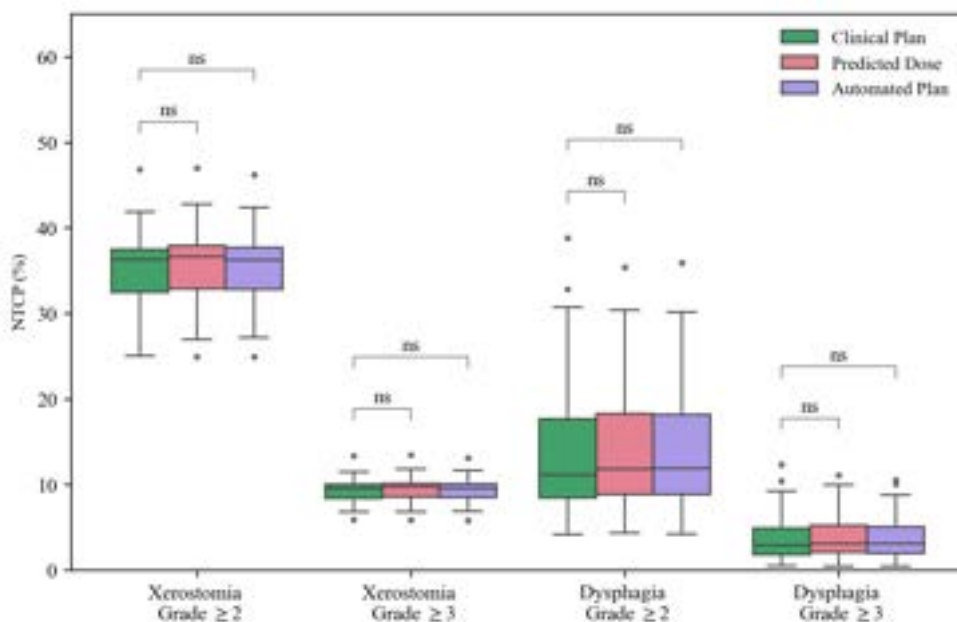


Figure 5.5: Comparison of NTCP estimates between clinical plans, predicted dose, and automated plans for xerostomia and dysphagia endpoints. *: $p < 0.05$, ns = not significant.

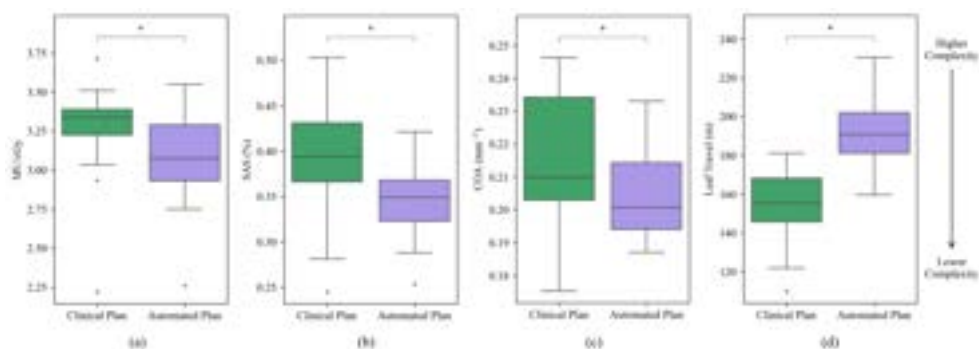


Figure 5.6: Comparison of plan complexity metrics between clinical and automated plans. For all metrics, lower values indicate lower plan complexity. *: $p < 0.05$.

6

Summary and Future Work

6.1 Summary

We investigated automated radiotherapy planning for head and neck cancer, with a primary focus on deep learning-based dose prediction and the generation of clinically deliverable VMAT plans through dose mimicking. We also explored generalizable medical image segmentation methods that are relevant to radiotherapy and broader medical imaging applications. Across this thesis, we addressed key components of automated radiotherapy planning, including accurate dose prediction, clinically guided model optimization, and integration into a commercial treatment planning system (TPS).

Chapter 2 presented MCP-MedSAM, a lightweight and modality-agnostic medical image segmentation foundation model. To improve segmentation performance across different imaging modalities while maintaining computational efficiency, we introduced modality prompts and content prompts that provide additional task-relevant information to the model with minimal overhead. We further adopted a modality-based data sampling strategy to alleviate data imbalance across modalities. The results demonstrated that MCP-MedSAM achieved strong segmentation performance on a large-scale benchmark while remaining trainable on a single A100 GPU within one day, providing a practical and scalable framework for general medical image segmentation.

Chapter 3 investigated key factors influencing deep learning-based 3D dose prediction for head and neck photon radiotherapy. We systematically evaluated the effects of input and dose-grid resolution, input representation, loss function, model architecture, and robustness to noise using both a public dataset and an in-house clinical dataset. The study showed that high-resolution and anatomically informative inputs, including CT, PTVs, and OARs, substantially improved prediction accuracy. We also found that combining voxel-wise and DVH-oriented losses improved clinically relevant performance, while robustness analyses revealed that most models were relatively robust to Poisson noise but more sensitive to adversarial perturbations. These findings provide practical guidance for designing accurate and reliable dose prediction models.

Chapter 4 proposed a clinical DVH metric loss (CDM loss) for 3D dose prediction,

aiming to better align model training with clinical plan evaluation criteria. Instead of relying only on voxel-wise supervision, the proposed loss directly incorporated differentiable formulations of commonly used clinical DVH metrics, including both *D-metrics* and surrogate formulations of *V-metrics*. In addition, we introduced a lossless bit-mask encoding scheme to efficiently represent multiple overlapping ROIs and thereby substantially reduce computational cost during training. The results showed that the CDM loss improved target coverage and satisfied all clinical constraints while maintaining comparable OAR sparing, demonstrating that clinically guided supervision can produce dose predictions that are better aligned with treatment planning requirements.

Chapter 5 integrated the trained dose prediction model into a commercial TPS (RayStation) and established a fully automated planning pipeline based on dose mimicking. Predicted dose distributions were imported into the TPS and used to guide dose mimicking optimization for the generation of clinically deliverable dual-arc VMAT plans. On an independent temporal test cohort, the automated plans achieved target coverage comparable to clinical plans, with similar or improved sparing for several OARs and no significant differences in NTCP estimates for xerostomia and dysphagia. In addition, the automatically generated plans showed lower modulation-related complexity for several plan complexity metrics. These results demonstrated the feasibility of combining deep learning-based dose prediction with dose mimicking to generate clinically deliverable head and neck radiotherapy plans in a fully automated manner.

6.2 Discussion and future work

The work presented in this thesis supports the feasibility of data-driven radiotherapy planning for head and neck cancer, from clinically guided dose prediction to deliverable VMAT plans. Taken together, these results point toward automated planning frameworks that more closely link patient anatomy, dose prediction, and delivery feasibility.

Although this work focuses on head and neck cancer, it has the potential to generalize to other treatment sites, particularly those with less complex anatomical and planning requirements. While such extensions would require additional data collection and validation, radiotherapy provides a favorable setting for data-driven approaches, as high-quality manual treatment plans have long been generated in routine clinical practice and can serve as training data. Even as automated planning becomes more widespread, clinician-reviewed and modified plans may continue to provide valuable supervision, enabling iterative model improvement over time.

From a practical perspective, multiple technical pathways exist for automated planning, including approaches such as iCycle [118] and ECHO [119]. The approach

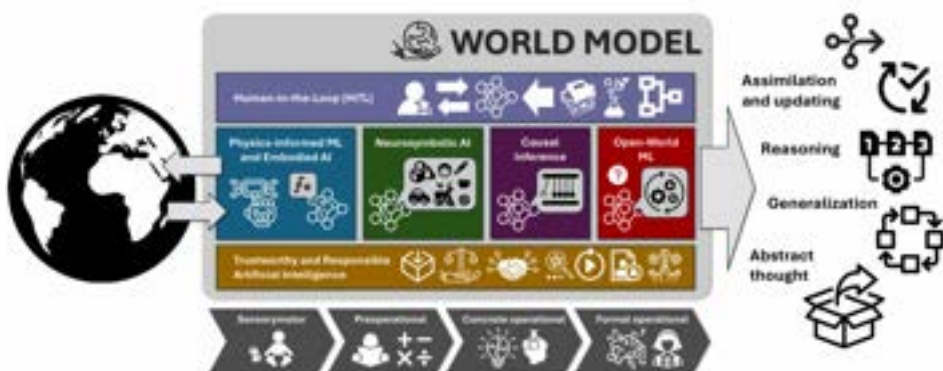


Figure 6.1: A world model built upon six research areas: (1) embodied AI and physics-informed machine learning, allowing AI systems to interact with the physical world; (2) neuro-symbolic approaches that integrate statistical learning with symbolic representations; (3) causal inference to capture action–effect relationships in observed data; (4) open-world machine learning to enable adaptation to unknown situations; (5) human-in-the-loop approaches that incorporate human knowledge and oversight; and (6) trustworthy and responsible AI to ensure accountability and reliability. Together, these research areas illustrate possible components of future reasoning-capable and human-centered AI systems. Figure reproduced from [117].

explored in this thesis, based on dose prediction and dose mimicking within a commercial TPS, offers advantages in terms of computational efficiency and clinical integration. In the head and neck cases studied here, fully automated plan generation can be achieved in approximately 15 minutes, suggesting that such methods may provide a pragmatic balance between plan quality, robustness, and speed in real-world clinical workflows.

From a broader artificial intelligence perspective, future planning systems may be viewed through the lens of “world model”, that is, integrated representations of the environment, its constraints, and the consequences of possible actions [117]. As illustrated in Figure 6.1, such systems may combine embodied and physics-informed learning, neuro-symbolic reasoning, causal inference, open-world learning, human-in-the-loop interaction, and trustworthy and responsible AI. This perspective provides a useful framework for interpreting the future directions discussed below.

In radiotherapy, the relevant “world” includes patient anatomy, the physical processes governing radiation transport and dose deposition, the mechanical constraints of treatment delivery systems, and the clinical judgments involved in balancing target coverage against organ sparing [3, 7, 18]. Within this context, end-to-end planning and personalized, interactive planning can be broadly viewed as complementary steps

toward a radiotherapy-specific world model, with particular relevance to embodied and physics-informed learning, human-in-the-loop guidance, and trustworthy and responsible AI.

6.2.1 Toward end-to-end radiotherapy planning

The framework developed in Chapter 5 reduces the need for manual trial-and-error, but it still consists of sequential components: although dose prediction can be substantially improved through appropriate input design and clinically guided loss functions (Chapters 3 and 4), the predicted dose distributions must still be translated into deliverable plans through dose mimicking within a commercial TPS. This separation limits the integration of learning-based models with the physics of treatment delivery. A natural next step is therefore the development of more unified end-to-end planning frameworks.

Radiotherapy planning is fundamentally governed by well-established physical models. In conventional systems, dose calculation is based on beamlet–voxel formulations, and optimization algorithms search for machine parameters that satisfy clinical objectives while respecting delivery constraints [18, 19]. These models are reliable and interpretable, but they are also computationally expensive and typically rely on iterative optimization. Integrating physics-based planning with modern machine learning remains a major challenge, because accurate dose calculation, deliverability constraints, and clinical objective trade-offs must be modeled jointly within a computationally efficient and clinically reliable framework.

The results in Chapters 3 and 4 suggest that learned dose prediction models can capture clinically relevant relationships between anatomy and achievable dose, particularly when anatomically informative inputs and clinically motivated supervision are used. A natural next step is to extend such models so that they not only predict voxel dose, but also reason about the physical and mechanical constraints of treatment delivery. This may involve combining image-derived anatomical information with representations of dose deposition and delivery constraints. Examples include differentiable or surrogate dose calculation models [120], physics-informed neural networks [26, 27], and learned optimization policies [8, 121]. Explicit modeling of delivery parameters such as MLC trajectories, gantry motion, and dose-rate modulation may allow future systems to reason jointly about dose quality and delivery feasibility, thereby reducing reliance on intermediate steps such as dose mimicking and enabling faster, potentially sub-minute, end-to-end plan generation.

A practical limitation, however, is that the pipeline in Chapter 5 depends on a commercial TPS, where key components such as dose calculation and optimization engines are implemented as closed-source software. As a result, researchers can interact with these systems mainly through predefined scripting interfaces and can

influence the optimization process only indirectly through objective settings and constraints. This limits the extent to which learning-based methods can be integrated with the underlying physical planning process.

These limitations are particularly relevant for applications such as online adaptive radiotherapy, where rapid replanning is required to account for daily anatomical variation [80, 9]. In such settings, the need for reliable and scalable anatomical delineation also becomes especially important, linking future end-to-end planning not only to the dose prediction work in Chapters 3–5, but also to the segmentation ideas developed in Chapter 2. Existing optimization pipelines remain computationally demanding, and many TPS engines still rely heavily on CPU-based iterative solvers. Advances in GPU-accelerated dose calculation, differentiable optimization, and hybrid physics-informed learning approaches may help make real-time or near-real-time automated replanning more practical. At the same time, any move toward end-to-end planning must still ensure physical consistency, interpretability, safety, and regulatory compliance.

6.2.2 Personalized and interactive planning through human preferences and human–AI collaboration

Although automation can reduce manual workload, radiotherapy planning remains inherently dependent on clinical judgment and preference variability. Different clinicians may prioritize target coverage, organ-at-risk sparing, toxicity risk, or quality-of-life considerations differently depending on patient condition, prognosis, and institutional practice [20, 21]. These variations are difficult to encode in a fixed set of optimization objectives, motivating the need for systems that can both represent clinician intent and support interactive decision-making.

Although Chapters 3 and 4 did not directly address inter-clinician preference variability, they show that automated planning performance depends not only on anatomy, but also on how clinical priorities are represented. Chapter 3 highlighted the importance of input and supervision design, while Chapter 4 showed that explicitly encoding clinical evaluation criteria in the loss function can improve the clinical relevance of predicted dose distributions.

One possible direction is preference-conditioned or prompt-based planning, in which clinicians provide high-level guidance to steer automated plan generation [122, 123]. Such prompts could include, for example, prioritizing specific organs-at-risk, specifying acceptable toxicity thresholds, selecting planning templates, or defining trade-offs between competing objectives (e.g., improved parotid sparing at the expense of target homogeneity). Extending this paradigm may enable more flexible and user-adaptive planning systems.

Another direction is learning from clinician interaction and feedback. In routine

workflows, automated plans are often reviewed and further refined by planners and other clinical staff involved in treatment planning. These refinements can be interpreted as signals of mismatch between model outputs and clinical expectations. Analogous to autonomous driving systems, where human interventions are treated as indicators of model failure and used to improve model performance, a similar feedback loop could be established in radiotherapy. Modifications to auto-generated plans could be systematically recorded and incorporated into model retraining, enabling continuous alignment with clinical practice.

Because radiotherapy planning is fundamentally a multi-criteria optimization problem, multiple Pareto-optimal solutions may be clinically acceptable [18]. A natural extension is therefore to generate multiple clinically plausible candidate plans that reflect different trade-offs between competing objectives. However, presenting multiple solutions introduces practical challenges. Clinicians are already subject to information overload, and displaying too many alternatives may reduce usability. Important questions include how many candidate plans should be presented, how to ensure that solutions are meaningfully diverse, and how to enable efficient navigation of the solution space. Possible directions include generating a limited set of representative solutions, using diversity-aware selection criteria, and developing interfaces that allow clinicians to navigate trade-offs efficiently without excessive cognitive burden [124].

To support effective human–AI collaboration, future systems may incorporate tools such as real-time DVH steering, rapid re-optimization under updated constraints, and uncertainty-aware plan comparison. Uncertainty may arise from multiple sources, including anatomical and delineation uncertainty, dose prediction model uncertainty, and optimization-related variability. Explicitly quantifying and communicating these uncertainties may improve interpretability and support more informed clinical decision-making.

Overall, future automated planning systems should not only generate high-quality plans, but also function as clinical decision-support tools that allow clinicians to explore, refine, and personalize treatment strategies in a controlled and interpretable manner.

6.3 General conclusions

This thesis demonstrates the potential of data-driven methods for automated radiotherapy planning in head and neck cancer, particularly for deep learning–based dose prediction and the generation of clinically deliverable VMAT plans through dose mimicking. It also highlights the broader relevance of generalizable medical image segmentation for scalable radiotherapy and medical imaging workflows.

Overall, the results show that clinically meaningful dose distributions can be learned from patient anatomy and translated into deliverable treatment plans within

a commercial TPS. These findings support the feasibility of planning workflows that are more efficient and consistent, with less reliance on manual trial-and-error. Further progress will depend on continued advances in machine learning, physics-informed modeling, and human–AI collaboration, together with close collaboration among clinicians, medical physicists, and researchers.

7

Nederlandse samenvatting

7.1 Samenvatting

In dit proefschrift onderzochten we geautomatiseerde radiotherapieplanning voor hoofd-halskanker, met een primaire focus op dosisvoorspelling op basis van deep learning en het genereren van klinisch uitvoerbare VMAT-plannen door middel van dose mimicking. Daarnaast onderzochten we generaliseerbare methoden voor segmentatie van medische beelden die relevant zijn voor zowel radiotherapie als bredere toepassingen binnen de medische beeldvorming. In dit proefschrift behandelden we verschillende essentiële onderdelen van geautomatiseerde radiotherapieplanning, waaronder nauwkeurige dosisvoorspelling, klinisch gestuurde modeloptimalisatie en integratie in een commercieel behandelplanningssysteem (TPS).

Hoofdstuk 2 presenteerde MCP-MedSAM, een lichtgewicht en modaliteitsonafhankelijk foundation model voor de segmentatie van medische beelden. Om de segmentatieprestaties voor verschillende beeldvormingsmodaliteiten te verbeteren en tegelijkertijd de computationele efficiëntie te behouden, introduceerden we modaliteitsprompts en inhoudsprompts, die met minimale extra rekenkosten aanvullende taakspecifieke informatie aan het model verstrekken. Daarnaast pasten we een op modaliteit gebaseerde datasamplingstrategie toe om de onbalans tussen de verschillende modaliteiten te verminderen. De resultaten toonden aan dat MCP-MedSAM sterke segmentatieprestaties behaalde op een grootschalige benchmark, terwijl het model binnen één dag op één A100-GPU kon worden getraind. Daarmee biedt MCP-MedSAM een praktisch en schaalbaar raamwerk voor de algemene segmentatie van medische beelden.

Hoofdstuk 3 onderzocht de belangrijkste factoren die van invloed zijn op deep-learninggebaseerde driedimensionale dosisvoorspelling voor fotonenradiotherapie bij hoofd-halskanker. Met behulp van zowel een openbare dataset als een interne klinische dataset evalueerden we systematisch de effecten van de resolutie van de invoer en het dosISRaster, de invoerrepresentatie, de verliesfunctie, de modelarchitectuur en de robuustheid tegen ruis. Uit het onderzoek bleek dat invoergegevens met een hoge resolutie en uitgebreide anatomische informatie, waaronder CT-beelden, PTV's en OAR's, de nauwkeurigheid van de voorspellingen aanzienlijk verbeterden. Daarnaast stelden we vast dat het combineren van voxelgewijze en DVH-georiënteerde verliesfuncties

de klinisch relevante prestaties verbeterde. Uit de robuustheidsanalyses bleek dat de meeste modellen relatief robuust waren tegen Poisson ruis, maar gevoeliger waren voor adversariële verstoringen. Deze bevindingen bieden praktische richtlijnen voor het ontwerpen van nauwkeurige en betrouwbare modellen voor dosisvoorspelling.

Hoofdstuk 4 introduceerde een verliesfunctie gebaseerd op klinische DVH-metrieken, de zogenoemde clinical DVH metric loss (CDM loss), voor driedimensionale dosisvoorspelling. Het doel hiervan was om de training van het model beter af te stemmen op de klinische criteria voor de beoordeling van behandelplannen. In plaats van uitsluitend gebruik te maken van voxelgewijze supervisie, integreerde de voorgestelde verliesfunctie direct differentieerbare formuleringen van veelgebruikte klinische DVH-metrieken, waaronder zowel *D-metrieken* als surrogaatformuleringen van *V-metrieken*. Daarnaast introduceerden we een verliesvrije bitmask-encoding waarmee meerdere overlappende ROI's efficiënt konden worden gerepresenteerd, waardoor de computationele kosten tijdens de training aanzienlijk werden verlaagd. De resultaten toonden aan dat de CDM loss de dekking van de doelvolumes verbeterde en aan alle klinische randvoorwaarden voldeed, terwijl een vergelijkbare sparing van de OAR's behouden bleef. Dit toont aan dat klinisch gestuurde supervisie kan leiden tot dosisvoorspellingen die beter aansluiten bij de vereisten van de behandelplanning.

Hoofdstuk 5 integreerde het getrainde dosisvoorspellingsmodel in een commercieel TPS (RayStation) en ontwikkelde een volledig geautomatiseerde planningspipeline op basis van dose mimicking. De voorspelde dosisverdelingen werden in het TPS geïmporteerd en gebruikt om de optimalisatie door middel van dose mimicking aan te sturen, met als doel klinisch uitvoerbare VMAT-plannen met twee arcs te genereren. In een onafhankelijk temporeel testcohort bereikten de geautomatiseerde plannen een dekking van de doelvolumes die vergelijkbaar was met die van de klinische plannen. Voor verschillende OAR's werd een vergelijkbare of verbeterde sparing bereikt en er werden geen significante verschillen gevonden in de NTCP-schattingen voor xerostomie en dysfagie. Daarnaast vertoonden de automatisch gegenereerde plannen voor verschillende maten van plancomplexiteit een lagere modulatiegerelateerde complexiteit. Deze resultaten tonen aan dat deep-learninggebaseerde dosisvoorspelling kan worden gecombineerd met dose mimicking om op volledig geautomatiseerde wijze klinisch uitvoerbare radiotherapieplannen voor hoofd-halskanker te genereren.

7.2 Algemene conclusies

Dit proefschrift toont het potentieel aan van datagestuurde methoden voor geautomatiseerde radiotherapieplanning bij hoofd-halskanker, in het bijzonder voor dosisvoorspelling op basis van deep learning en het genereren van klinisch uitvoerbare VMAT-plannen door middel van dose mimicking. Daarnaast benadrukt het proefschrift het bredere belang van generaliseerbare segmentatie van medische beelden voor

schaalbare workflows binnen de radiotherapie en medische beeldvorming.

Over het geheel genomen laten de resultaten zien dat klinisch betekenisvolle dosisverdelingen kunnen worden afgeleid uit de anatomie van de patiënt en binnen een commercieel TPS kunnen worden vertaald naar uitvoerbare behandelplannen. Deze bevindingen ondersteunen de haalbaarheid van planningsworkflows die efficiënter en consistenten zijn en minder afhankelijk zijn van handmatige trial-and-error. Verdere vooruitgang vereist voortdurende ontwikkelingen op het gebied van machine learning, fysicageïnformeerde modellering en samenwerking tussen mens en artificiële intelligentie, evenals nauwe samenwerking tussen klinici, klinisch fysici en onderzoekers.

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List of Publications

Journal articles

Jun Ma, Feifei Li, Sumin Kim, Reza Asakereh, Bao-Hiep Le, Dang-Khoa Nguyen-Vu, Alexander Pfefferle, Muxin Wei, **Ruochen Gao**, Donghang Lyu, et al., "Efficient MedSAMs: Segment Anything in Medical Images on Laptop", arXiv preprint arXiv:2412.16085, 2024.

Ruochen Gao, Prerak Mody, Chinmay Rao, Frank Dankers, and Marius Staring, "On Factors that Influence Deep Learning-Based Dose Prediction of Head and Neck Tumors", *Physics in Medicine & Biology*, vol. 70, no. 11, p. 115006, 2025.

Donghang Lyu, **Ruochen Gao**[†], and Marius Staring, "MCP-MedSAM: A Powerful Lightweight Medical Segment Anything Model Trained with a Single GPU in Just One Day", *Machine Learning for Biomedical Imaging*, vol. 3, pp. 135–151, May 2025.

Ruochen Gao, Marius Staring, and Frank Dankers, "Clinical DVH metrics as a loss function for 3D dose prediction in head and neck radiotherapy", accepted for publication in the *International Journal of Computer Assisted Radiology and Surgery (IJCARS)*, CARS 2026 Special Issue.

Ruochen Gao, Noud van Ruremonde, Alice Onderwater, Martin de Jong, Marius Staring, and Frank Dankers, "Automated Head and Neck Radiotherapy Planning from Deep Learning-Based Dose Prediction via Dose Mimicking", submitted, 2026.

International conference proceedings

Ruochen Gao, Donghang Lyu, and Marius Staring, "Swin-LiteMedSAM: A lightweight box-based segment anything model for large-scale medical image datasets", in *Medical Image Segmentation Challenge*, pp. 70–82, Cham: Springer Nature Switzerland, 2024.

Ruochen Gao, Prerak Mody, Chinmay Rao, Frank Dankers, and Marius Staring, "Deep Learning-Based Dose Prediction for Head and Neck Tumors: Influence of Loss Function and Model Architecture", *Radiotherapy and Oncology*, vol. 206, pp. S2792–S2794, 2025.

Portfolio

Leiden University Medical Center Graduate School

Graduate School Courses

- Introductory Meeting of Leiden University
- Scientific Conduct for PhDs
- Code Management with Git
- Basic Methods and Reasoning in Biostatistics

ASCI School Courses

- A Programmer's Guide for Modern High-Performance Computing
- Computer Vision by Learning

Summer School

- AI4Health Summer School, Paris (2025)

International Conferences

- Medical Imaging with Deep Learning (MIDL), Zurich (2022)
- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Online (2024)
- European Society for Radiotherapy and Oncology (ESTRO), Vienna (2025)
- Computer Assisted Radiology and Surgery (CARS), Nagoya (2026)

Awards

- Segment Anything in Medical Images on Laptop CVPR 2024 Challenge; Team Leader, 4th Place
- Generalizable Dose Prediction for Heterogeneous Multi-Cohort and Multi-Site Radiotherapy Planning AAPM 2025 Challenge; Team Leader, 3rd Place

Supervision

- Co-supervision of a Bachelor's student, Vrije Universiteit Amsterdam (VU Amsterdam)
- Co-supervision of a Master's student, Technische Universiteit Delft (TU Delft)

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Curriculum Vitae

Ruochen Gao was born in Tongling, Anhui Province, China, in August 1996. In September 2011, he began his studies at Tongling No. 1 High School and graduated in June 2014. In September 2014, he began his undergraduate studies in Communication Engineering at Hunan University and graduated in June 2018. In September 2018, he began his master's studies in Computer Science at the University of Chinese Academy of Sciences and graduated in June 2021. From April 2021 to December 2021, he spent some time with a startup and got a brief taste of startup life until the company came to an end in late 2021. In February 2022, he began his PhD studies at the Laboratorium voor Klinische en Experimentele Beeldverwerking (LKEB), Leiden University Medical Center, in the Netherlands. His PhD research focuses on automated radiotherapy planning for head and neck cancer using deep learning techniques.

